

5G NR and 5G Core, explained

Fourteen topics from the air interface to the core network, each with diagrams, a step-by-step explanation and the 3GPP references.

Written from the 3GPP specifications. Diagrams are simplified and not to scale. Clause numbers refer to the Release 15 to 18 versions of the specifications named; check the current version before relying on a detail.

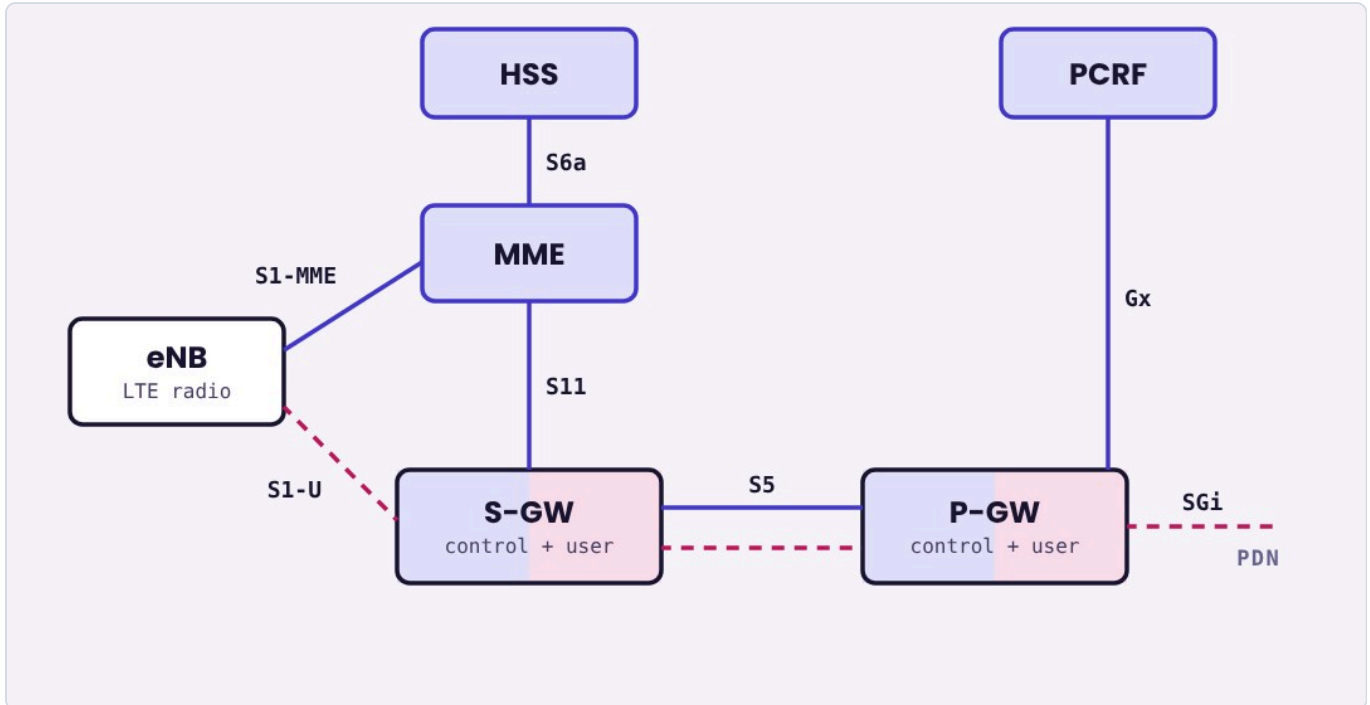
Contents

01	From EPC to the 5G Core	7 steps
02	5G Core Network Functions	9 pages
03	5G Protocol Stacks	7 pages
04	Inside the gNB	8 steps
05	5G Deployment Options	9 steps
06	Dual Connectivity	8 steps
07	Secondary Node Addition	8 steps
08	The 5G QoS Model	8 steps
09	5G QoS: Rules, Parameters and 5QI	8 pages
10	RM, CM and RRC States	8 steps
11	RRC Inactive	8 steps
12	5G Access Control	8 steps
13	5G SA Initial Registration	13 steps
14	5G NR Bands, NR-ARFCN and Bandwidth	9 steps

From EPC to the 5G Core

How MME, S-GW and P-GW became AMF, SMF and UPF, in two steps.

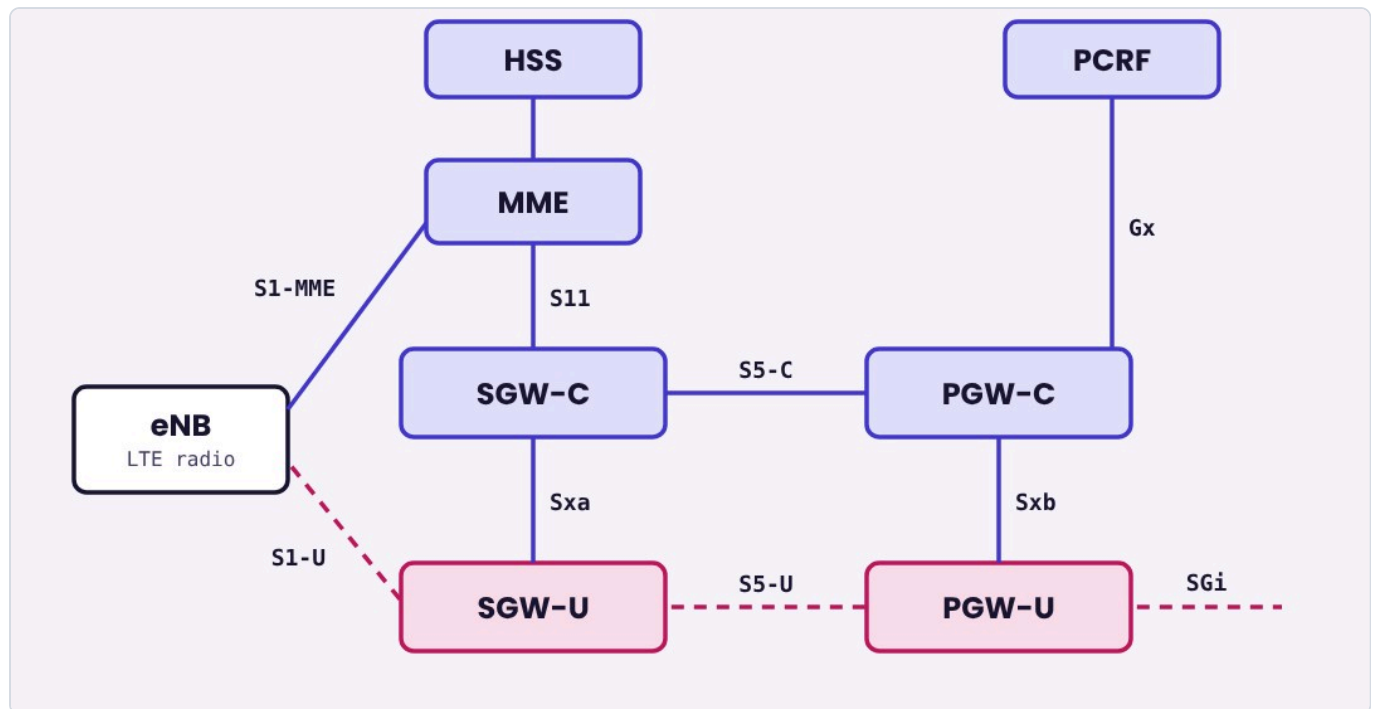
1.1 The 4G core today



In the EPC, the MME handles signalling, mobility and session control. The S-GW and P-GW each do two jobs at once: they take part in the signalling and they forward the user data.

TS 23.401 §4.2

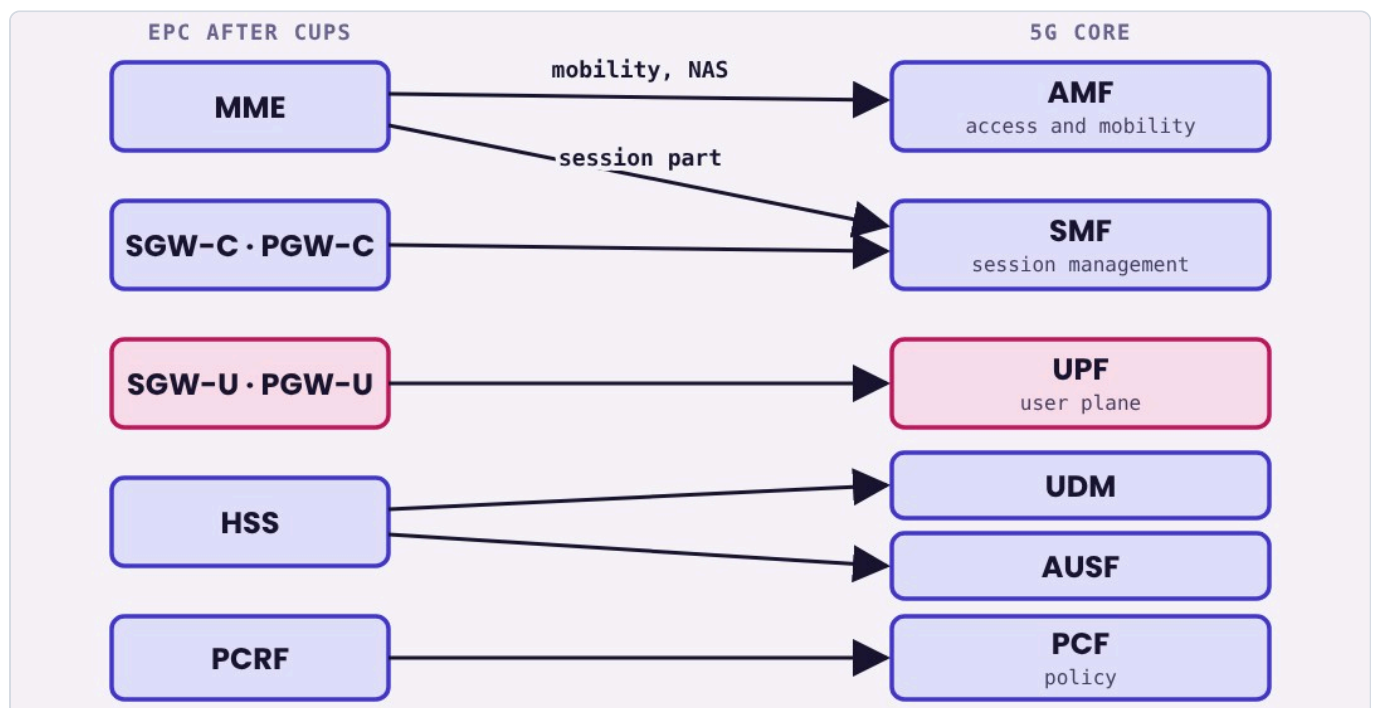
1.2 Step one: CUPS



Release 14 split each gateway in two. SGW-C and PGW-C keep the control part. SGW-U and PGW-U only forward packets. The control part steers the user part over the new Sx interfaces, so user plane nodes can be scaled and placed close to the user.

TS 23.214 · TS 29.244

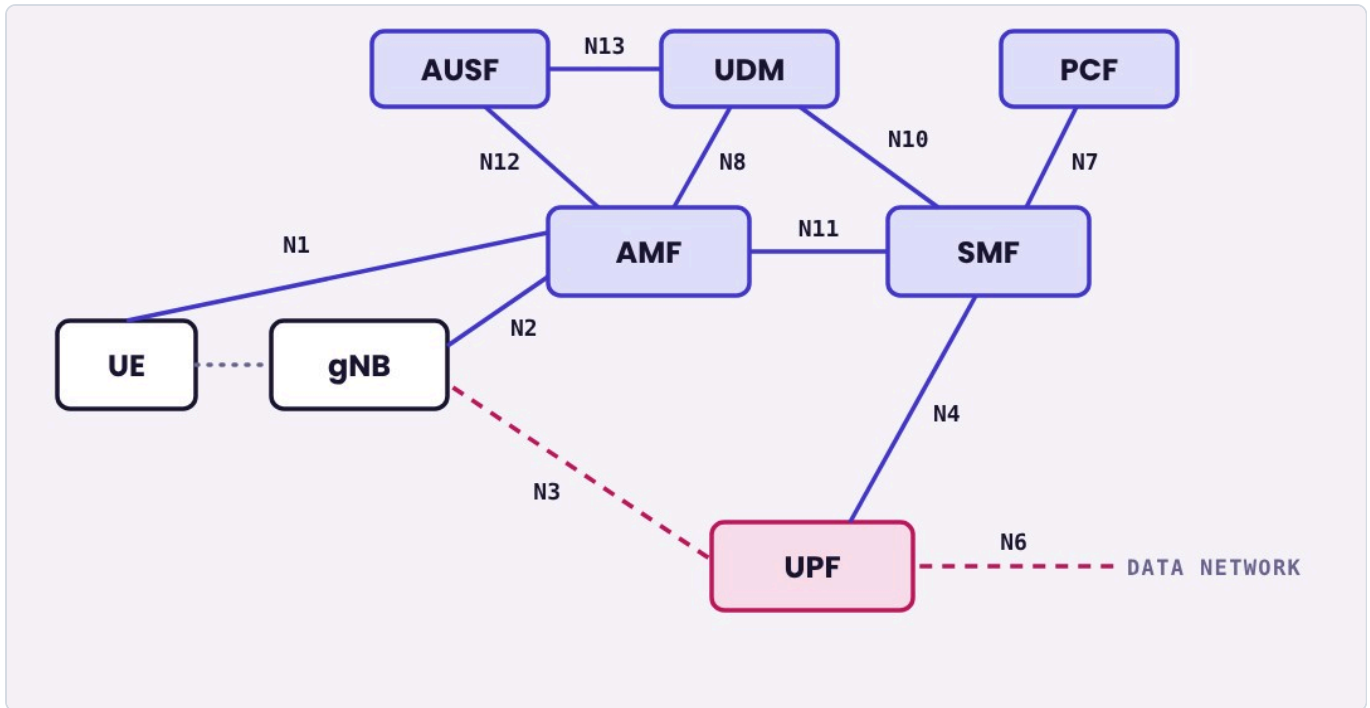
1.3 Step two: regroup the functions



5G regroups these pieces. The mobility part of the MME becomes the AMF. Its session part joins SGW-C and PGW-C in the SMF. The two user plane gateways merge into the UPF. The HSS becomes UDM and AUSF, and the PCRF becomes the PCF.

TS 23.501 §4.2 · TS 23.214

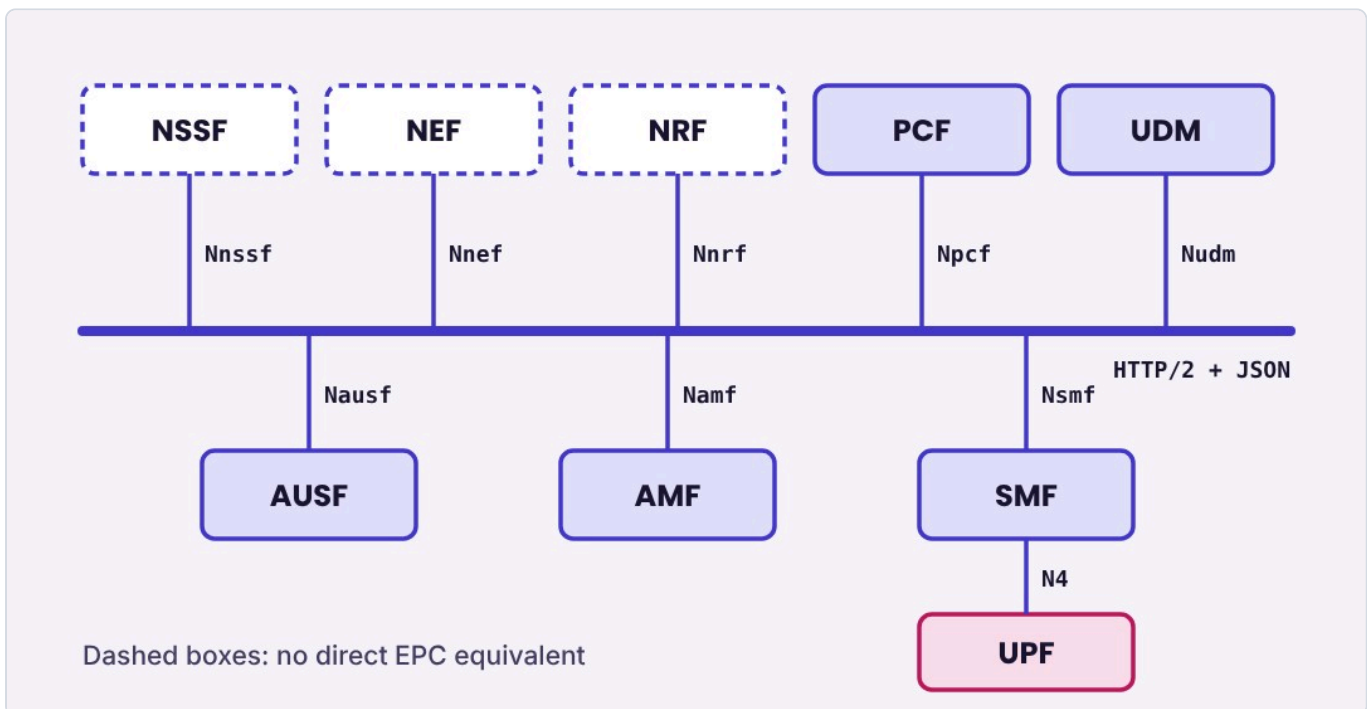
1.4 The 5G core as reference points



N1 and N2 carry signalling from the UE and the gNB to the AMF. N3 carries user data to the UPF. The SMF controls the UPF over N4, using PFCP, the same protocol CUPS introduced for Sx.

TS 23.501 §4.2.3 · TS 29.244

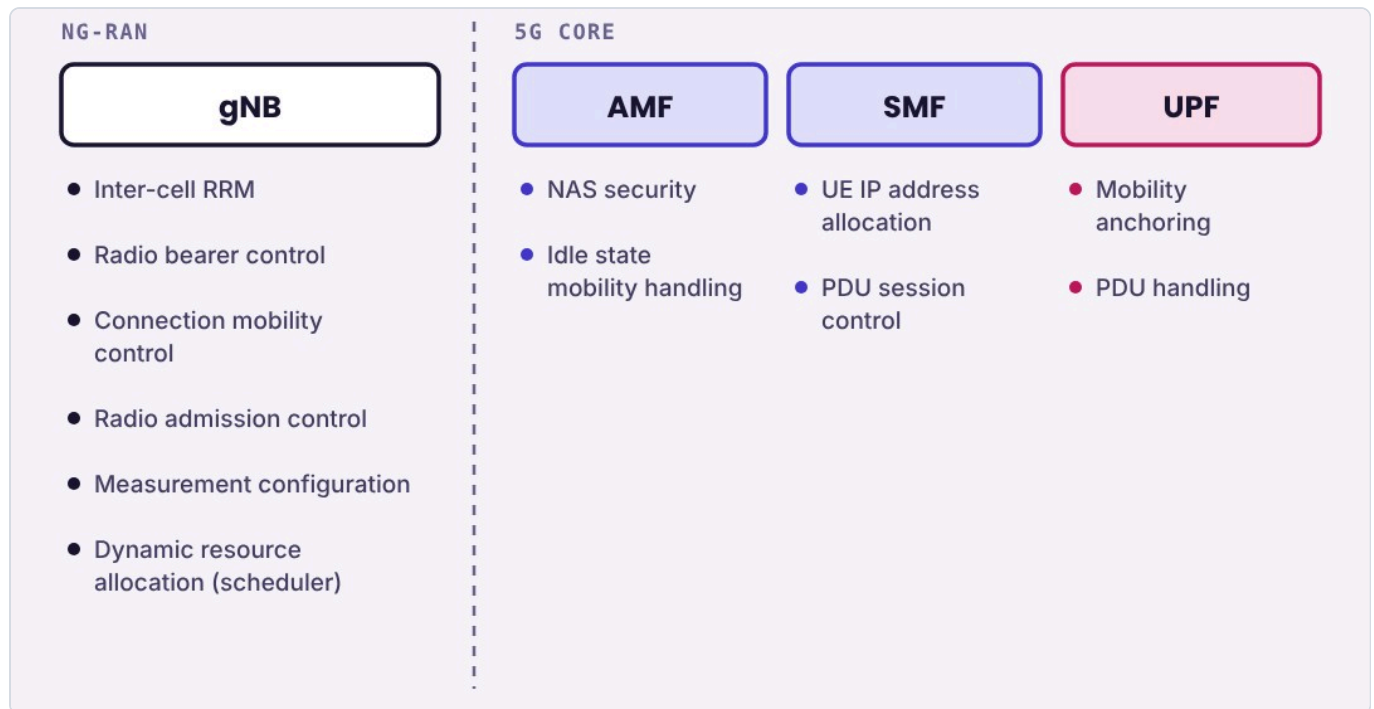
1.5 The 5G core as services



The control functions no longer need a dedicated interface for every pair. Each one offers services over HTTP/2 and the others call them. NRF, NSSF and NEF are new: service discovery, slice selection and exposure to applications.

TS 23.501 §4.2.6 · TS 29.500

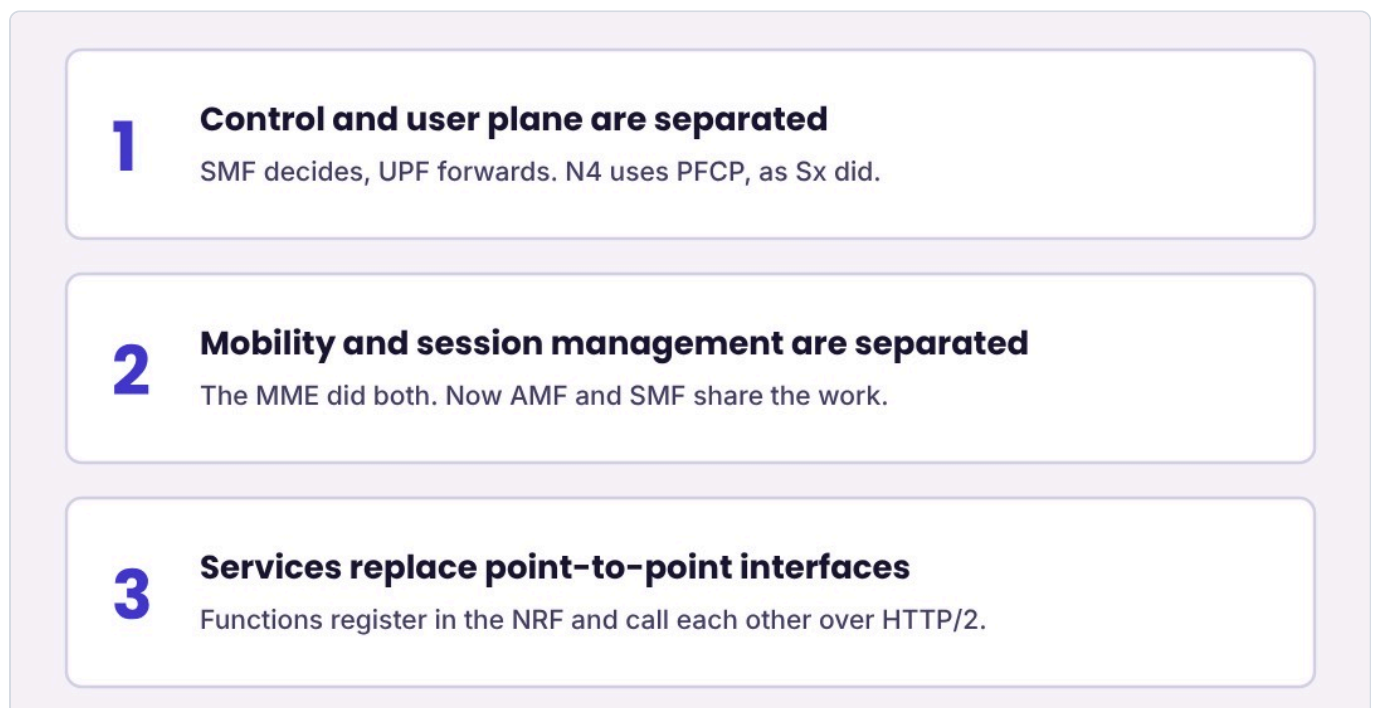
1.6 Who does what: NG-RAN and 5GC



The radio side keeps radio resource management and scheduling. The AMF handles NAS security and idle mode mobility. The SMF allocates the UE IP address and controls PDU sessions. The UPF anchors mobility and handles the packets.

TS 38.300 §4.2

1.7 Three changes to remember



The 5G core is the EPC taken apart and put back together: control separated from user plane, mobility separated from sessions, and services in place of fixed interfaces.

TS 23.501 · TS 23.214 · TS 38.300

5G Core Network Functions

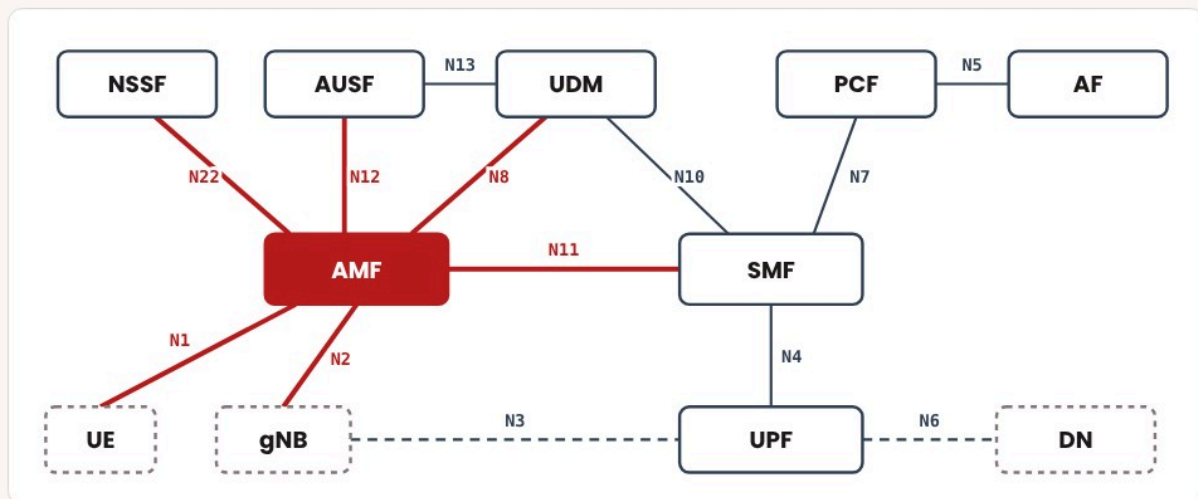
Who does what in the 5G core: one page per network function, then the reference points and the 4G to 5G vocabulary.

5G NR SERIES · CORE NETWORK

NETWORK FUNCTIONS

AMF

Access and Mobility Management Function



The front door of the core. Every UE has one AMF, and all its signalling passes through it.

- Terminates **N2** from the RAN and **N1**, the NAS signalling from the UE
- **NAS ciphering** and integrity protection
- **Registration, connection, reachability and mobility** management
- Access **authentication and authorization**; holds the security anchor (SEAF)
- **Transports session management messages** between the UE and the SMF

4G EQUIVALENT

MME (mobility part)

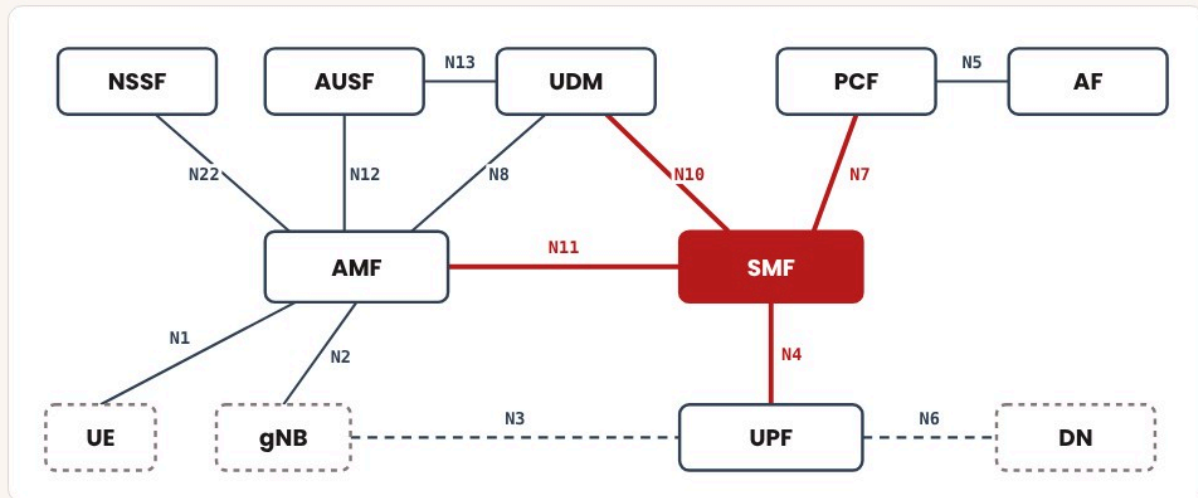
REFERENCE POINTS

N1 · N2 · N8 · N11 · N12 · N22

TS 23.501 §6.2.1

SMF

Session Management Function



Sets up and looks after PDU sessions. It decides; the UPF carries out.

- PDU session **establishment, modification and release**
- **UE IP address** allocation and management
- **Selects and controls the UPF**, and configures traffic steering there
- Handles the **session management part of NAS**
- Terminates the interface to the **policy function**; supports charging
- **Downlink data notification** when data arrives for an idle UE

4G EQUIVALENT

MME (session part) + SGW-C + PGW-C

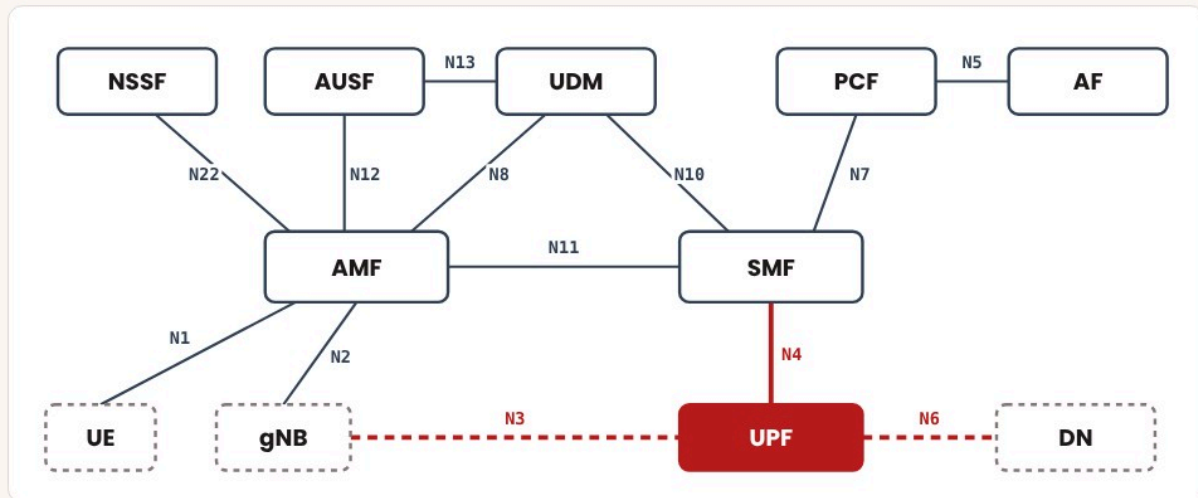
REFERENCE POINTS

N4 • N7 • N10 • N11

TS 23.501 §6.2.2

UPF

User Plane Function



The only core function that touches user data. Everything else is signalling.

- **Anchor point** for intra-RAT and inter-RAT mobility
- PDU session **point of interconnect** to the data network
- **Packet routing, forwarding** and inspection
- **QoS handling** for the user plane: rate enforcement, packet marking
- **Traffic usage reporting** for charging
- **Downlink buffering** and triggering the data notification

4G EQUIVALENT

SGW-U + PGW-U

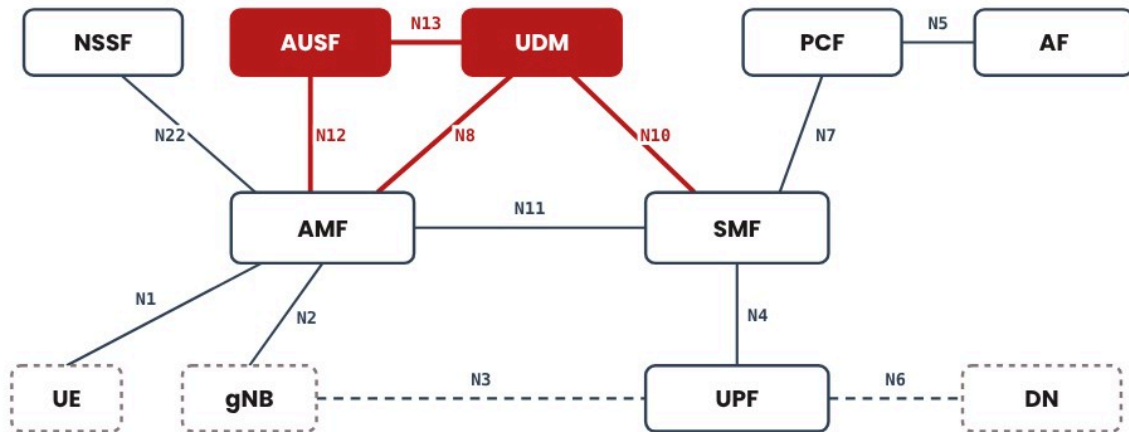
REFERENCE POINTS

N3 • N4 • N6 • N9

TS 23.501 §6.2.3

UDM and AUSF

Subscriber data and authentication



UDM Unified Data Management

The subscriber database front end.

- Generates the **3GPP AKA authentication credentials**
- **User identification:** handles the SUPI and de-conceals the SUCI
- **Access authorization** from subscription data, such as roaming restrictions
- Keeps track of the **serving AMF and SMF** of each UE

4G equivalent: HSS · TS 23.501 §6.2.7

AUSF Authentication Server Function

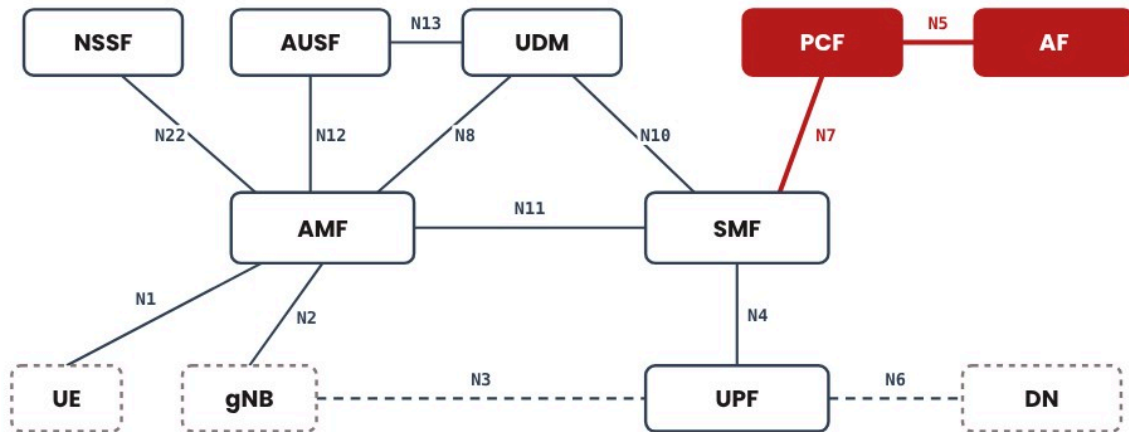
Runs the authentication of the UE for the home network.

- **Authentication** for 3GPP and untrusted non-3GPP access
- Compares RES* with XRES* in 5G AKA and delivers the anchor key

4G equivalent: authentication part of the HSS · TS 23.501 §6.2.8 · TS 33.501

PCF and AF

Policy, and the applications that influence it



PCF Policy Control Function

One policy framework for the whole network.

- Supports a **unified policy framework** to govern network behaviour
- Provides **policy rules** to the control plane functions that enforce them
- Reads subscription information for policy decisions from the **UDR**

4G equivalent: PCRF • TS 23.501 §6.2.4

AF Application Function

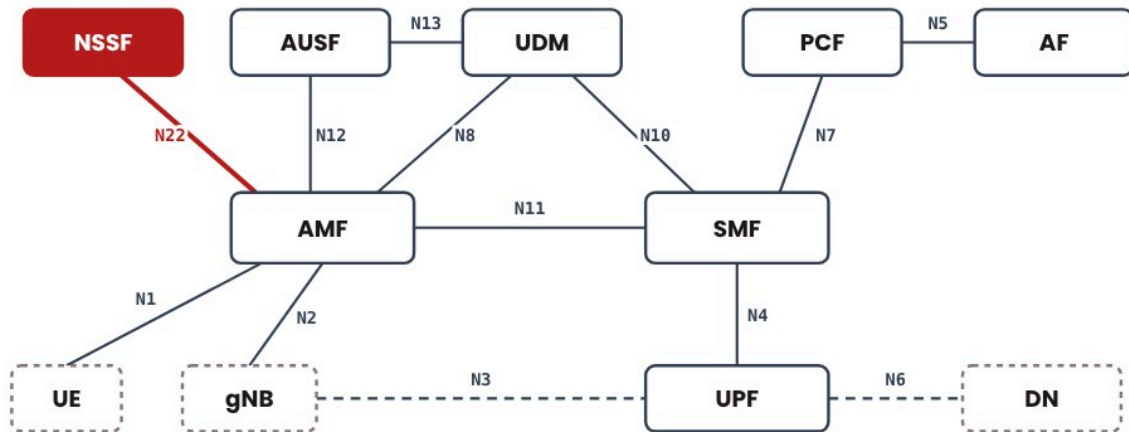
An application, such as IMS, talking to the core.

- **Application influence** on traffic routing
- Interacts with the **policy framework** for policy control
- Reaches the core directly if trusted, otherwise through the NEF

4G equivalent: AF • TS 23.501 §6.2.10

NSSF, NRF and NEF

New in 5G: slices, discovery and exposure



NSSF Network Slice Selection Function

Picks the slices for a UE.

- Selects the **network slice instances** serving the UE
- Determines the **Allowed NSSAI** and the **AMF set** to serve the UE

TS 23.501 §6.2.14

NRF Network Repository Function

The directory of the service-based core.

- Functions **register their profile**; others **discover** them here

TS 23.501 §6.2.6

NEF Network Exposure Function

The controlled way in from outside.

- **Exposes capabilities and events** to applications and third parties

Closest 4G equivalent: SCEF • TS 23.501 §6.2.5

Reference points

Which N-number connects what

POINT	BETWEEN	CARRIES
N1	UE – AMF	NAS signalling
N2	RAN – AMF	NGAP: RAN control plane
N3	RAN – UPF	User data in GTP-U tunnels
N4	SMF – UPF	PCF: session rules for the UPF
N5	PCF – AF	Application requests to policy
N6	UPF – Data network	User data to the outside
N7	SMF – PCF	Session policy
N8	AMF – UDM	Subscription and registration data
N9	UPF – UPF	User data between UPFs
N10	SMF – UDM	Session subscription data
N11	AMF – SMF	PDU session control
N12	AMF – AUSF	Authentication
N13	AUSF – UDM	Authentication vectors
N14	AMF – AMF	UE context transfer
N15	AMF – PCF	Access and mobility policy
N22	AMF – NSSF	Slice selection

TS 23.501 §4.2.7

Same idea, **new name**

4G terms and their 5G counterparts

4G / EPS	5G / 5GS	NOTE
APN	DNN	Data Network Name
PDN connection	PDU session	Now also Ethernet and unstructured types
EPS bearer	QoS flow	Finest level of QoS, identified by QFI
QCI	5QI	Standardised QoS characteristics
Attach	Registration	And TAU becomes a registration update
IMSI	SUPI	Sent concealed, as the SUCI
GUTI	5G-GUTI	Temporary identity from the AMF
S1-MME / S1-U	N2 / N3	RAN to core
S1AP	NGAP	RAN control protocol
Tracking area list	Registration area	List of tracking areas

TS 23.501 • TS 23.502 • TS 23.401

Summary

AMF is the entry point: registration, mobility and NAS security. It forwards session messages to the SMF.

SMF controls PDU sessions and the UE IP address, and tells the **UPF** what to do over N4.

UPF is the only function in the user data path: N3 from the RAN, N6 to the data network.

UDM and **AUSF** together replace the HSS. **PCF** replaces the PCRF.

NSSF, NRF and NEF are new: slice selection, service discovery and exposure.

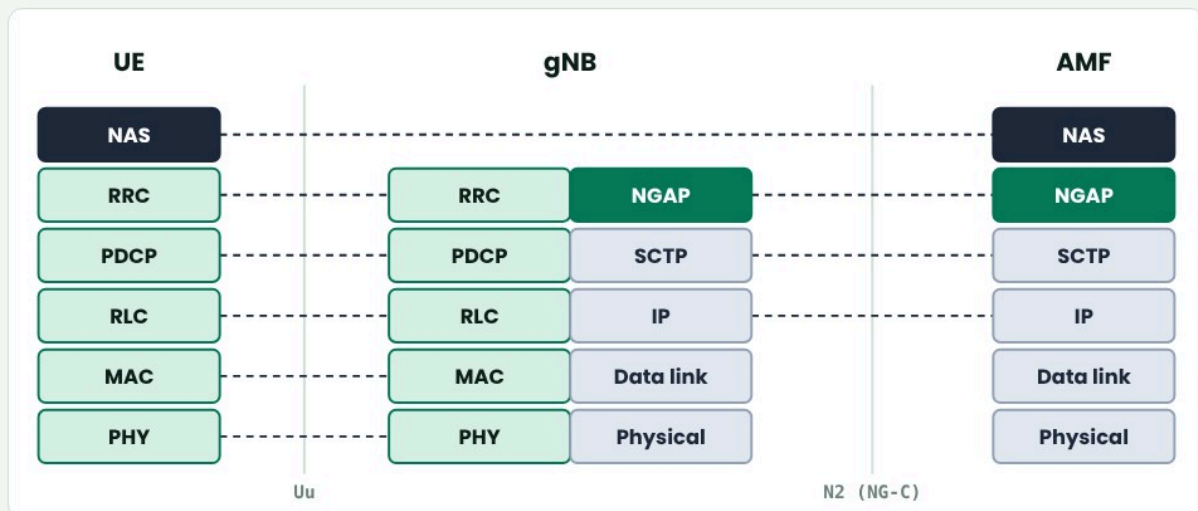
References: 3GPP TS 23.501 §4.2 and §6.2 · TS 23.502 · TS 33.501

5G Protocol Stacks

Which protocol runs on which interface: control plane, user plane, and the stack of every RAN and core interface.

Control plane

UE to gNB to AMF



NAS runs end to end between the UE and the AMF. The gNB does not read it: it arrives inside RRC and leaves inside NGAP.

RRC ends in the gNB. It configures the radio and carries the NAS messages over the air.

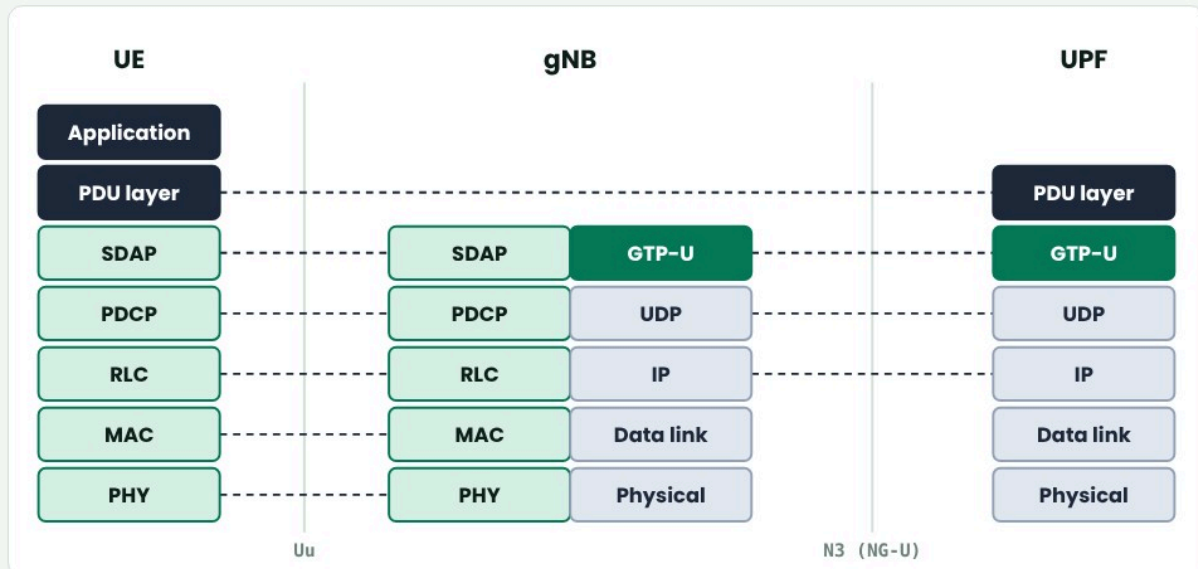
NGAP runs over SCTP, which gives reliable, in-order delivery of the signalling.

NAS has two parts: **NAS-MM** ends in the AMF, and **NAS-SM** is passed on to the SMF.

TS 38.300 §4.4.2 · TS 23.501 §8.2 · TS 38.413 · TS 24.501

User plane

UE to gNB to UPF



■ End to end: NAS, user data □ Radio protocols ■ Interface protocol □ Transport

The PDU layer is the user's own traffic: IPv4, IPv6, Ethernet or unstructured. It ends in the UPF.

SDAP is new in 5G. It maps QoS flows to data radio bearers and marks packets with the QFI.

Between the gNB and the UPF the packets travel in a **GTP-U** tunnel over UDP, one tunnel per PDU session.

TS 38.300 §4.4.1 · TS 23.501 §8.3 · TS 37.324 · TS 29.281

What each **layer** does

From the top of the stack down

NAS Registration, authentication and session management between the UE and the core.
TS 24.501

RRC System information, connection control, radio bearers, measurements and mobility.
TS 38.331

SDAP Maps QoS flows to data radio bearers. Marks the QFI. User plane only.
TS 37.324

PDCP Cipherring and integrity protection, header compression, reordering and duplication.
TS 38.323

RLC Segmentation and, in acknowledged mode, retransmission (ARQ). Modes TM, UM and AM.
TS 38.322

MAC Scheduling, multiplexing of logical channels, HARQ and random access.
TS 38.321

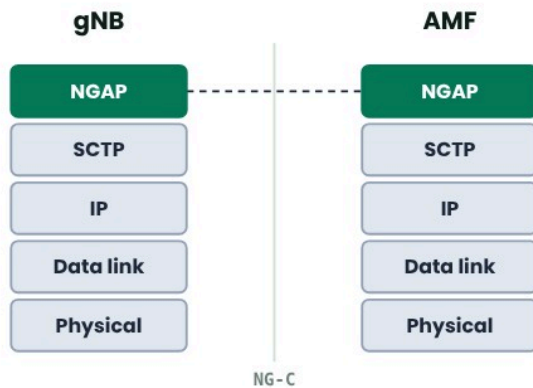
PHY Channel coding, modulation, OFDM and beamforming.
TS 38.201 • TS 38.211 to 38.214

NG and Xn

gNB to core, and gNB to gNB

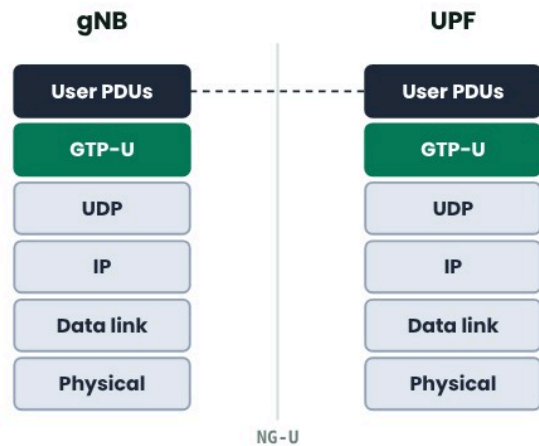
NG-C ^{N2}

Signalling to the AMF.



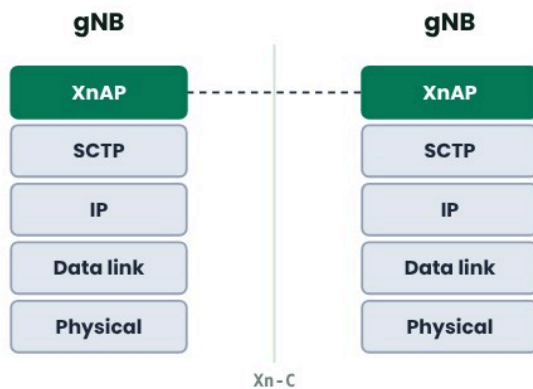
NG-U ^{N3}

User data to the UPF.



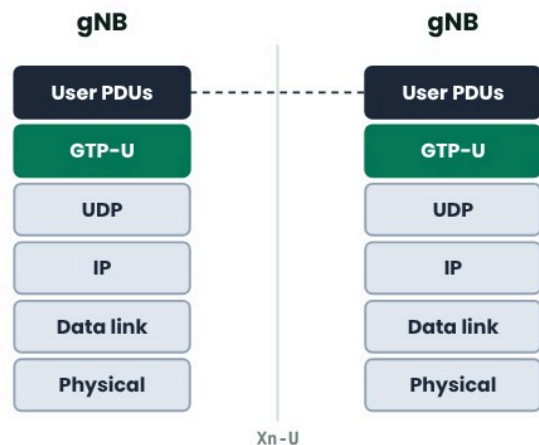
Xn-C

Handover and dual connectivity signalling.



Xn-U

Data forwarding between nodes.



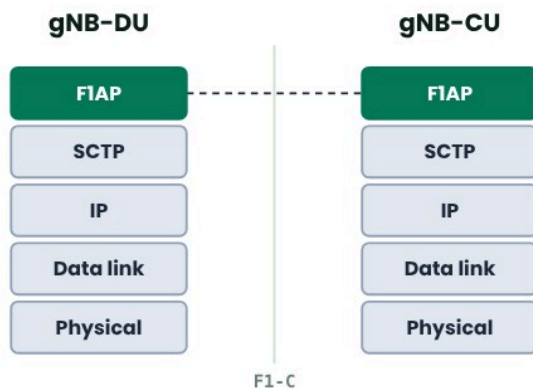
TS 38.410 to 38.414 (NG) · TS 38.420 to 38.425 (Xn)

F1, E1 and N4

Inside a split gNB, and between SMF and UPF

F1-C

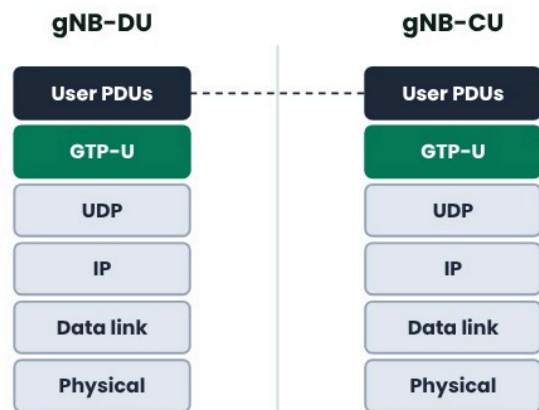
CU to DU signalling. Carries RRC messages.



F1-C

F1-U

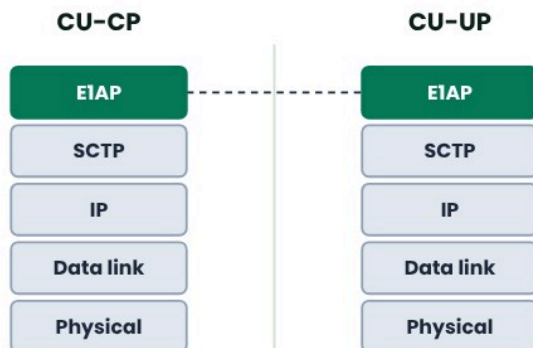
User data between DU and CU.



F1-U

E1

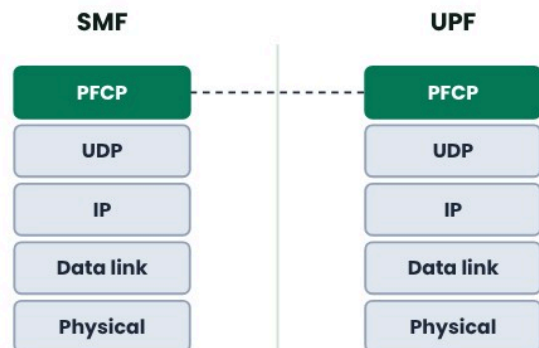
CU control plane to CU user plane. No user data.



E1

N4

SMF tells the UPF how to treat packets.

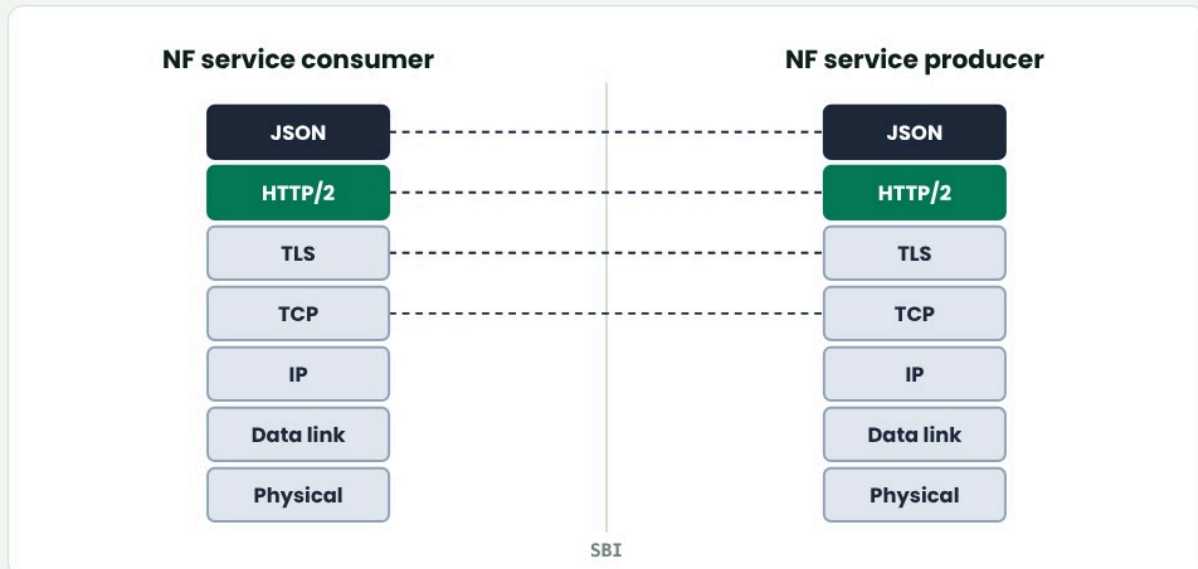


N4

TS 38.470 to 38.475 (F1) · TS 38.460 to 38.463 (E1) · TS 29.244 (PCF)

Service-based interface

Between the control functions of the 5G core



The same web stack for every function. AMF, SMF, UDM, PCF and the others call each other with HTTP/2 requests, with the content in JSON.

TCP in place of SCTP. The telecom-specific transport is used towards the RAN; inside the core it is ordinary IT transport, with TLS for protection.

APIs are described in OpenAPI, one specification per function, such as TS 29.518 for the AMF.

TS 29.500 · TS 29.501 · TS 23.501 §7

One-page summary

Interface, protocol, transport and specification

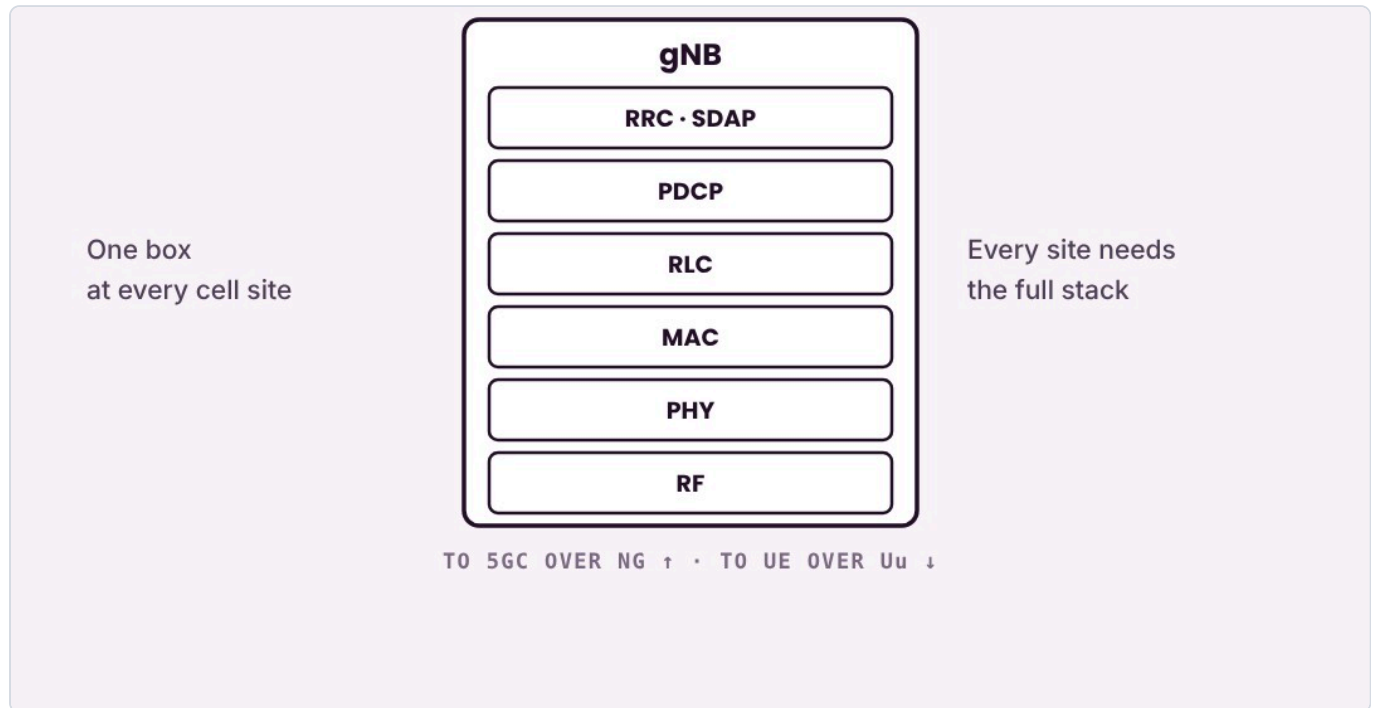
INTERFACE	PROTOCOL	TRANSPORT	SPEC
N1	NAS	inside RRC and NGAP	TS 24.501
Uu	RRC	PDCP / RLC / MAC / PHY	TS 38.331
N2 (NG-C)	NGAP	SCTP	TS 38.413
Xn-C	XnAP	SCTP	TS 38.423
F1-C	F1AP	SCTP	TS 38.473
E1	E1AP	SCTP	TS 38.463
N4	PCF	UDP	TS 29.244
SBI	HTTP/2 + JSON	TCP with TLS	TS 29.500
N3 • Xn-U • F1-U	GTP-U	UDP	TS 29.281

Rule of thumb: RAN signalling runs over SCTP, user data runs in GTP-U over UDP, and the core talks HTTP/2 over TCP.

Inside the gNB

CU, DU and RU: where the base station is cut, and what F1 and E1 carry.

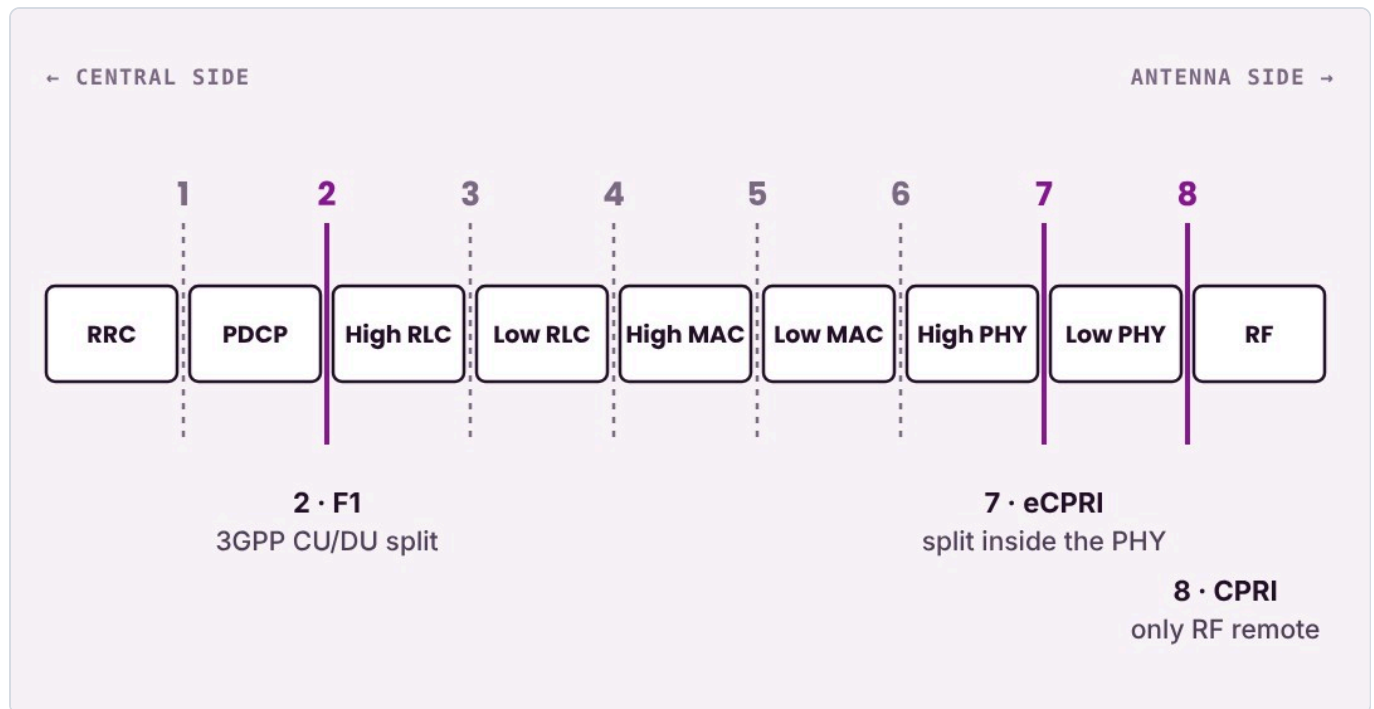
4.1 The classic base station



A gNB can be one box that holds the whole radio protocol stack, from RRC down to RF. That is simple, but every site then needs the full processing, and sites cannot share it.

TS 38.300 §4 · TS 38.401

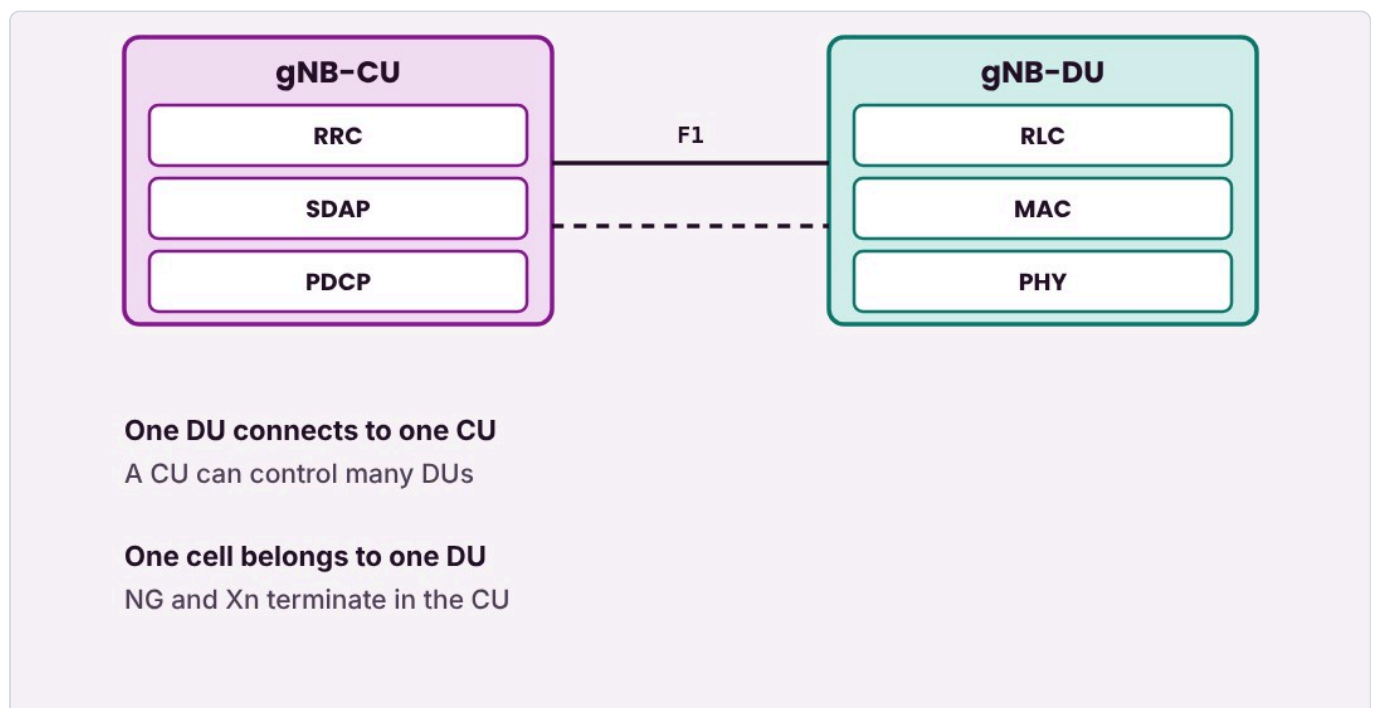
4.2 Eight places to cut



3GPP studied eight options for splitting the stack between a central and a remote part. The further right the cut, the more is centralised. Three matter in practice: option 2, option 7 and option 8.

TR 38.801 §11

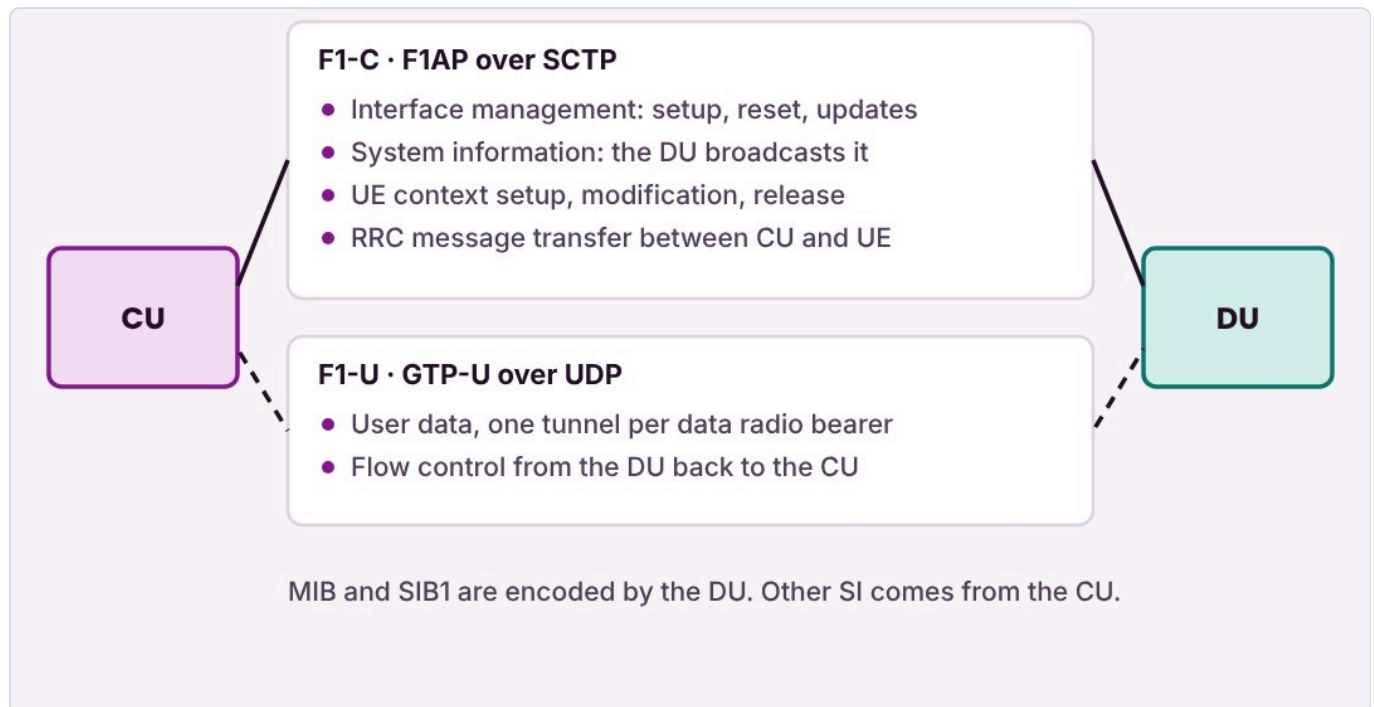
4.3 The 3GPP split: CU and DU



3GPP standardised option 2. The gNB-CU hosts RRC, SDAP and PDCP. The gNB-DU hosts RLC, MAC and PHY. The F1 interface connects them. To the core and to other gNBs the pair still looks like one gNB.

TS 38.401 §6.1.1

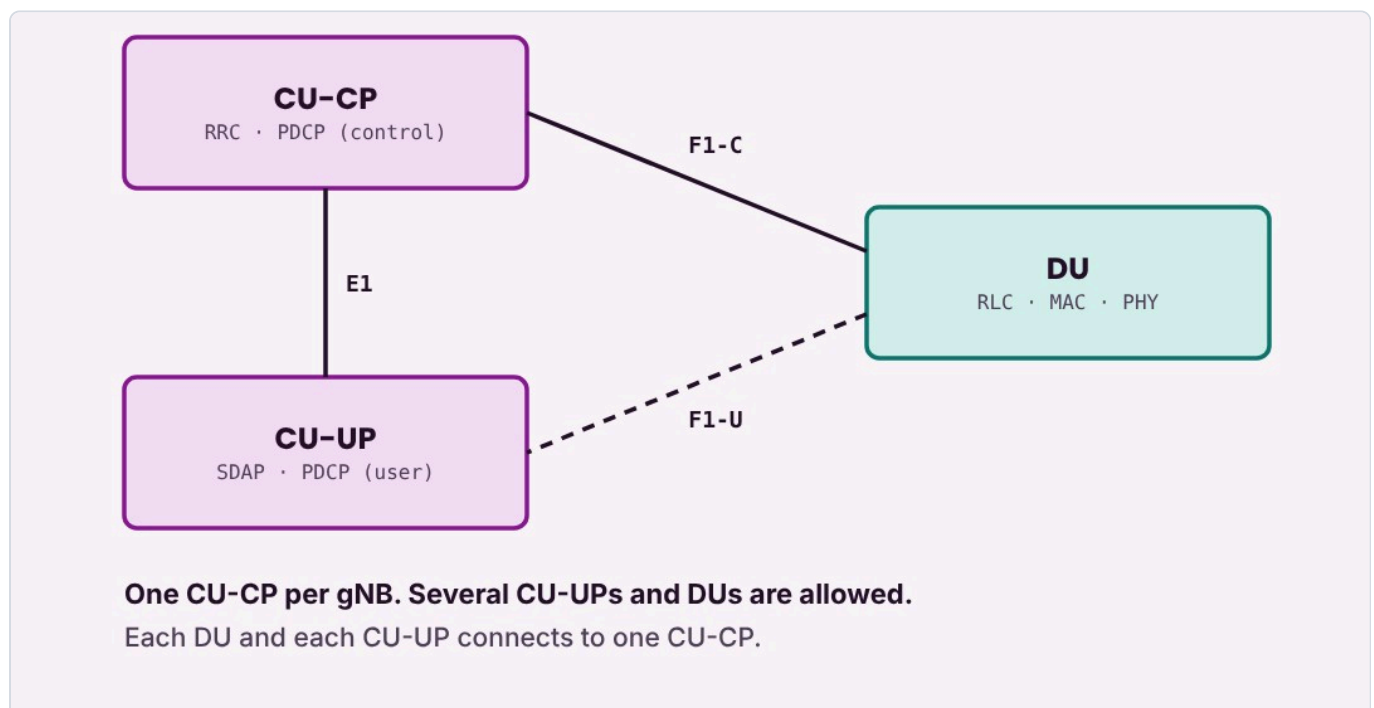
4.4 What F1 carries



F1-C carries F1AP signalling: interface management, UE context management and the RRC messages, which the DU passes on without reading. F1-U carries the user data in GTP-U tunnels, with flow control so the CU does not overfill the DU.

TS 38.470 · TS 38.473 · TS 38.425

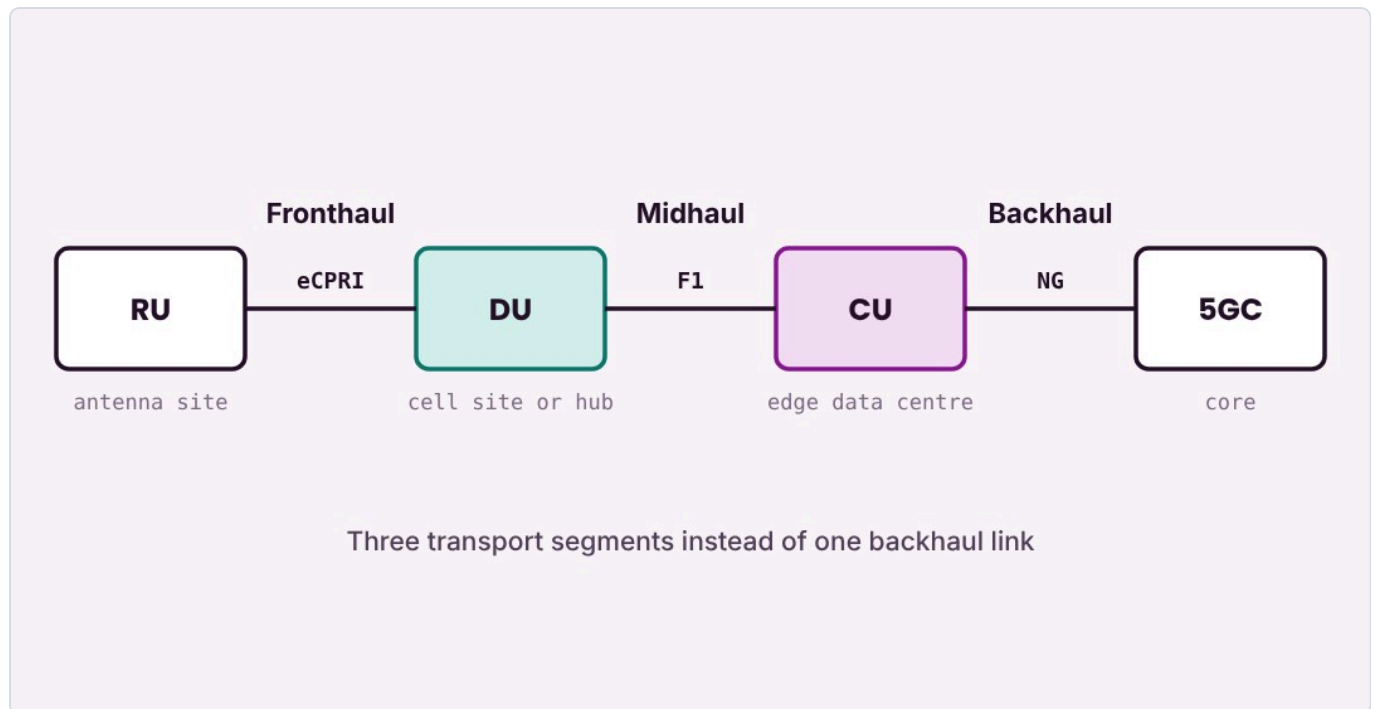
4.5 Splitting the CU as well



The CU can be split again into a control part and a user part. CU-CP hosts RRC and the PDCP for signalling. CU-UP hosts SDAP and the PDCP for user data. E1 connects the two, and it carries no user data.

TS 38.401 §6.1.2 · TS 38.460

4.6 Fronthaul, midhaul, backhaul



A split gNB needs three transport segments. Fronthaul runs from the radio unit to the DU. Midhaul is F1, from the DU to the CU. Backhaul is NG, from the CU to the core.

TS 38.401 · eCPRI and O-RAN specifications

4.7 Why the cut position matters



A high layer split is easy on transport: its bitrate follows the user traffic and it tolerates delay. A low layer split needs far more capacity and very low latency, but allows close coordination between cells.

TR 38.801 §11.1

4.8 Who hosts what

UNIT	HOSTS	INTERFACES
CU-CP	RRC, PDCP for signalling	E1 · F1-C · NG-C · Xn-C
CU-UP	SDAP, PDCP for user data	E1 · F1-U · NG-U · Xn-U
DU	RLC, MAC, PHY	F1 · fronthaul
RU	lower PHY, RF	fronthaul

3GPP defines CU, DU, F1 and E1. The RU and its fronthaul come from CPRI and O-RAN.

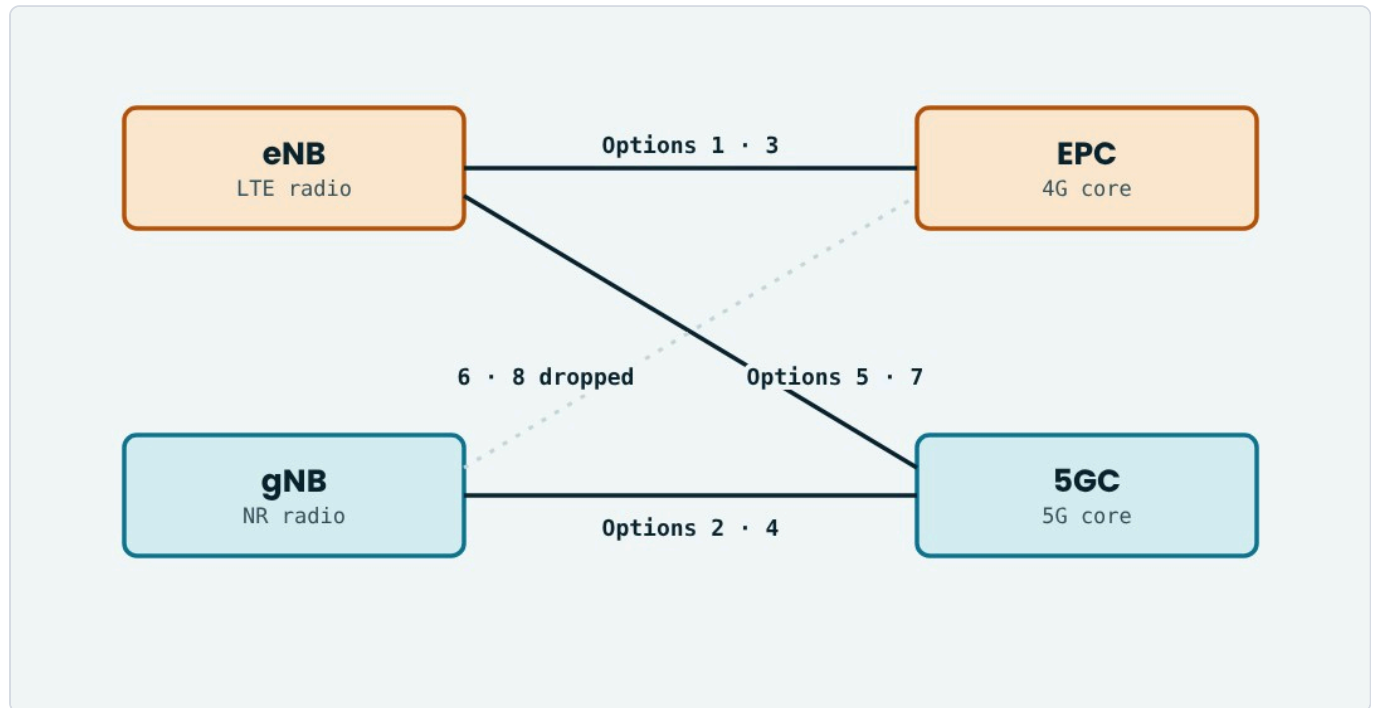
CU-CP for signalling, CU-UP for user data, DU for the real-time layers and RU for the radio. F1 and E1 are open 3GPP interfaces, so the parts can come from different vendors.

TS 38.401 §6.1 · TS 38.470 · TS 38.460

5G Deployment Options

Six ways to combine the LTE and NR radios with the EPC and the 5G core, and where SA and NSA fit.

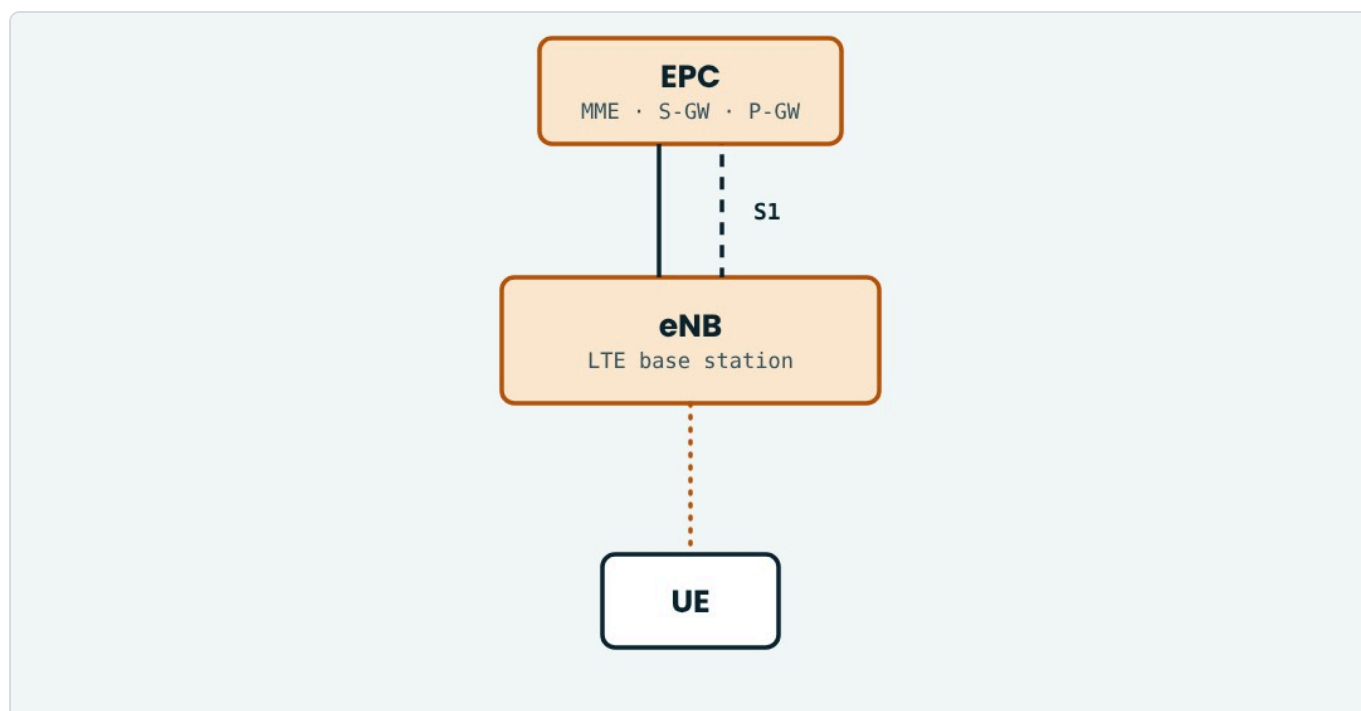
5.1 Two radios, two cores



5G brought a new radio, NR, and a new core, 5GC. Each can be paired with the LTE radio or the LTE core, EPC. 3GPP numbered these combinations as options. Options 6 and 8, NR anchored on the EPC alone, were dropped, so six remain.

TR 38.801 · TS 38.300 §4.1 · TS 37.340

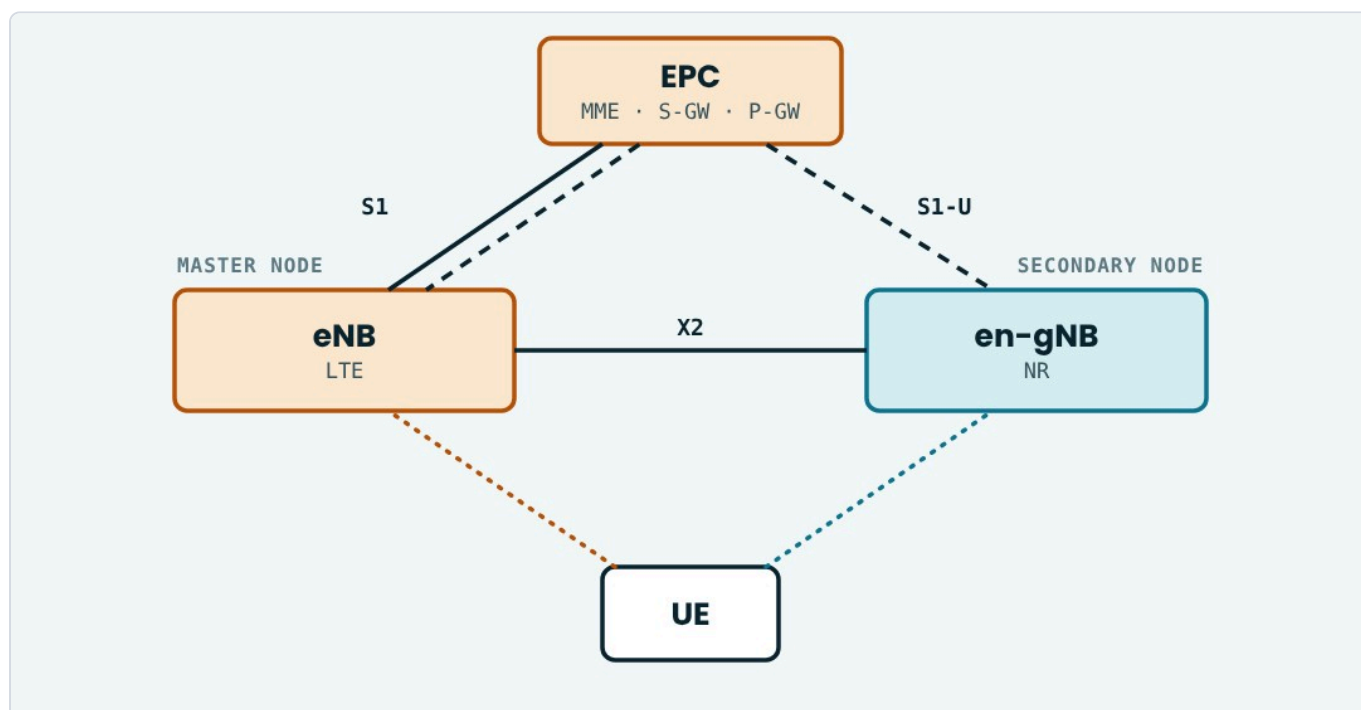
5.2 Option 1: LTE as it is



Option 1 is the 4G network as it is: an LTE eNB connected to the EPC over S1. It is the starting point of every migration path.

TS 36.300

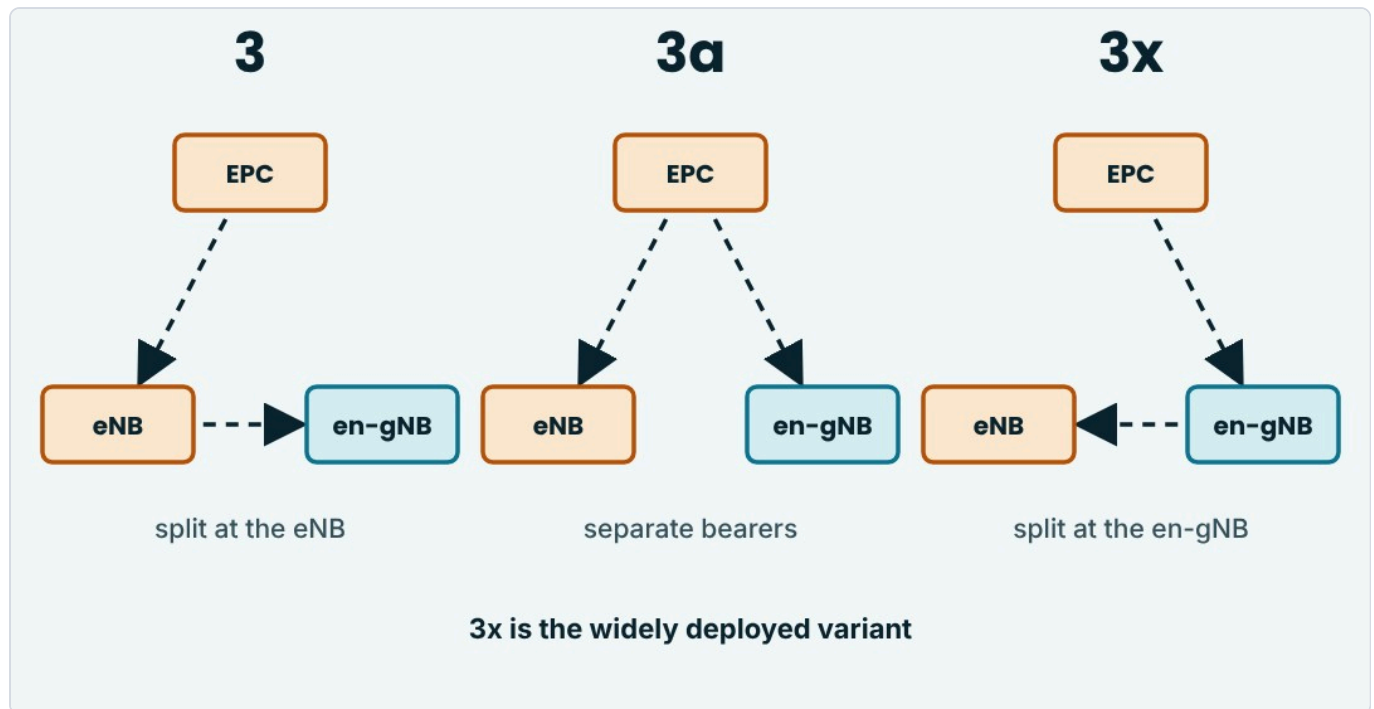
5.3 Option 3: EN-DC, the NSA launch



Option 3 adds NR without a new core. The LTE eNB is the master node and holds the control plane to the EPC. An NR node, the en-gNB, is the secondary node and adds capacity. This is EN-DC, the non-standalone 5G most networks launched with.

TS 37.340 §4.1.2

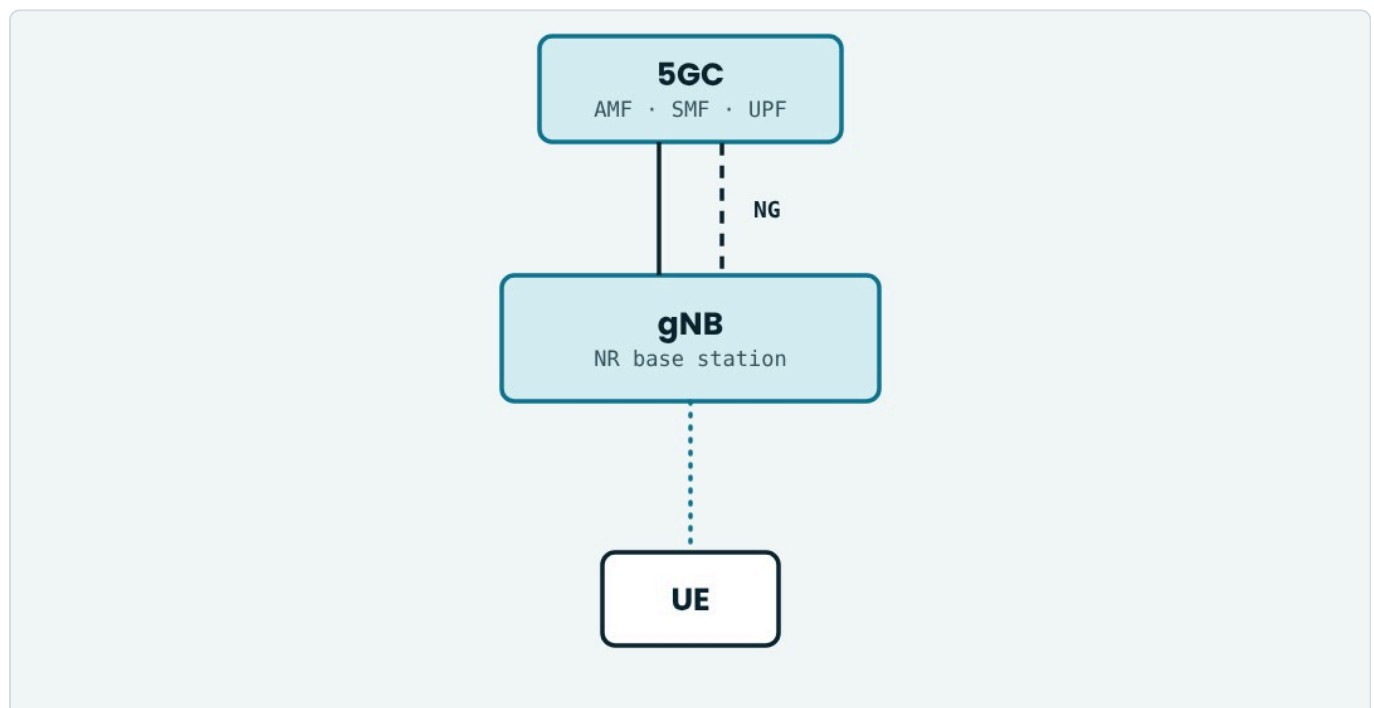
5.4 Option 3, 3a and 3x



The three variants differ in where user data splits. In 3, the eNB splits it. In 3a, the core sends separate bearers to each node. In 3x, the en-gNB splits it, so the NR node carries the load.

TS 37.340 §4.2 · MCG, SCG and split bearers

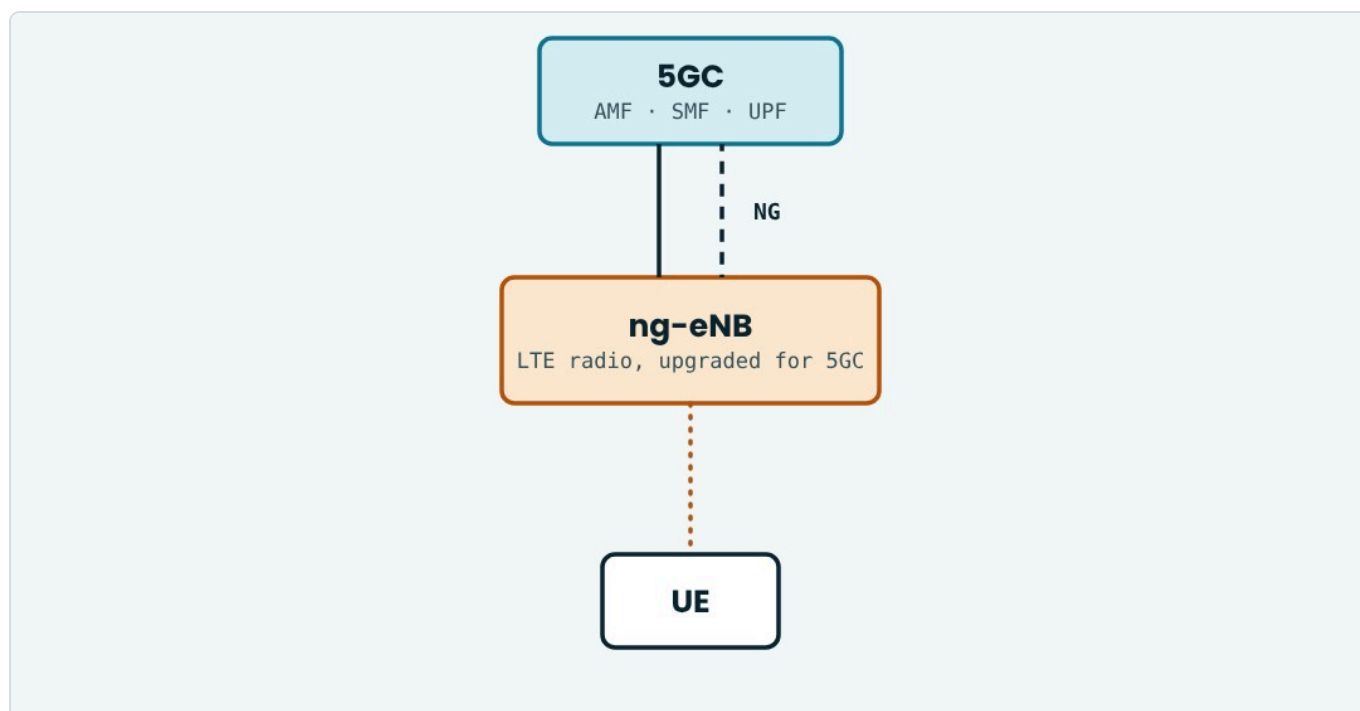
5.5 Option 2: standalone 5G



Option 2 is standalone 5G: an NR gNB connected to the 5G core over NG, with no LTE involved. The 5G core brings network slicing and the service-based architecture.

TS 38.300 §4.1 · TS 23.501

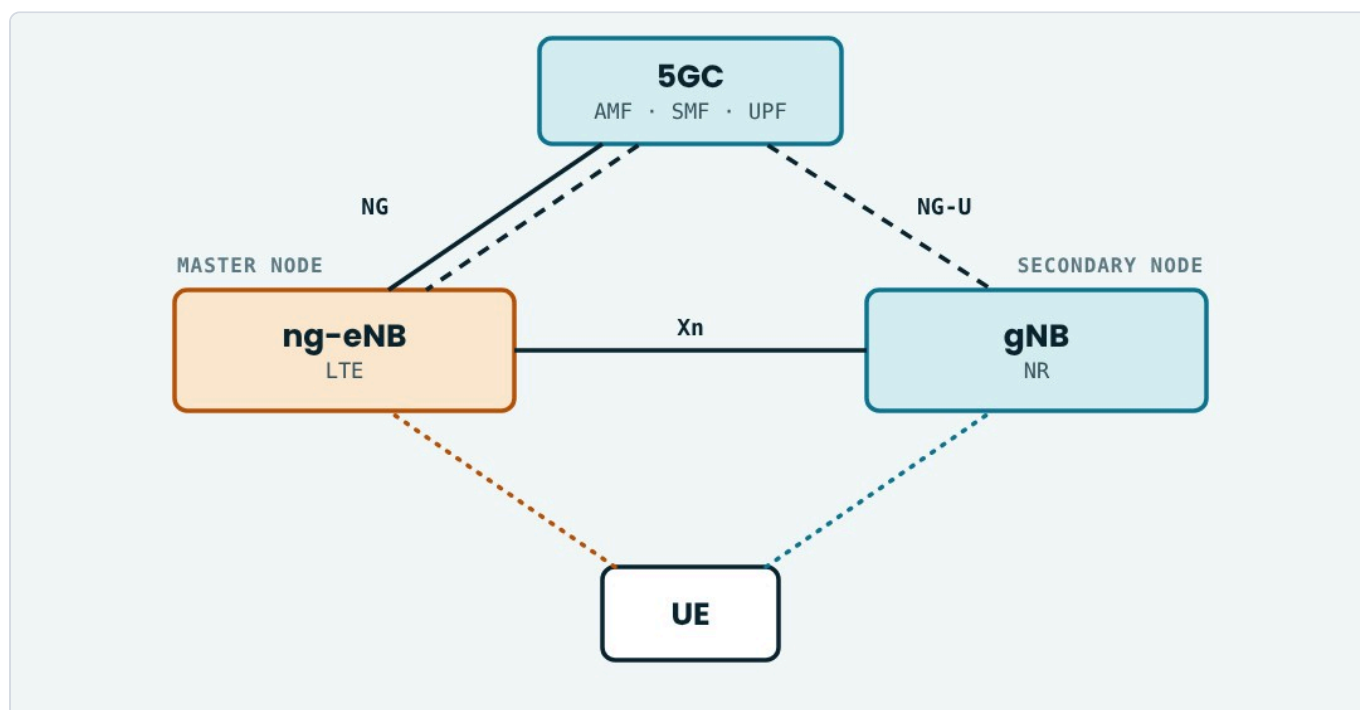
5.6 Option 5: LTE on the 5G core



Option 5 keeps the LTE radio but connects it to the 5G core. The upgraded eNB is called an ng-eNB and uses the NG interface. The UE needs 5G NAS to use it.

TS 38.300 §4.1

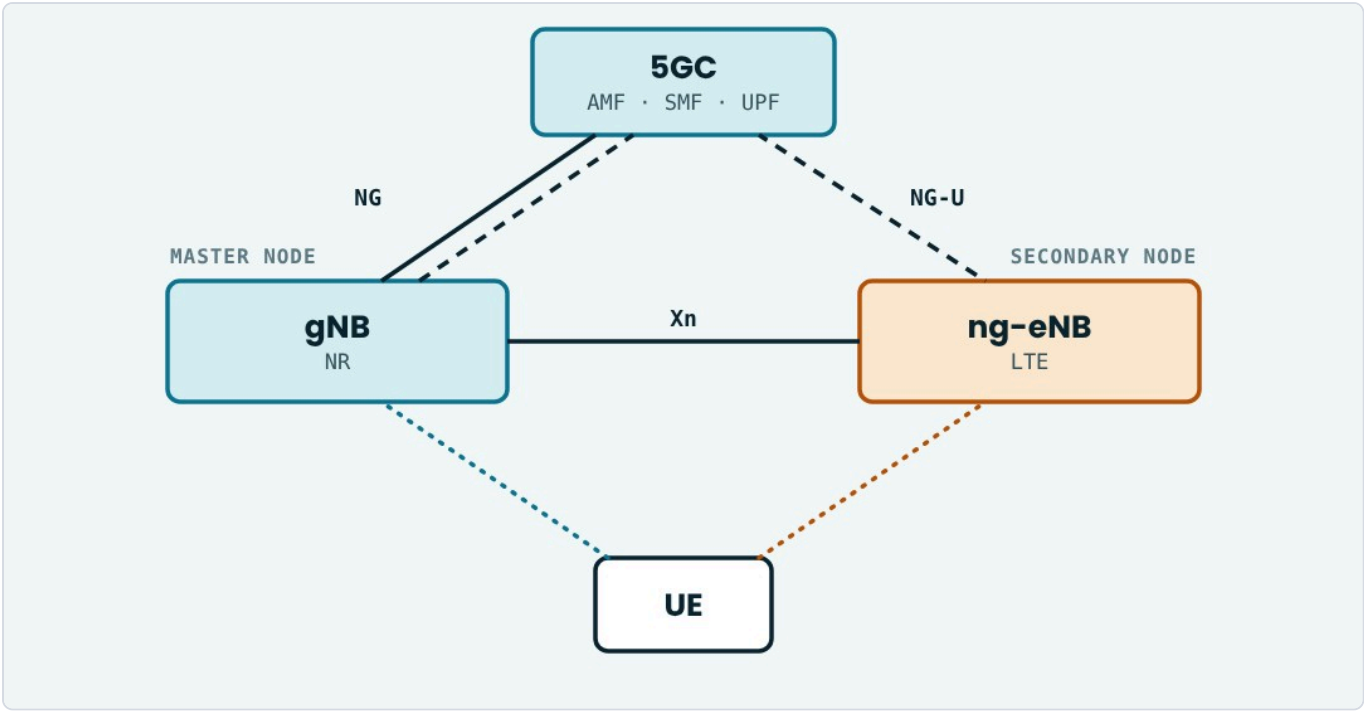
5.7 Option 7: NGEN-DC



Option 7 is dual connectivity on the 5G core with LTE as the anchor. The ng-eNB is the master node and an NR gNB is the secondary node. This is NGEN-DC.

TS 37.340 §4.1.3

5.8 Option 4: NE-DC



Option 4 turns it around. The NR gNB is the master node and an ng-eNB is the secondary node. This is NE-DC: NR leads and LTE adds capacity.

TS 37.340 §4.1.3

5.9 The six options side by side

OPTION	NAME	MASTER	SECONDARY	CORE
1	LTE	eNB	–	EPC
3	EN-DC (NSA)	eNB	en-gNB	EPC
2	NR standalone (SA)	gNB	–	5GC
5	LTE on 5GC	ng-eNB	–	5GC
7	NGEN-DC	ng-eNB	gNB	5GC
4	NE-DC	gNB	ng-eNB	5GC

Typical path: option 3 first, then option 2

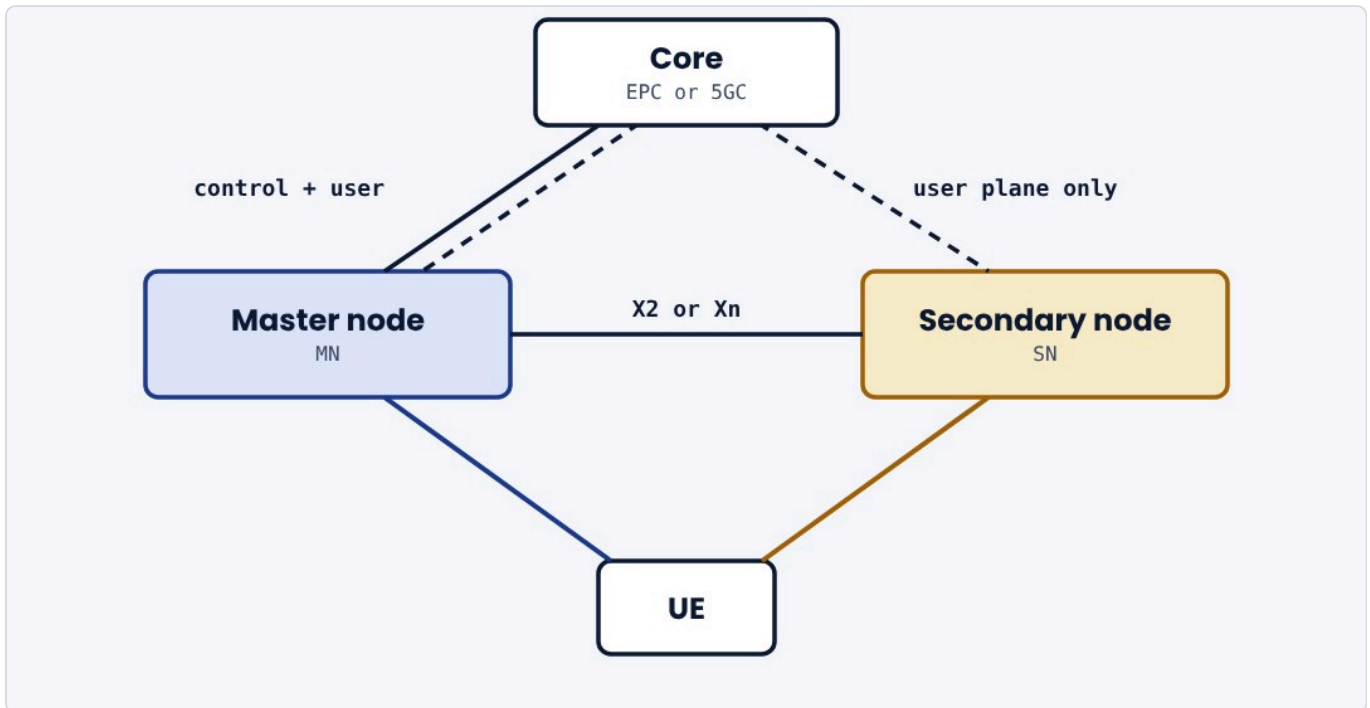
Most operators started with option 3 and are moving to option 2. Options 4, 5 and 7 are in the specifications but rarely deployed. Two NR nodes on the 5G core is called NR-DC.

TS 37.340 §4.1 · TS 38.300 §4.1

Dual Connectivity

One UE, two base stations at once: cell groups, signalling bearers and the three kinds of data bearer.

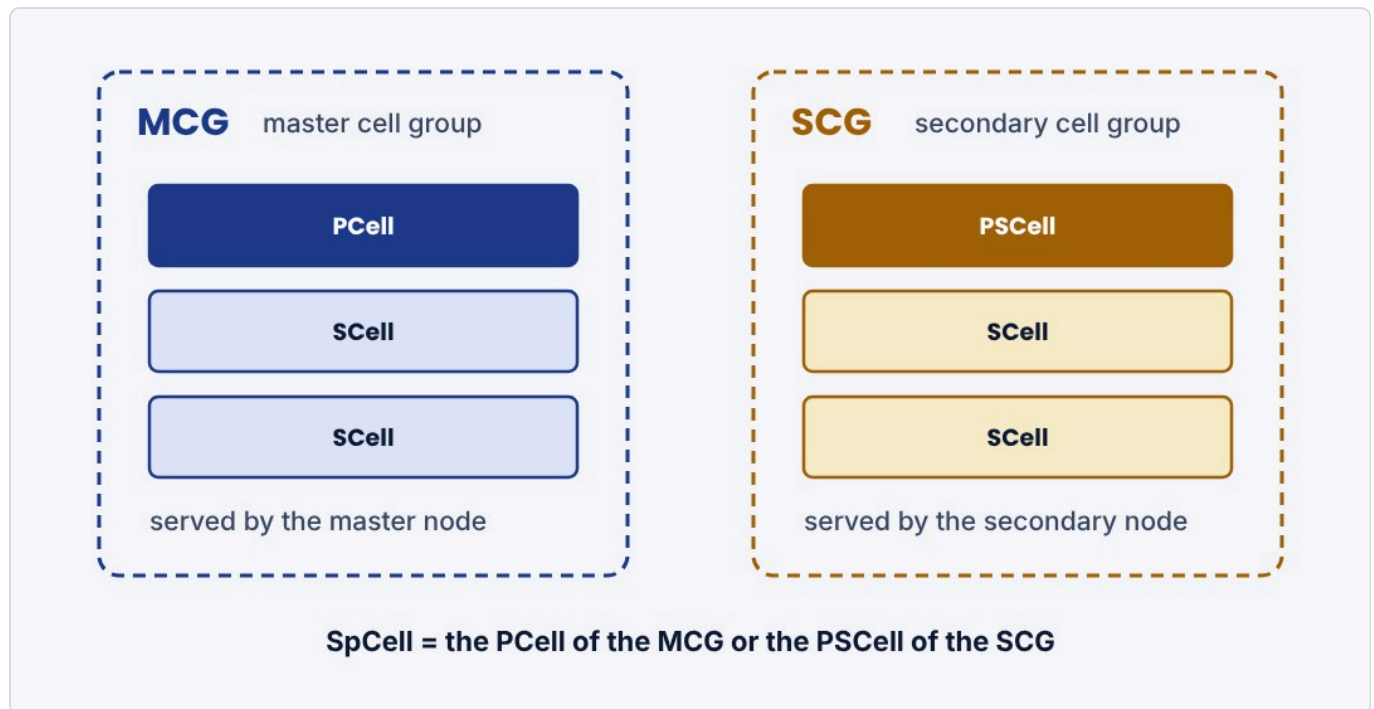
6.1 One UE, two nodes



In dual connectivity the UE uses radio resources of two base stations at the same time. The master node holds the control plane connection to the core. The secondary node adds capacity. The two nodes talk over X2 or Xn.

TS 37.340 §4.1

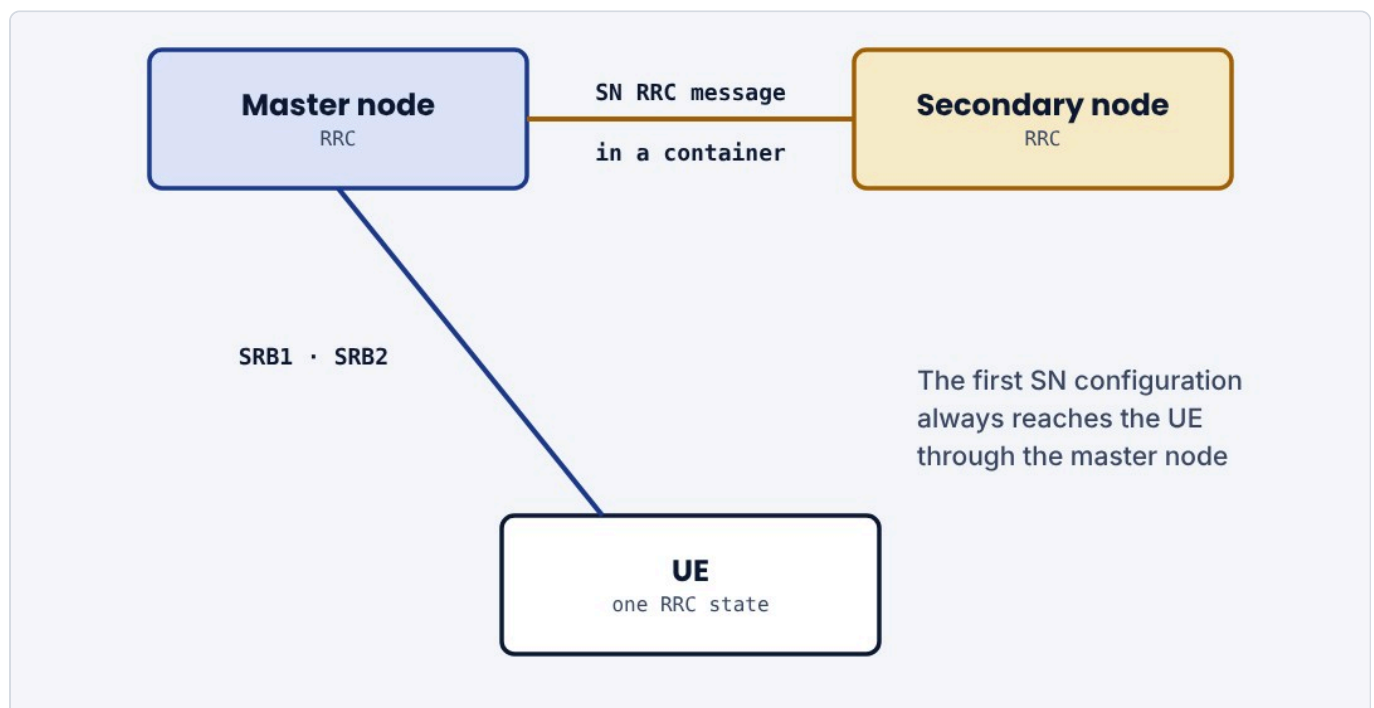
6.2 Two cell groups



The cells of the master node form the master cell group, with a PCell. The cells of the secondary node form the secondary cell group, with a PCell. Inside each group, carrier aggregation can add SCells.

TS 37.340 §3.1 · TS 38.331

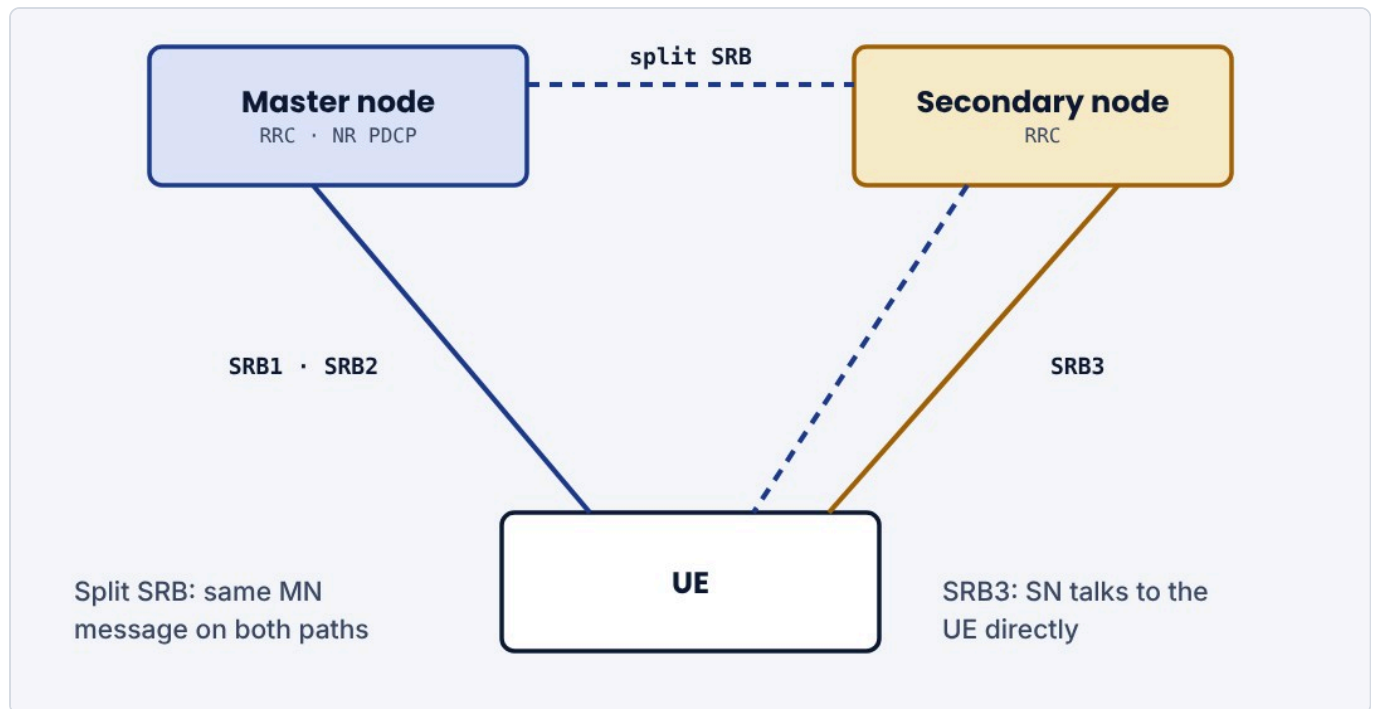
6.3 Control plane: one RRC state



The UE has a single RRC state, set by the master node. Each node still has its own RRC entity. The secondary node writes its own configuration, and the master node carries it to the UE inside its own RRC message.

TS 37.340 §4.2.1

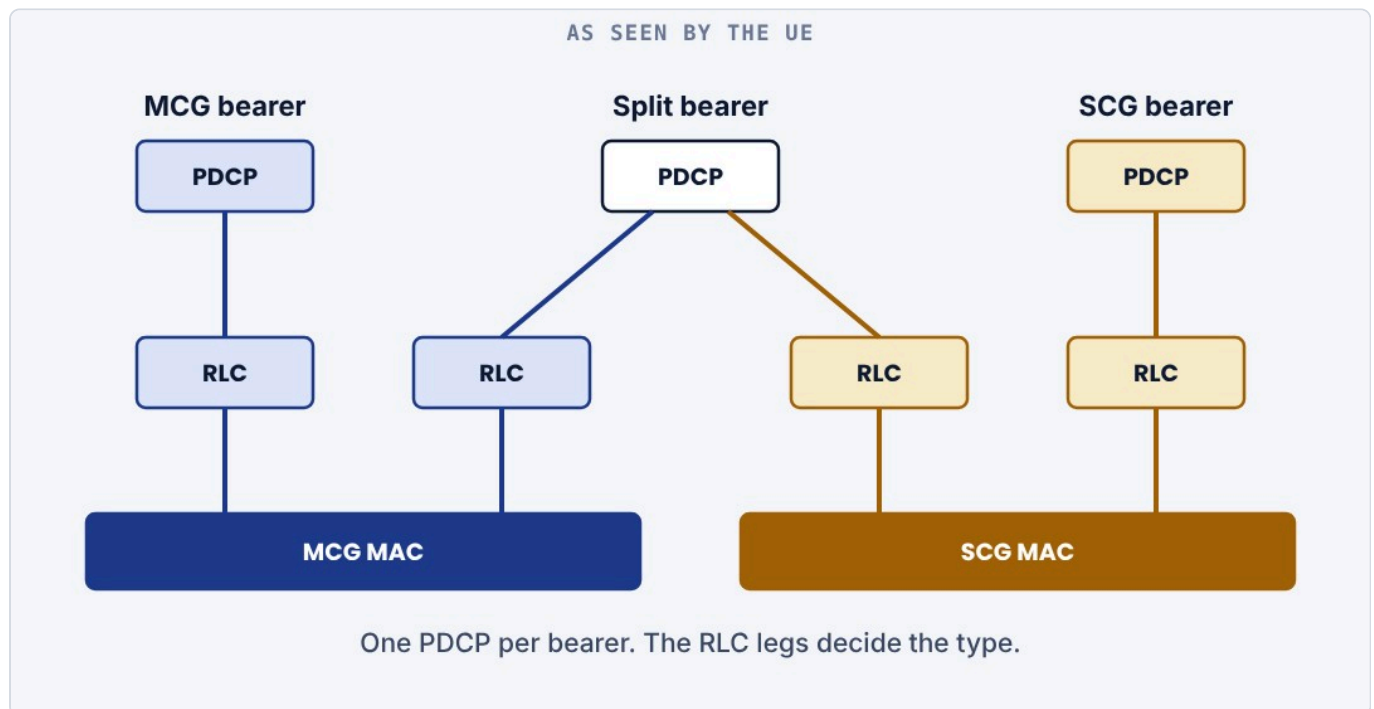
6.4 Split SRB and SRB3



With a split SRB, the master node sends the same RRC message over both radio paths, which protects it when one path is weak. With SRB3, a gNB acting as secondary node exchanges RRC messages with the UE directly, for example its own measurements and reconfigurations.

TS 37.340 §4.2.1

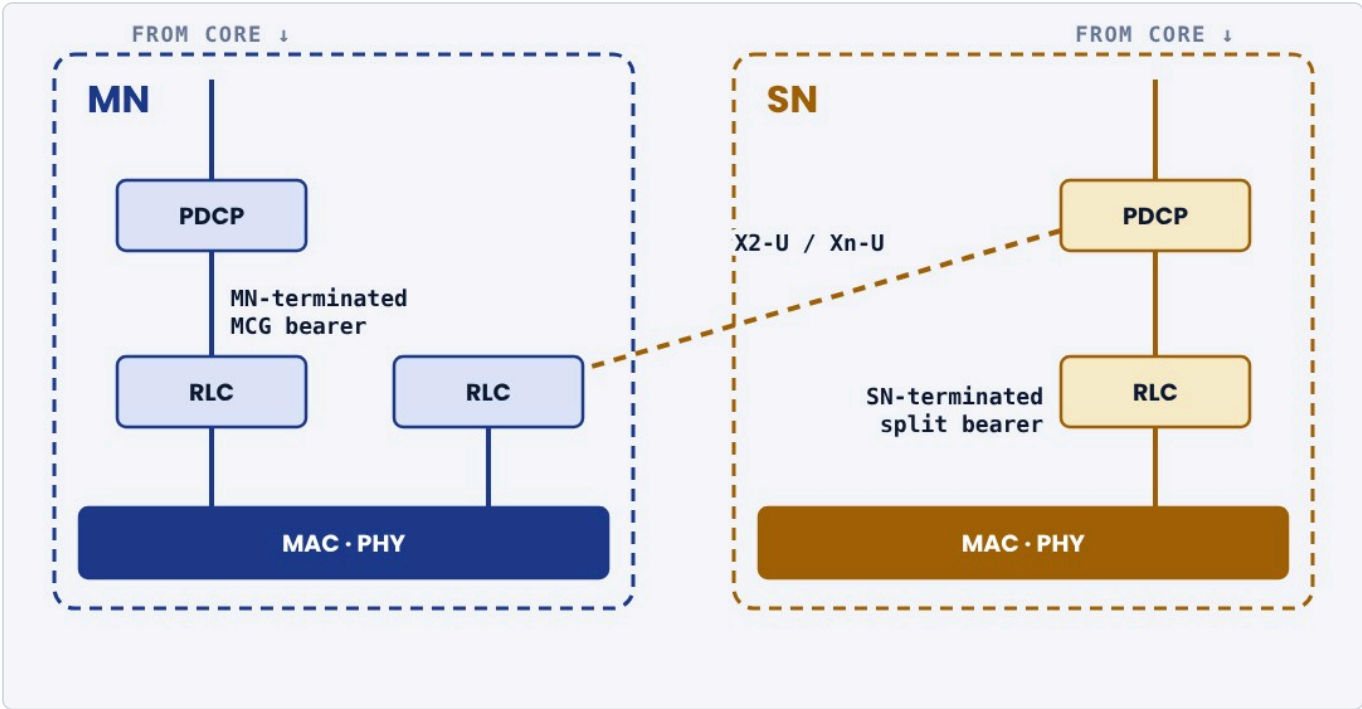
6.5 Three bearer types



From the UE side there are three kinds of data bearer. An MCG bearer uses only the master cell group. An SCG bearer uses only the secondary cell group. A split bearer has one PDCP and an RLC leg in each group.

TS 37.340 §4.2.2

6.6 Where PDCP sits in the network



On the network side each bearer is terminated in one node: the one that hosts its PDCP and its connection to the core. For a split bearer, that node sends part of the data to the other node over X2-U or Xn-U.

TS 37.340 §4.2.2

6.7 Carrier aggregation or dual connectivity

	CARRIER AGGREGATION	DUAL CONNECTIVITY
Nodes	one	two: MN and SN
MAC entities	one	one per cell group
Data splits at	MAC	PDCP
Between carriers	same node	X2 or Xn, can be non-ideal
PUCCH on	PCell	PCell and PSCell
RRC	one	one per node, one UE state

Both add carriers, but in different places. Carrier aggregation works inside one node and splits the data at MAC. Dual connectivity uses two nodes and splits at PDCP, so it tolerates a slower link between them. The two can be combined.

TS 38.300 §5.4.1 · TS 37.340

6.8 Three things to remember

1

Two nodes, two cell groups

The MN owns the control plane. Each group has its own special cell.

2

Signalling: SRB1, SRB2, split SRB and SRB3

MN messages can go on both paths. The SN can reach the UE directly.

3

Data: MCG, SCG and split bearers

PDCP sits in one node. A split bearer uses the RLC of both.

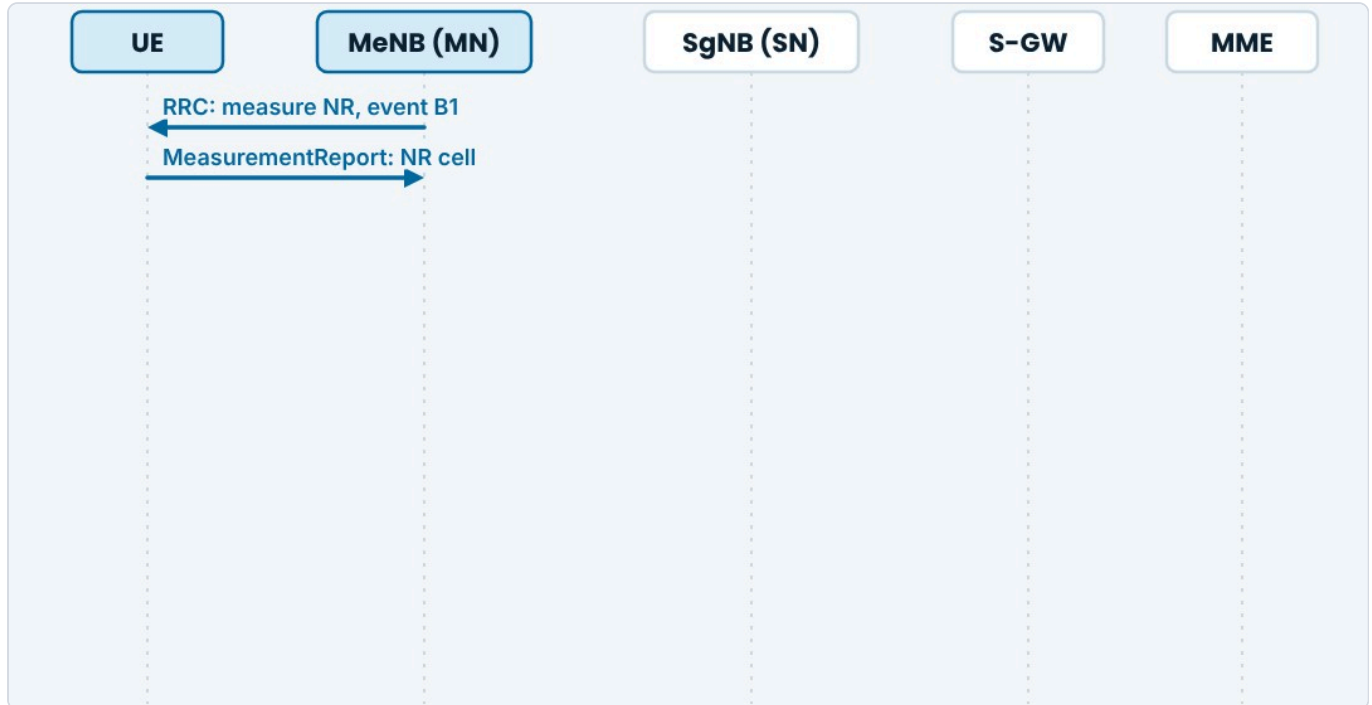
Dual connectivity is two nodes sharing one UE: the master node leads, the secondary node adds capacity, and PDCP is the layer that ties the two paths together.

TS 37.340 §4

Secondary Node Addition

How a UE on LTE gets its NR leg in EN-DC, message by message.

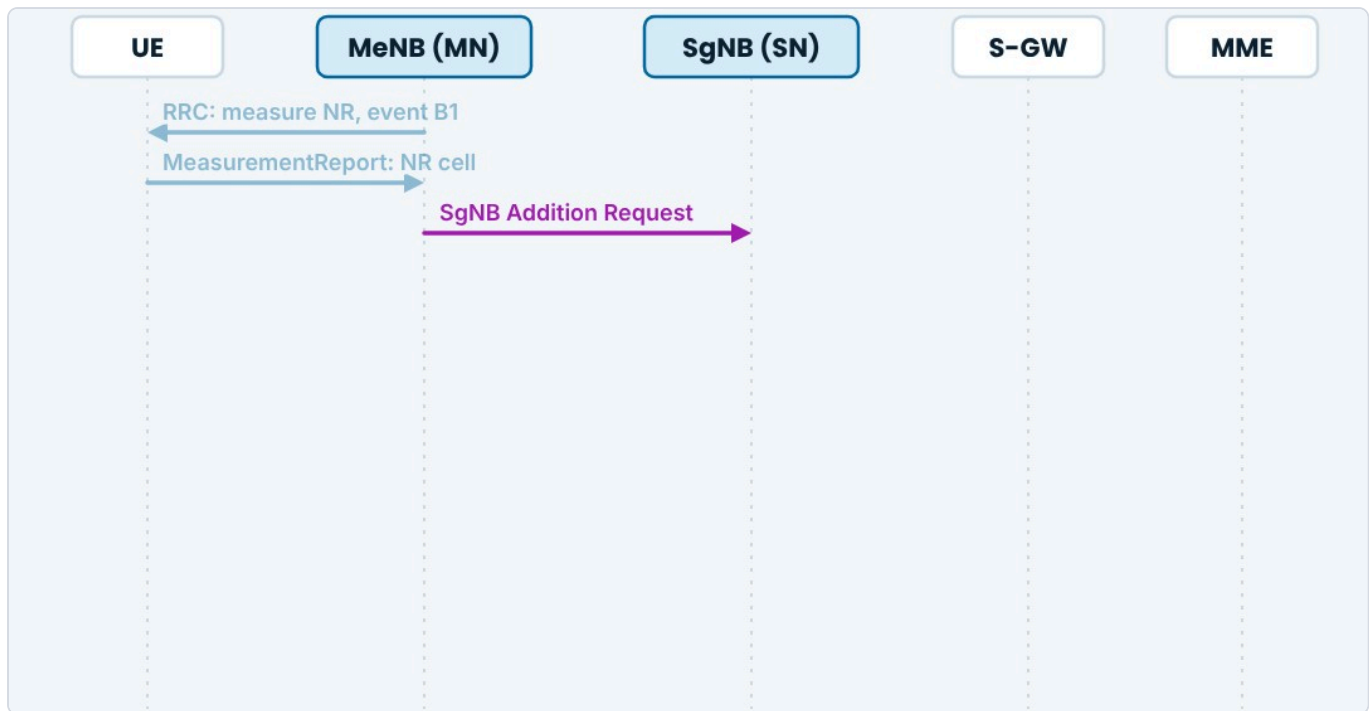
7.1 The trigger



The UE is connected on LTE. The MeNB asks it to measure NR with event B1. When the UE reports an NR cell that is strong enough, the MeNB decides to add an NR secondary node.

TS 36.331 · TS 37.340 §10.2.1

7.2 SgNB Addition Request



The MeNB asks the SgNB to allocate resources. The request carries the bearers to serve, the UE capabilities, the measurement results and the security key for the SgNB, S-KgNB.

TS 37.340 §10.2.1 step 1 · TS 36.423

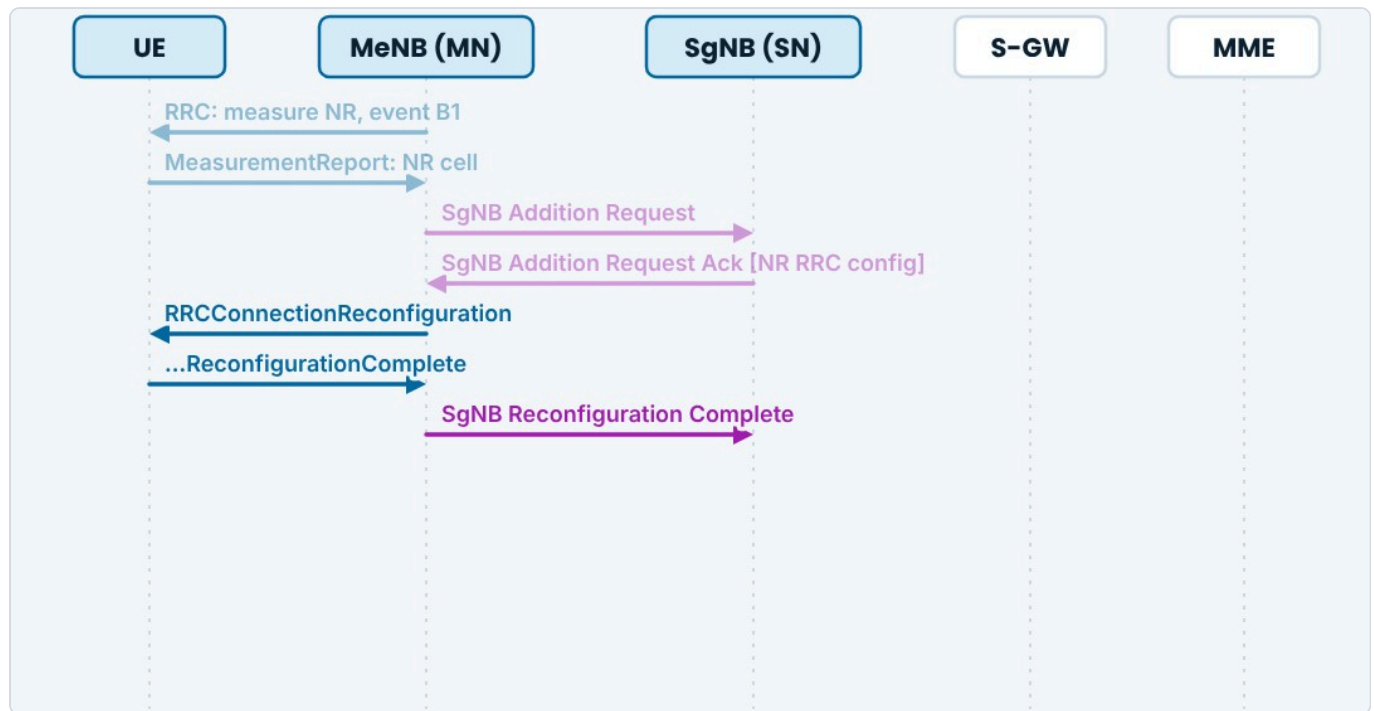
7.3 The SgNB accepts



The SgNB admits the bearers, picks the PSCell and reserves radio resources. Its answer contains the NR RRC configuration for the UE, the secondary cell group configuration, packed in a container.

TS 37.340 §10.2.1 step 2

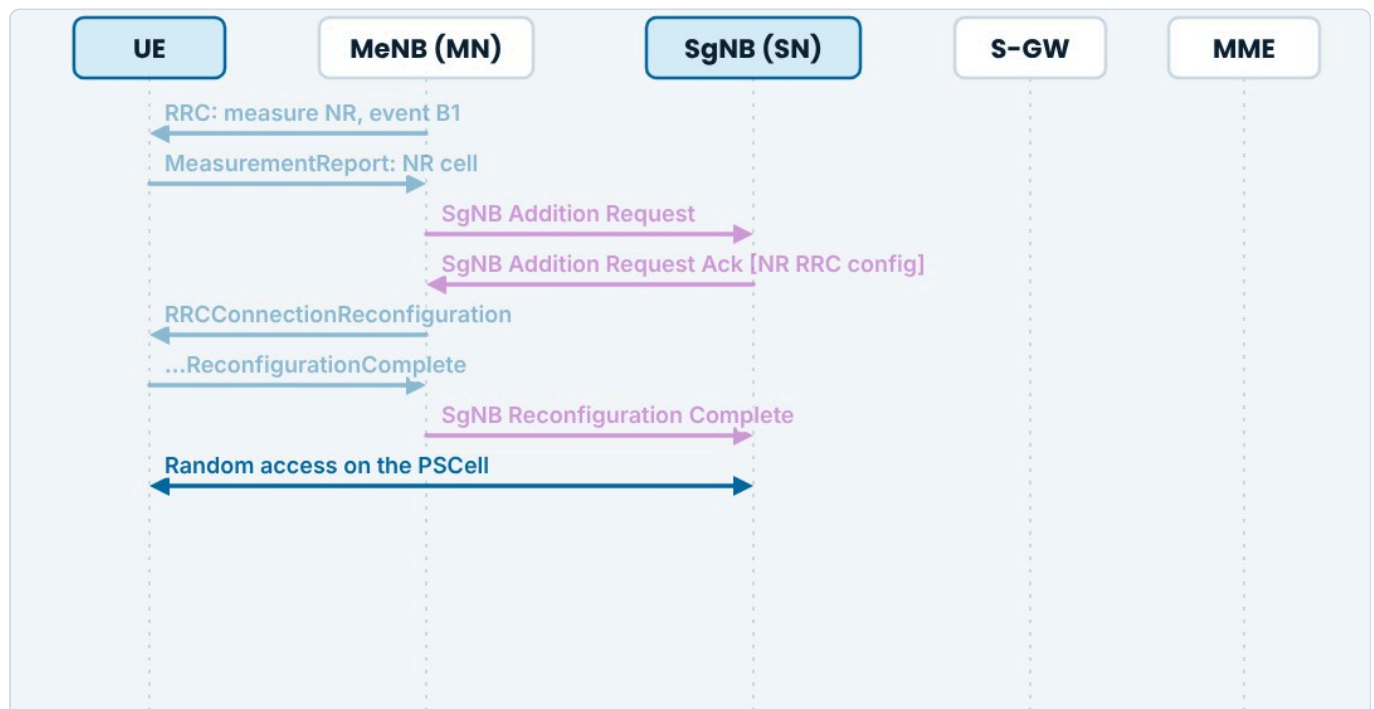
7.4 The UE is reconfigured



The MeNB sends RRCConnectionReconfiguration and puts the NR configuration inside it unchanged. The UE applies both parts and answers Complete. The MeNB then tells the SgNB that the UE has accepted.

TS 37.340 §10.2.1 steps 3–5 · TS 36.331

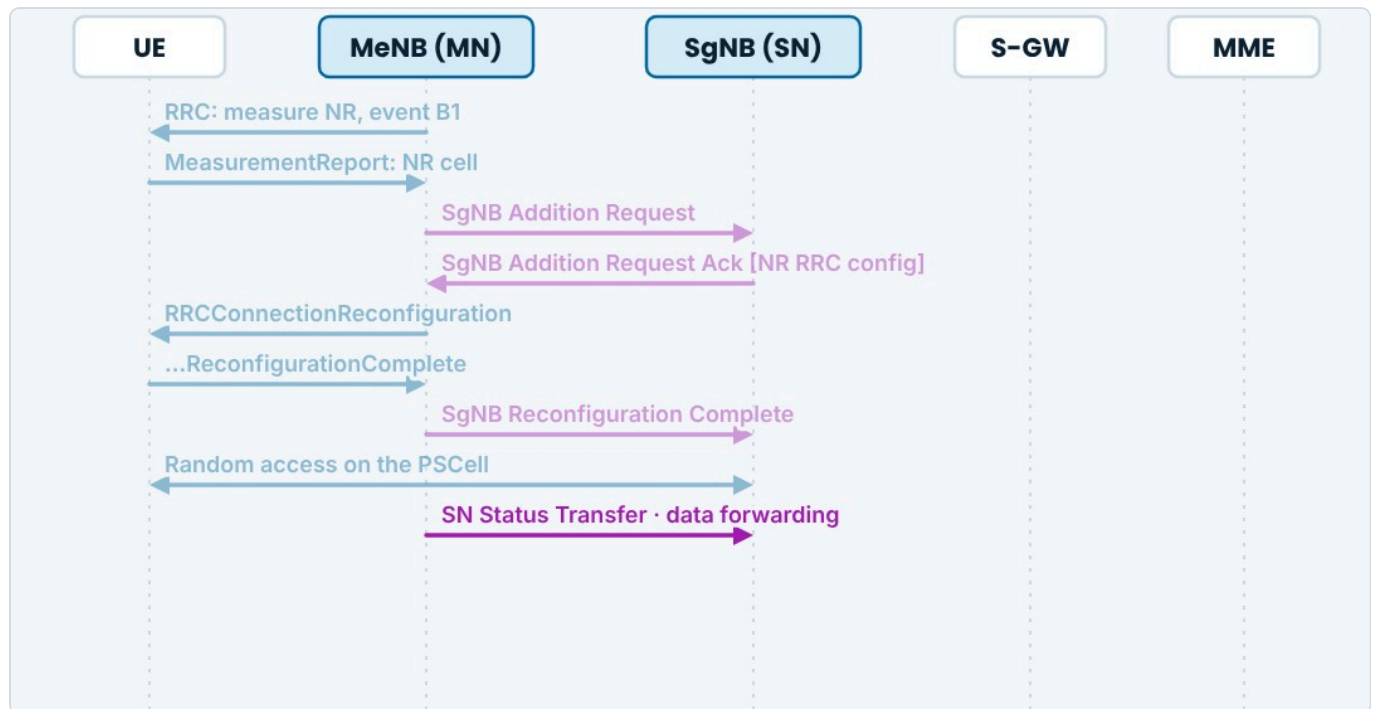
7.5 Random access on NR



The UE synchronises to the PSCell of the SgNB and performs random access there. From now on it has two radio links: LTE to the MeNB and NR to the SgNB.

TS 37.340 §10.2.1 step 6 · TS 38.321

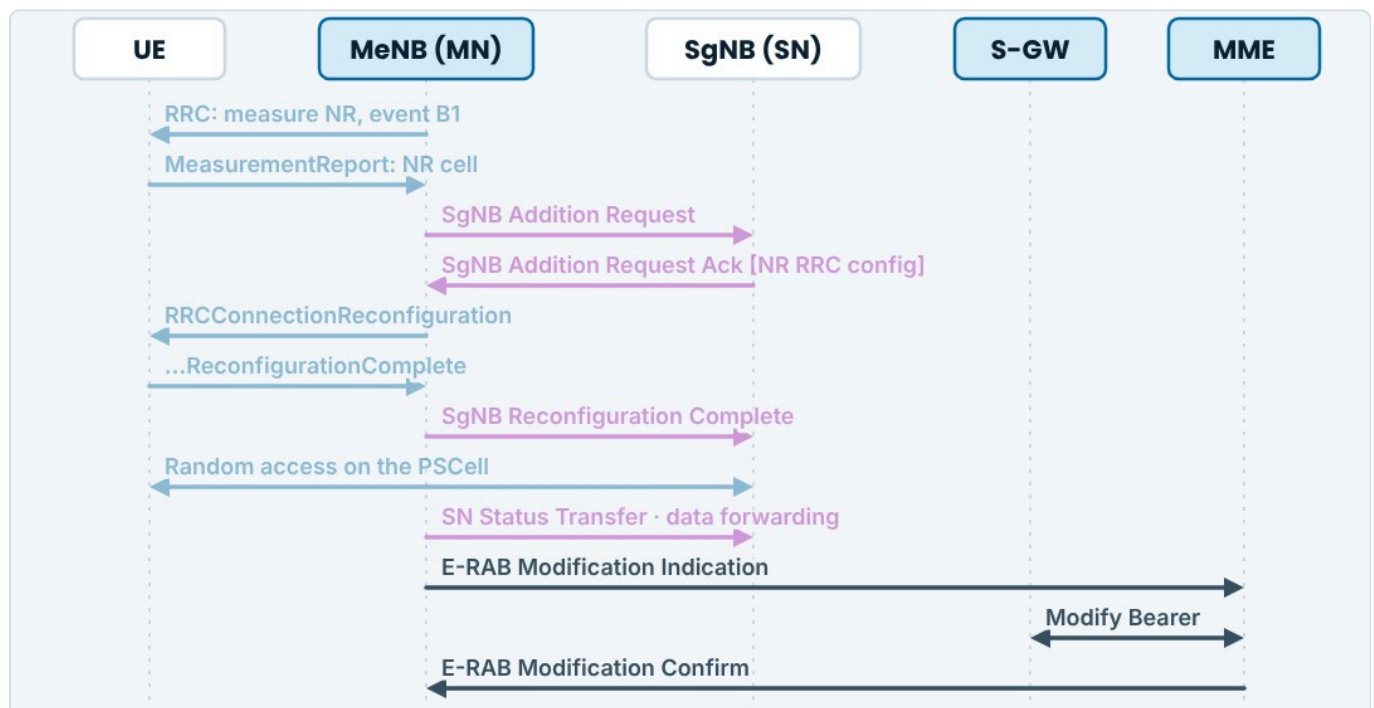
7.6 Bearer state moves over



If a bearer moves to the SgNB, the MeNB sends its PDCP sequence number status in SN Status Transfer and forwards the buffered data, so nothing is lost or duplicated.

TS 37.340 §10.2.1 steps 7–8

7.7 The core path is updated



For bearers that now end in the SgNB, the MeNB tells the MME the new downlink tunnel address. The S-GW switches the path and sends an end marker on the old one. User data now reaches the UE through NR.

TS 37.340 §10.2.1 steps 9–12 · TS 36.413

7.8 The same procedure on the 5G core

EN-DC · X2AP AND S1AP	WITH 5GC · XnAP AND NGAP
SgNB Addition Request	S-Node Addition Request
SgNB Addition Request Ack	S-Node Addition Request Ack
RRCConnectionReconfiguration	RRCReconfiguration (if MN is a gNB)
SgNB Reconfiguration Complete	S-Node Reconfiguration Complete
SN Status Transfer	SN Status Transfer
E-RAB Modification Indication	PDU Session Resource Modify Indication
MME and S-GW	AMF, SMF and UPF

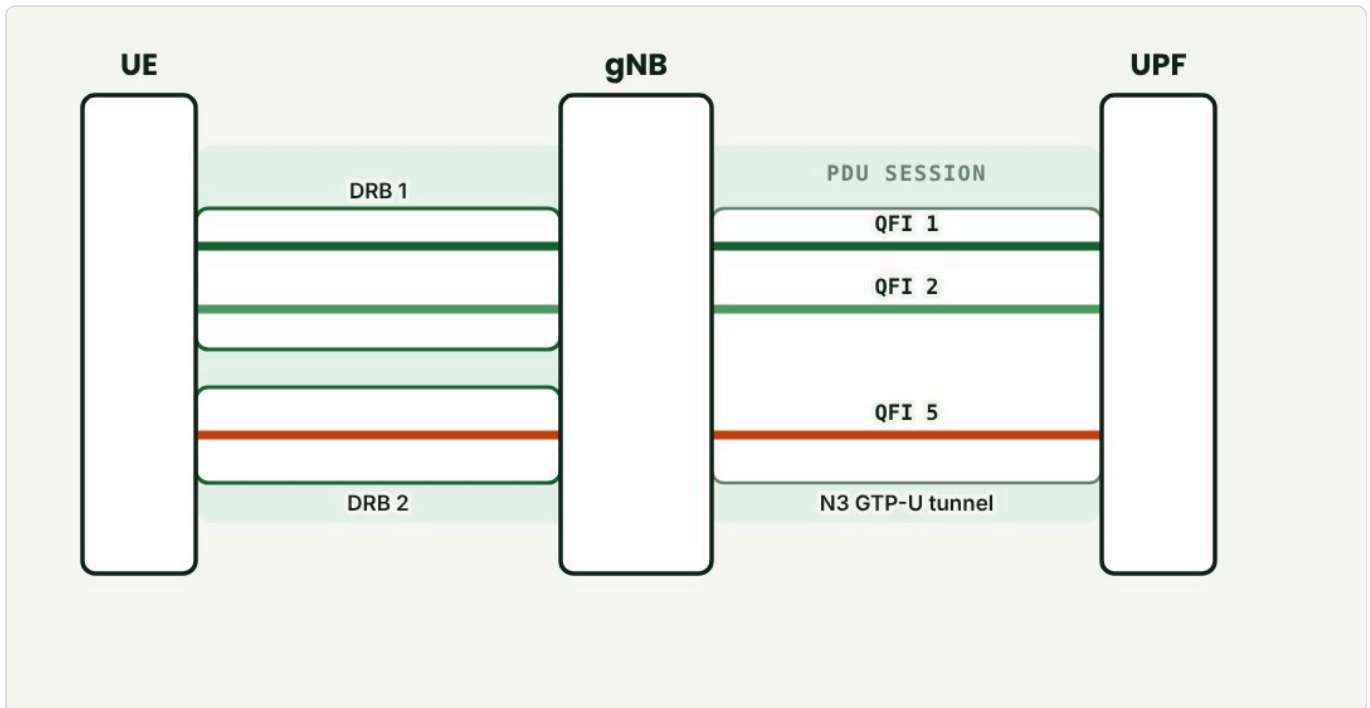
With a 5G core the steps are the same and only the names change. X2AP becomes XnAP, E-RABs become PDU sessions and QoS flows, and the path update goes to the AMF.

TS 37.340 §10.2.2 · TS 38.423 · TS 38.413

The 5G QoS Model

PDU session, QoS flow, QFI and radio bearer: how a packet gets its treatment from the core to the air.

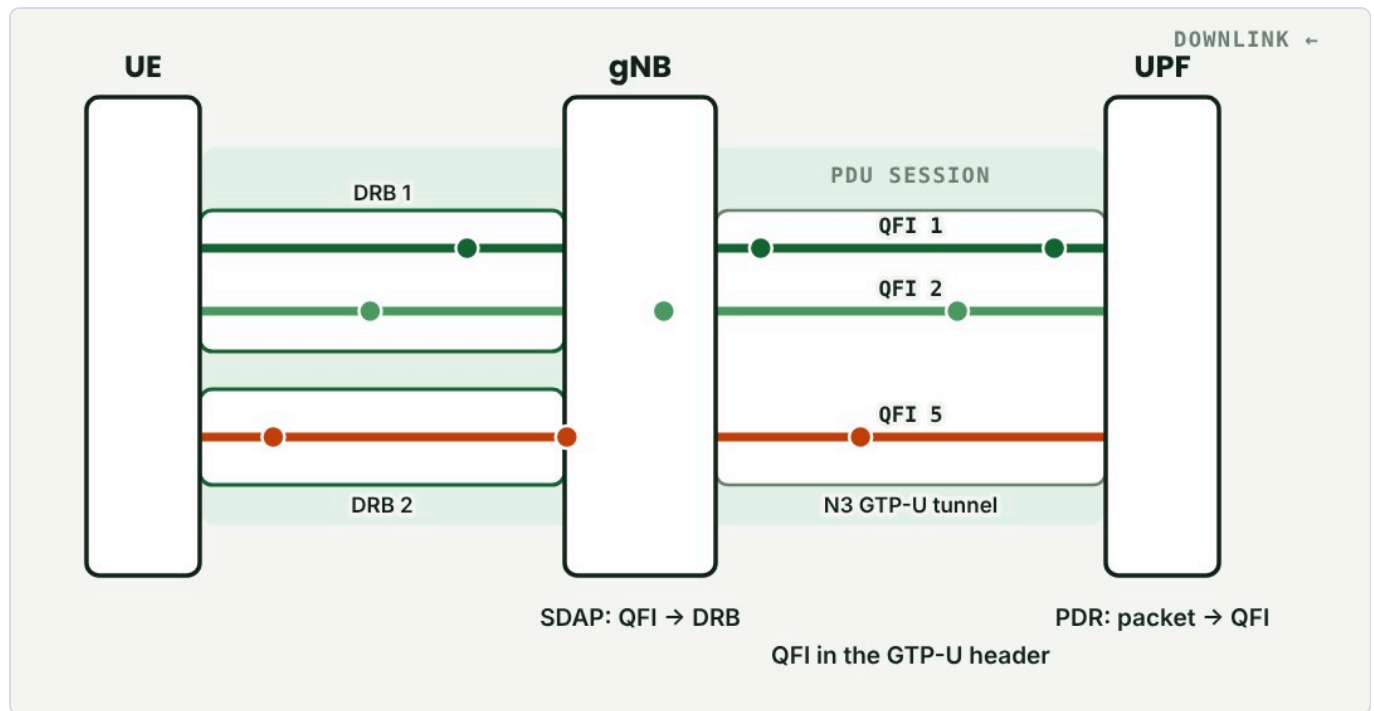
8.1 Three levels



A PDU session carries all traffic to one data network. Inside it, QoS flows are the finest unit of treatment, each with its own identifier, the QFI. On the radio, the gNB packs the flows into data radio bearers.

TS 23.501 §5.7.1 · TS 38.300 §12

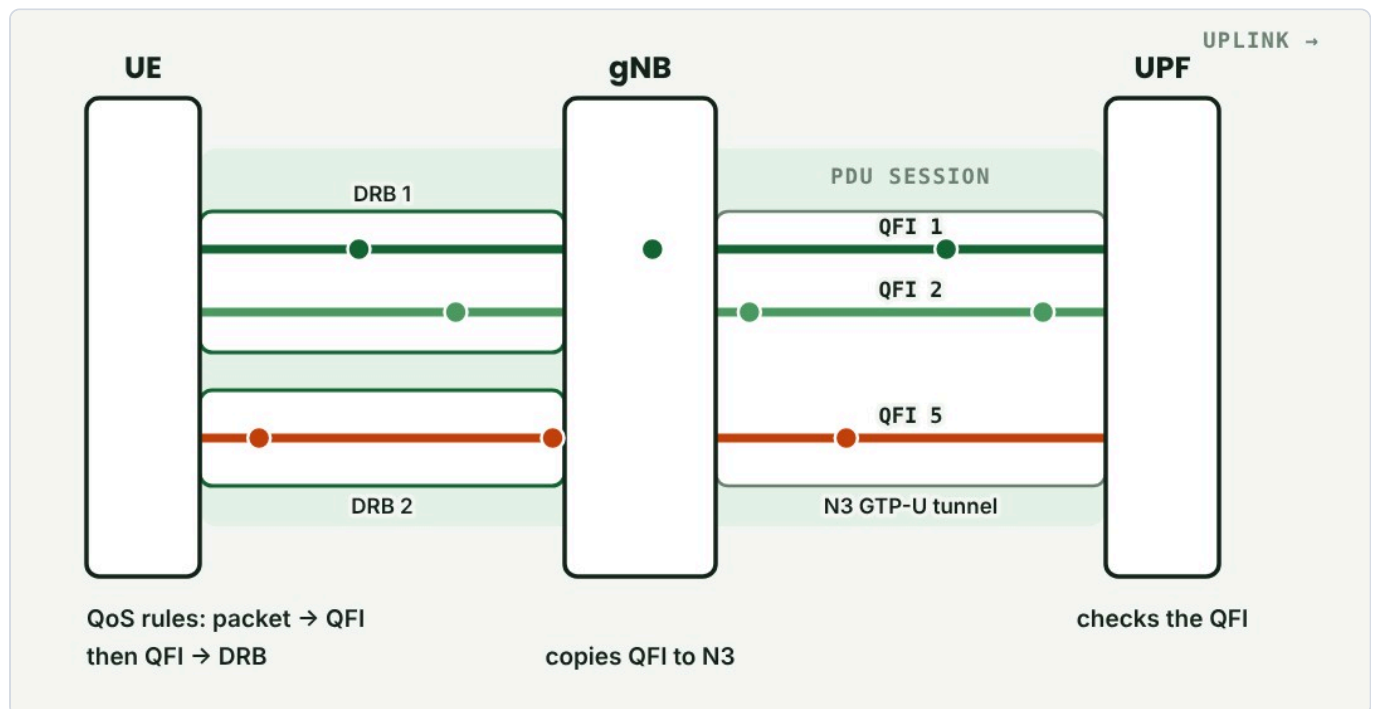
8.2 Downlink: marking in the UPF



The UPF classifies each downlink packet with its packet detection rules and writes the QFI into the GTP-U header. The gNB reads the QFI and SDAP places the packet on the right radio bearer.

TS 23.501 §5.7.1 · TS 29.281 · TS 37.324

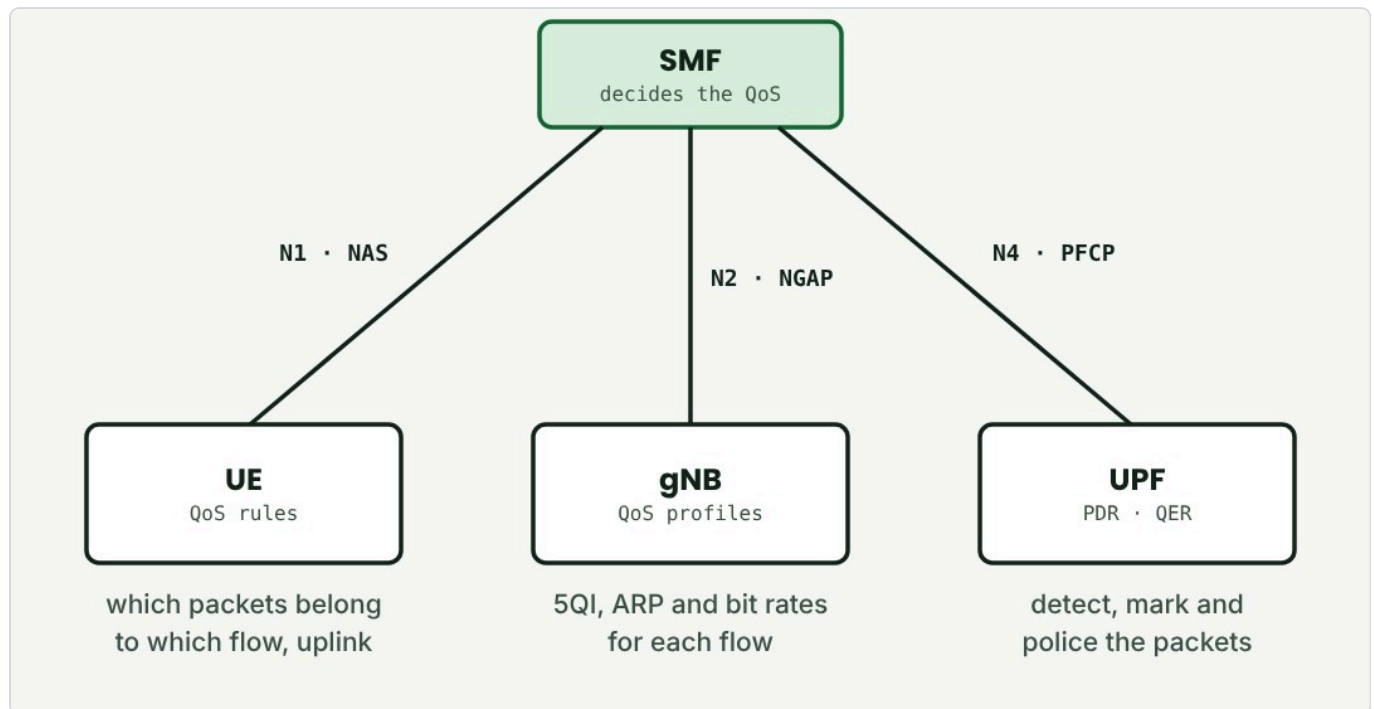
8.3 Uplink: QoS rules in the UE



In the uplink there are two steps in the UE. NAS-level QoS rules decide which QoS flow a packet belongs to. Then the access stratum decides which radio bearer carries that flow. The gNB passes the QFI on to the UPF.

TS 23.501 §5.7.1.4 · TS 38.300 §12

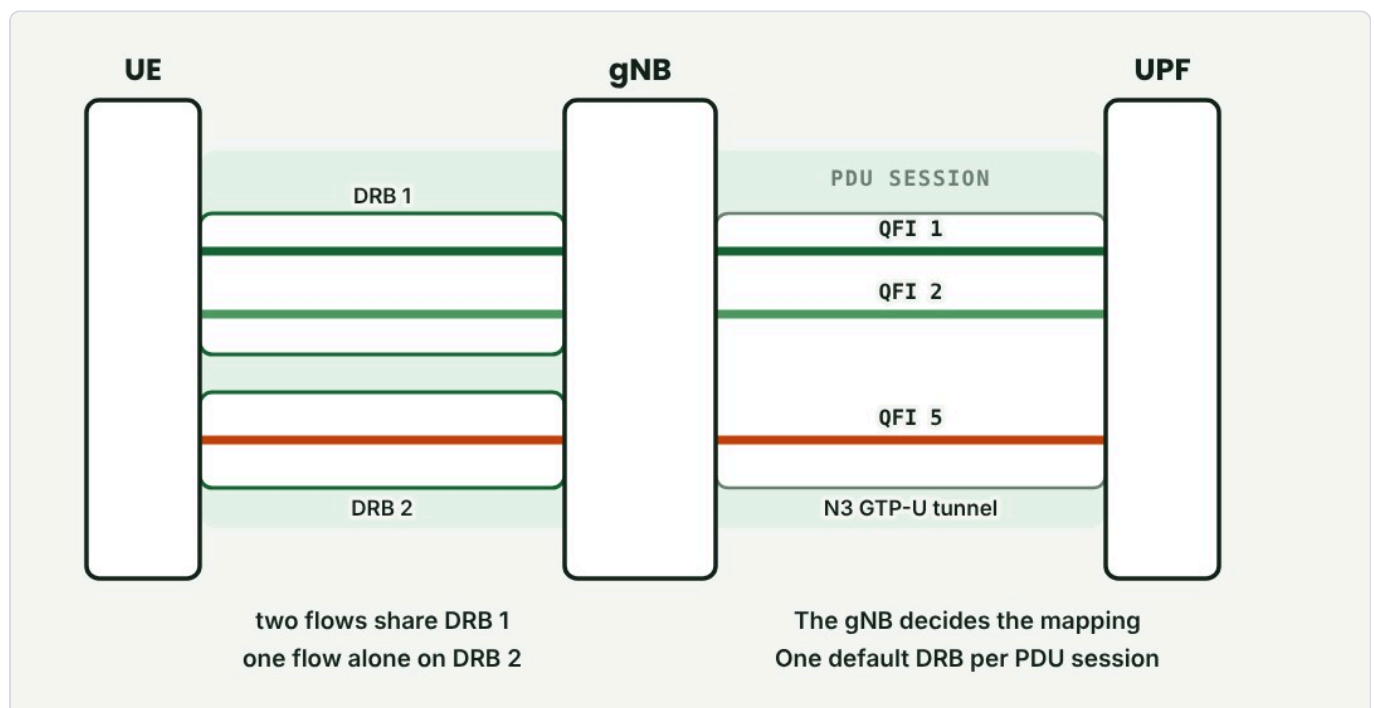
8.4 Who gets what from the SMF



The SMF decides the QoS and tells three parties. The UE gets QoS rules over NAS. The gNB gets a QoS profile for each flow over N2. The UPF gets detection and enforcement rules over N4.

TS 23.501 §5.7.1.2 · TS 23.502 §4.3.2

8.5 Flows onto radio bearers

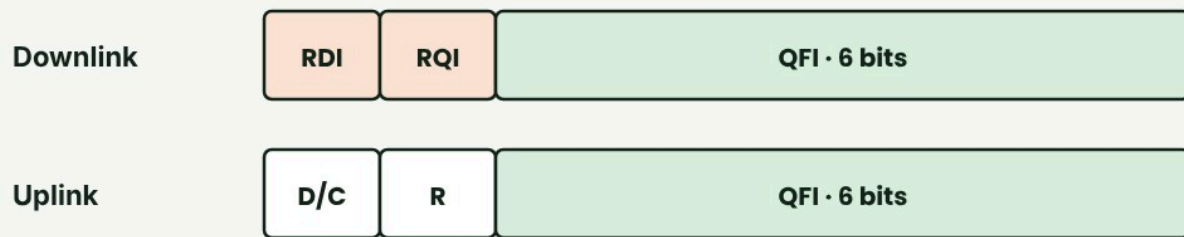


How flows map to radio bearers is up to the gNB. Flows with similar needs can share a bearer, and a demanding flow can have one to itself. Uplink packets that match no mapping rule go to the default bearer.

TS 38.300 §12 · TS 37.324

8.6 The SDAP header

ONE OCTET IN FRONT OF EVERY PACKET, IF CONFIGURED



RQI: reflective QoS

The UE copies the downlink mapping of packets to QoS flows for its uplink

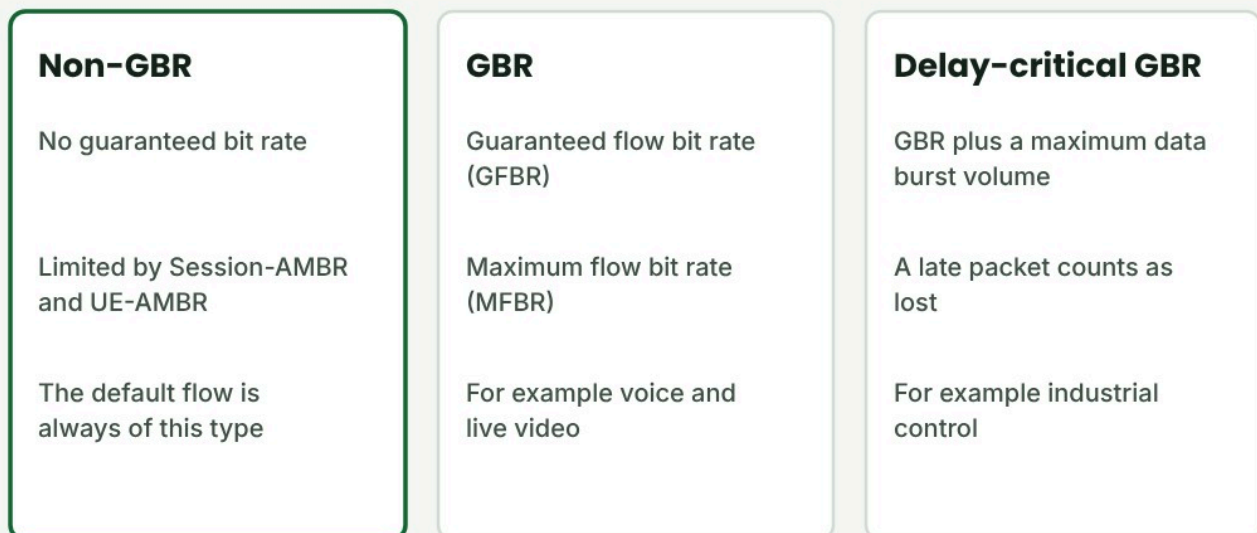
RDI: reflective mapping

The UE sends this QoS flow on the same DRB it arrived on

SDAP is the new layer that does this mapping. Its header is a single octet with the QFI. Two downlink bits switch on reflective behaviour, so the UE can learn its uplink mapping from the downlink packets without extra signalling.

TS 37.324 §6.2.2

8.7 GBR and non-GBR flows



Every PDU session has a default QoS flow, which is non-GBR and stays for the life of the session. GBR flows are added for traffic that needs a guaranteed rate, and delay-critical GBR flows for traffic with hard deadlines.

TS 23.501 §5.7.2 · §5.7.3

8.8 What changed from 4G

	4G · EPS	5G · 5GS
Finest QoS unit	EPS bearer	QoS flow
Identifier	EPS bearer ID	QFI
QoS class	QCI	5QI
To radio bearer	one to one	many to one, by SDAP
Uplink mapping	TFT	QoS rules, reflective QoS
Core tunnel	one per bearer	one per PDU session
New layer	none	SDAP

In 4G one EPS bearer meant one radio bearer and one core tunnel. In 5G the core tunnel is per PDU session, the QoS flow is the unit of QoS, and the gNB is free to group flows on the radio.

TS 23.401 · TS 23.501 §5.7

5G QoS: Rules, Parameters and 5QI

The reference side of QoS: the QoS rule, the parameters and characteristics, and the standardised 5QI tables.

The QoS rule

Sent to the UE by the SMF, or derived by reflective QoS

A QoS rule contains

QoS rule identifier	unique within the PDU session
QFI	the QoS flow that matching packets must use
Packet filter set	which traffic the rule applies to
Precedence value	the order in which rules are evaluated

IP packet filter set

Source and destination IP address or prefix, port numbers, protocol ID, type of service or traffic class, flow label, security parameter index, direction

Ethernet packet filter set

Source and destination MAC address, Ethertype, VLAN tags (C-TAG, S-TAG), IP packet filter fields when the payload is IP, direction

Default QoS rule

Every PDU session has one. It is the only rule whose packet filter set may match all uplink packets, and it is evaluated last, so traffic that fits no other rule still has a flow.

TS 23.501 §5.7.1.4 • §5.7.6 • TS 24.501 §9.11.4.13

QoS parameters

What is signalled for a QoS flow

5QI all flows

Points to the QoS characteristics: either a standardised value or a set signalled with it.

ARP all flows

Allocation and retention priority: who is admitted first when resources are short, and who may be pre-empted.

GFBR and MFBR GBR flows

Guaranteed and maximum flow bit rate, each for uplink and downlink.

Notification control GBR flows

Asks the gNB to tell the core when the guaranteed rate can no longer be met.

RQA non-GBR flows

Reflective QoS attribute: some traffic on this flow may use reflective QoS.

Session-AMBR per PDU session

Limit on the total bit rate of all non-GBR flows of the session. Enforced in the UPF and the UE.

UE-AMBR per UE

Limit on the total bit rate of all non-GBR flows of the UE. Enforced in the gNB.

TS 23.501 §5.7.2

QoS characteristics

What a 5QI stands for

Resource type

Non-GBR, GBR or delay-critical GBR. Decides whether a bit rate is reserved for the flow.

Priority level

Scheduling priority between flows. A lower number means higher priority.

Packet delay budget PDB

Upper bound for the delay of a packet between the UE and the UPF.

Packet error rate PER

Upper bound for the share of packets lost for reasons other than congestion.

Averaging window GBR only

The time over which GFBP and MFBR are calculated. Default 2000 ms.

Maximum data burst volume delay-critical GBR only

The largest amount of data the radio network must deliver within one packet delay budget.

Signalling shortcut: for a standardised 5QI the characteristics are not sent. The number alone is enough.

TS 23.501 §5.7.3

Standardised 5QI: GBR

Averaging window 2000 ms for all of these

5QI	PRI	DELAY	ERROR	EXAMPLE SERVICES
1	20	100 ms	10^{-2}	Conversational voice
2	40	150 ms	10^{-3}	Conversational video, live streaming
3	30	50 ms	10^{-3}	Real-time gaming, V2X messages
4	50	300 ms	10^{-6}	Non-conversational video, buffered streaming
65	7	75 ms	10^{-2}	Mission-critical push-to-talk voice
66	20	100 ms	10^{-2}	Non-mission-critical push-to-talk voice
67	15	100 ms	10^{-3}	Mission-critical video
71	56	150 ms	10^{-6}	Live uplink streaming
72	56	300 ms	10^{-4}	Live uplink streaming
73	56	300 ms	10^{-8}	Live uplink streaming
74	56	500 ms	10^{-8}	Live uplink streaming
75	25	50 ms	10^{-2}	V2X messages
76	56	500 ms	10^{-4}	Live uplink streaming

TS 23.501 Table 5.7.4-1

Standardised 5QI: **non-GBR**

No guaranteed bit rate, limited by the AMBR values

5QI	PRI	DEL	ERR	SER	EXAMPLE SERVICES
5	10	100 ms	10^{-6}		IMS signalling
6	60	300 ms	10^{-6}		Buffered video, TCP-based services
7	70	100 ms	10^{-3}		Voice, live video, interactive gaming
8	80	300 ms	10^{-6}		Buffered video, TCP-based services
9	90	300 ms	10^{-6}		Buffered video, TCP-based services (often the default)
10	90	1100 ms	10^{-6}		Buffered video and TCP over satellite access
69	5	60 ms	10^{-6}		Mission-critical delay-sensitive signalling
70	55	200 ms	10^{-6}		Mission-critical data
79	65	50 ms	10^{-2}		V2X messages
80	68	10 ms	10^{-6}		Low-latency eMBB, augmented reality

TS 23.501 Table 5.7.4-1

Standardised 5QI: **delay-critical GBR**

A packet later than the delay budget counts as lost

5QI	PRI	DEL	ERR	BURST	EXAMPLE SERVICES
82	19	10 ms	10^{-4}	255 B	Discrete automation
83	22	10 ms	10^{-4}	1354 B	Discrete automation, V2X
84	24	30 ms	10^{-5}	1354 B	Intelligent transport systems
85	21	5 ms	10^{-5}	255 B	Electricity distribution, remote driving
86	18	5 ms	10^{-4}	1354 B	V2X advanced driving
87	25	5 ms	10^{-3}	500 B	Interactive motion tracking
88	25	10 ms	10^{-3}	1125 B	Motion tracking, split AI/ML
89	25	15 ms	10^{-4}	17000 B	Cloud and edge rendering
90	25	20 ms	10^{-4}	63000 B	Cloud and edge rendering

Burst is the default maximum data burst volume. 5QI 87 to 90 were added in later releases for XR and media.

TS 23.501 Table 5.7.4-1

Where the QoS is signalled

The same flow, described on four interfaces

INTERFACE	MESSAGE	CARRIES
N1 • NAS	PDU Session Establishment Accept	Authorized QoS rules, QoS flow descriptions, Session-AMBR
N2 • NGAP	PDU Session Resource Setup Request	QoS flow list with QFI and QoS flow level QoS parameters
F1 • F1AP	UE Context Setup Request	DRBs to set up, with the flows mapped to each DRB and their QoS
E1 • E1AP	Bearer Context Setup Request	PDU session resources and flows for the CU-UP
N4 • PFCP	Session Establishment Request	PDR, QER and FAR: detect, police and forward
Uu • RRC	RRCReconfiguration	DRB configuration and the SDAP mapping

TS 24.501 • TS 38.413 • TS 38.473 • TS 38.463 • TS 29.244 • TS 38.331

Summary

A QoS rule is QFI, packet filter set and precedence. The default rule catches everything else.

Parameters are per flow: 5QI and ARP always, bit rates for GBR flows, AMBR limits for the non-GBR ones.

Characteristics are what the 5QI stands for: resource type, priority, delay budget and error rate.

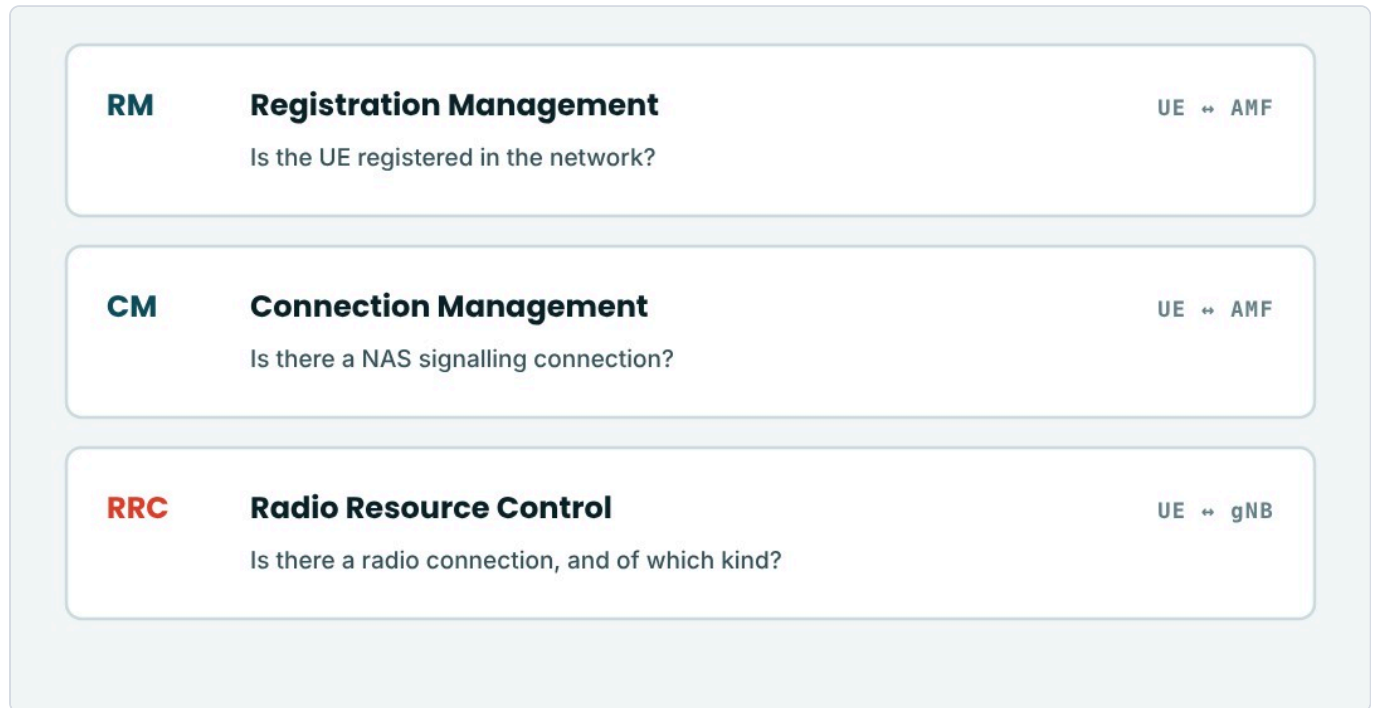
5QI 1, 5 and 9 are the ones in most logs: voice, IMS signalling and default data.

References: 3GPP TS 23.501 §5.7 • TS 24.501 • TS 38.413

RM, CM and RRC States

Three state machines run in every 5G UE at the same time. What each one tracks, and how they fit together.

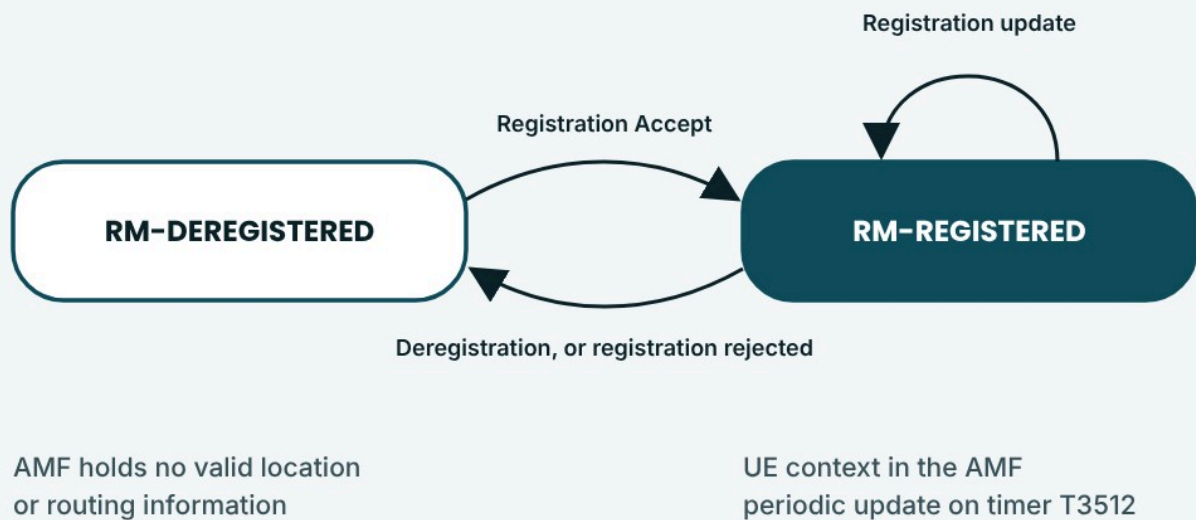
10.1 Three state machines



A 5G UE is always in three states at once. RM says whether it is registered. CM says whether it has a signalling connection to the core. RRC says what kind of radio connection it has. The first two are NAS states. The third is a radio state.

TS 23.501 §5.3.2 · §5.3.3 · TS 38.331 §4.2.1

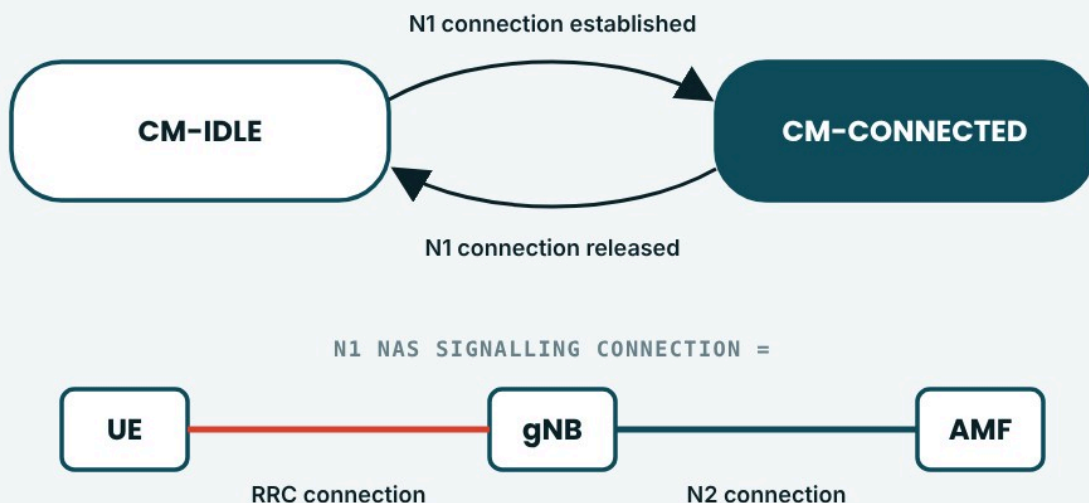
10.2 Registration management



In RM-DEREGISTERED the network cannot reach the UE. A successful registration moves it to RM-REGISTERED, where the AMF holds its context. The UE then refreshes its registration when it moves to a new area and periodically on timer T3512.

TS 23.501 §5.3.2

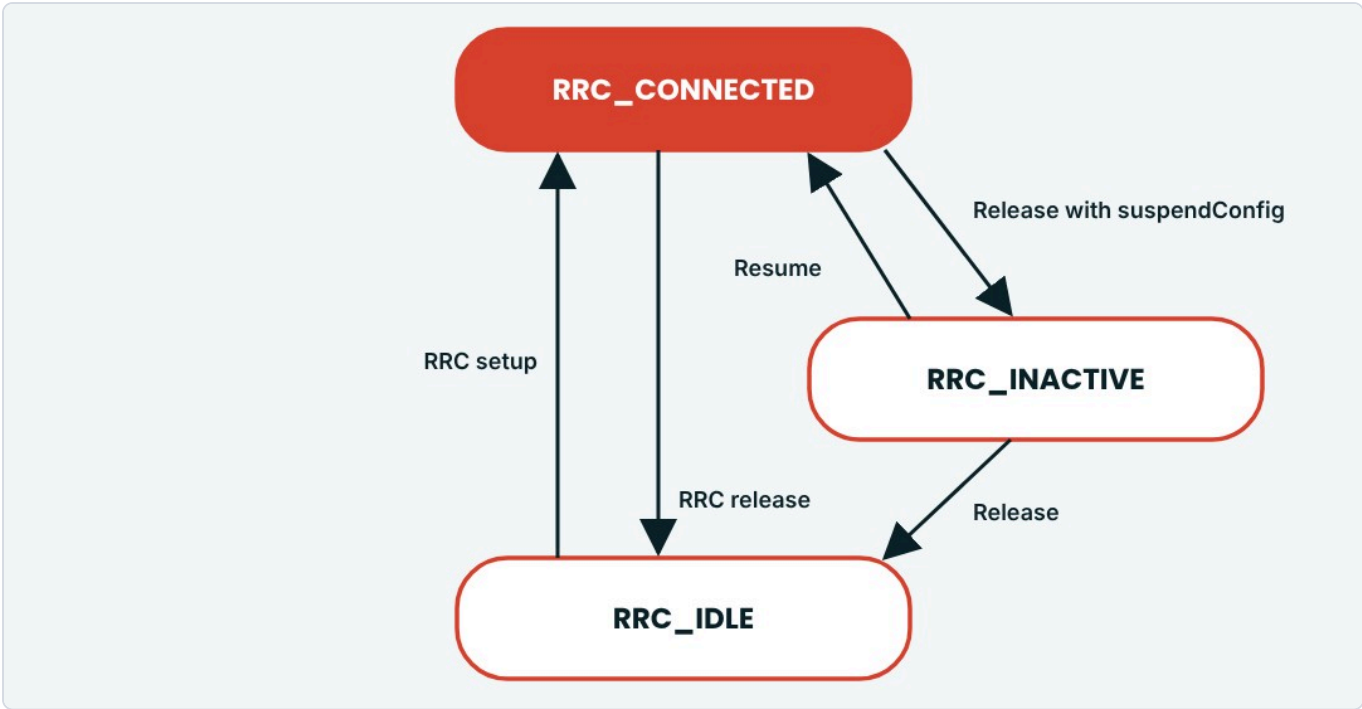
10.3 Connection management



CM-CONNECTED means the UE has an N1 signalling connection to the AMF. That connection is made of two parts: the RRC connection on the radio side and the N2 connection between the gNB and the AMF. In CM-IDLE neither exists.

TS 23.501 §5.3.3

10.4 Three RRC states



NR has three RRC states. RRC setup takes the UE from idle to connected. A plain release sends it back to idle. A release with suspendConfig sends it to RRC_INACTIVE, and a resume brings it back to connected.

TS 38.331 §4.2.1 · §5.3

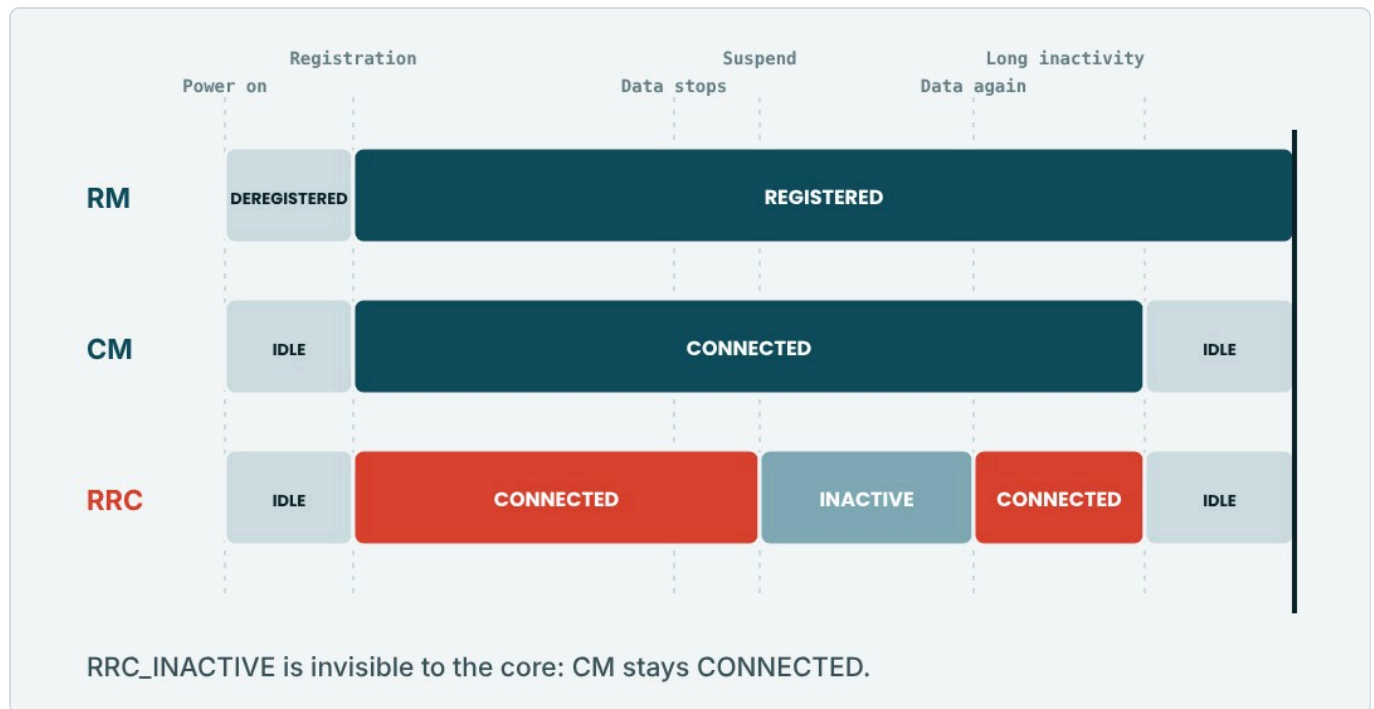
10.5 What changes in each RRC state

	RRC_IDLE	RRC_INACTIVE	RRC_CONNECTED
Mobility	cell reselection	cell reselection	handover
Paging from	the core (AMF)	the RAN (gNB)	not needed
UE context in gNB	no	yes, stored	yes
Location known per	tracking area list	RAN notification area	cell
Data transfer	no	no, resume first	yes
Core sees the UE as	CM-IDLE	CM-CONNECTED	CM-CONNECTED

Idle and inactive look alike for the UE: it reselects cells on its own and listens for paging. The difference is in the network. In inactive the gNB keeps the UE context, pages the UE itself, and the core still sees it as connected.

TS 38.331 §4.2.1 · TS 38.300 §7.2

10.6 One UE over time



Follow one UE. It registers and all three states change together. When the data stops, the gNB suspends the UE: RRC goes to inactive, but CM stays connected. When data returns, a resume is enough. After long inactivity everything goes back to idle.

TS 23.501 §5.3.3.2.5 · TS 38.300 §9.2.2

10.7 Why RRC_INACTIVE exists

From RRC_IDLE

RRC setup, security activation, context from the core, reconfiguration

slower, more signalling

From RRC_INACTIVE

RRC resume: the stored context and security keys are reused

faster, less signalling

The UE saves battery like in idle, but returns to data like from connected.

From idle, getting back to data needs a full setup with the core involved. From inactive, the UE resumes with the context and keys it already has. That saves signalling and time for devices that send small amounts of data often.

TS 38.300 §9.2.2 · TS 38.331 §5.3.13

10.8 Three things to remember

1

RM: registered or not

Changes with registration and deregistration.

2

CM: N1 signalling connection or not

CM-CONNECTED needs an RRC connection and an N2 connection.

3

RRC: idle, inactive or connected

Inactive keeps the context in the gNB and the core connection up.

RM and CM are tracked by the UE and the AMF. RRC is tracked by the UE and the gNB. RRC_INACTIVE is the one state where the radio side and the core side see the UE differently.

TS 23.501 §5.3 · TS 38.331 §4.2

RRC Inactive

Suspend, the RAN notification area, resume and RAN paging: how the gNB keeps a UE on standby.

11.1 Suspend: what the UE is given

RRCRelease WITH suspendConfig

fullI-RNTI · **shortI-RNTI** the identity of the stored UE context: 40 and 24 bits

ran-NotificationAreaInfo where the UE may move without telling the network

ran-PagingCycle how often the UE listens for RAN paging

t380 timer for the periodic RNA update

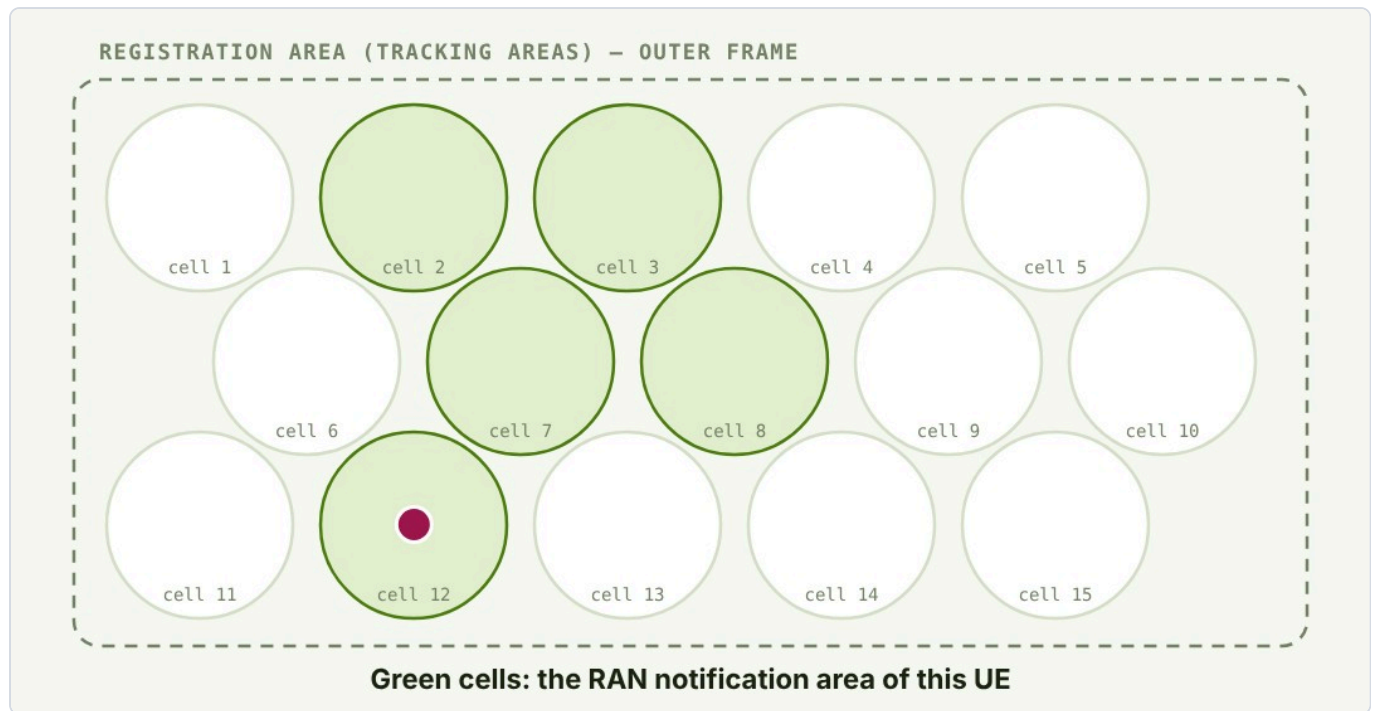
nextHopChainingCount used to derive the new key at resume

The UE and the gNB both store the UE context: bearers, security and configuration.

The gNB moves a UE to RRC_INACTIVE with an RRCRelease that contains suspendConfig. It gives the UE an identity for its stored context, the I-RNTI, an area to move in, a paging cycle and a timer.

TS 38.331 §5.3.8 · SuspendConfig

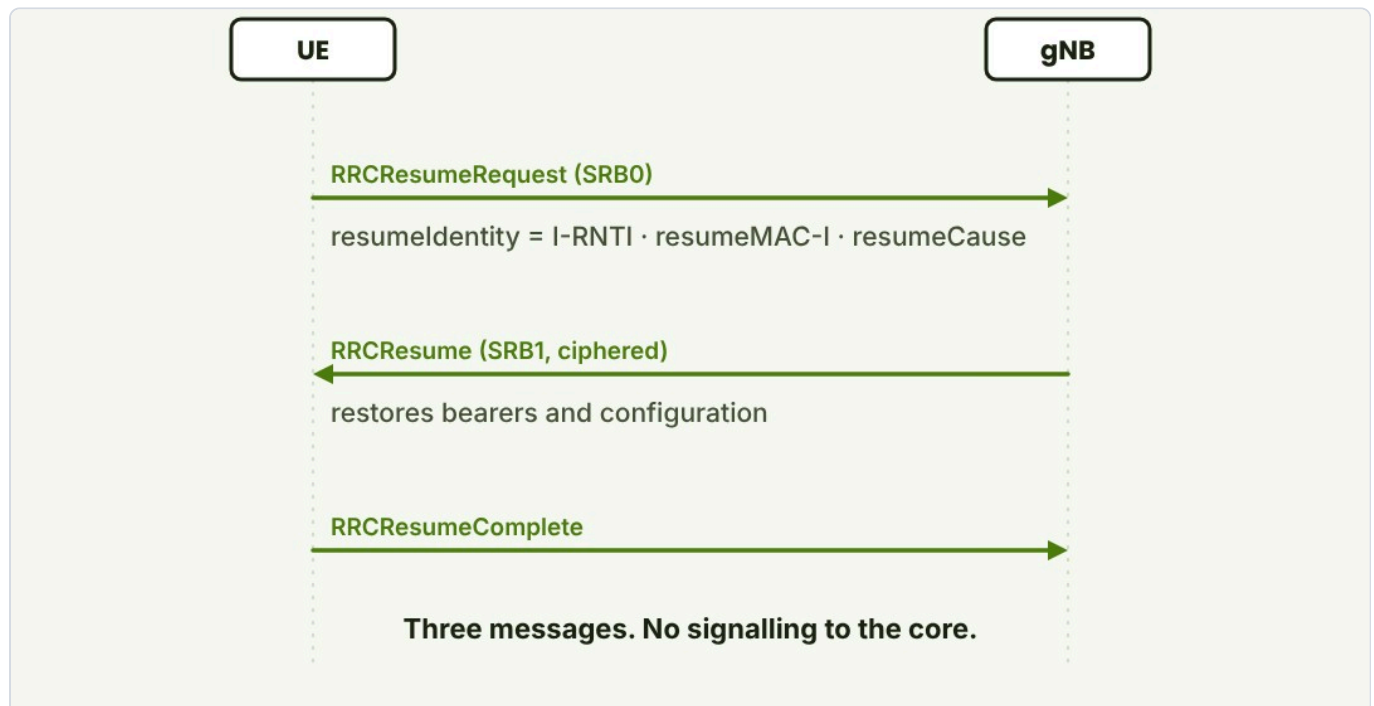
11.2 The RAN notification area



The RAN notification area is a set of cells, given as a cell list or as RAN area codes. It lies inside the registration area. Within it the UE reselects cells freely and tells no one.

TS 38.300 §9.2.2.3

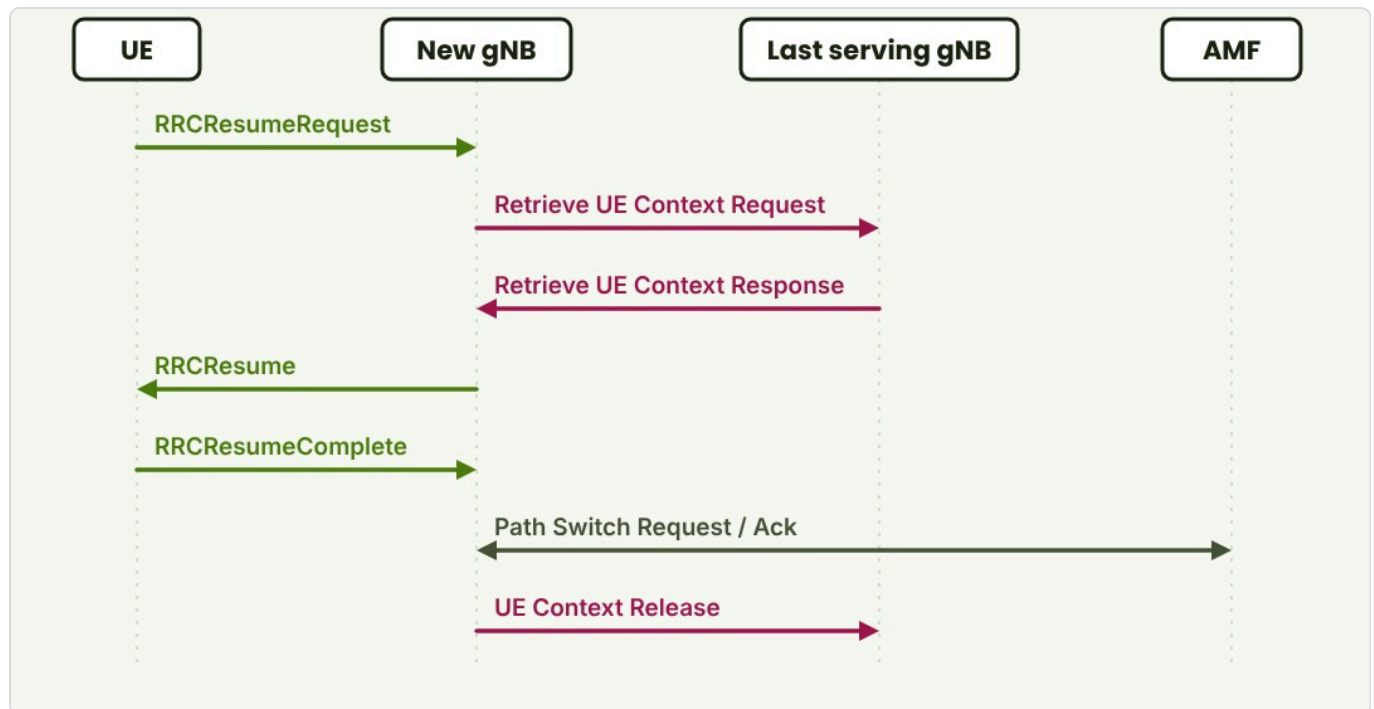
11.3 Resume at the same gNB



To return to connected, the UE sends RRCResumeRequest with its I-RNTI and a short integrity check. The gNB finds the stored context and answers RRCResume. The bearers come back as they were.

TS 38.331 §5.3.13 · TS 38.300 §9.2.2.4.1

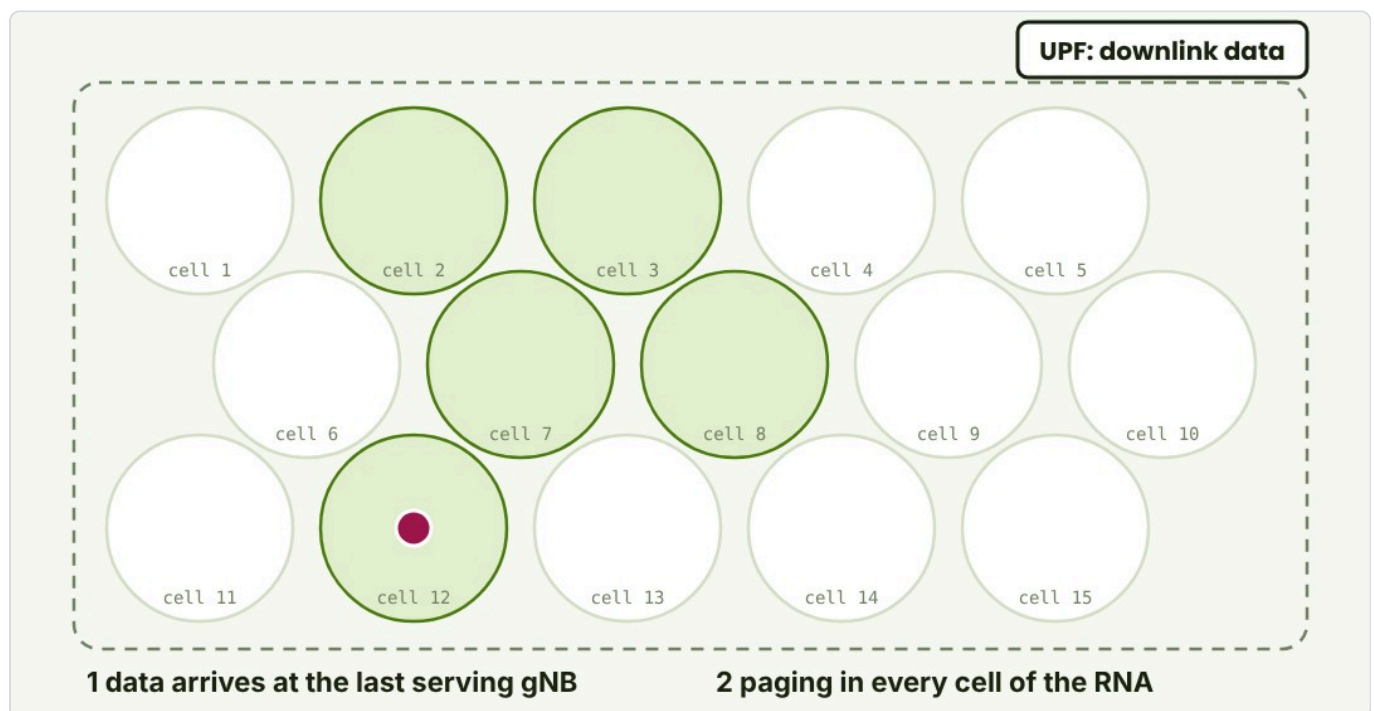
11.4 Resume at a different gNB



If the UE has moved, the new gNB reads the I-RNTI to find the last serving gNB and fetches the UE context from it over Xn. After the resume it switches the path at the core and releases the old context.

TS 38.300 §9.2.2.4.1 · TS 38.423

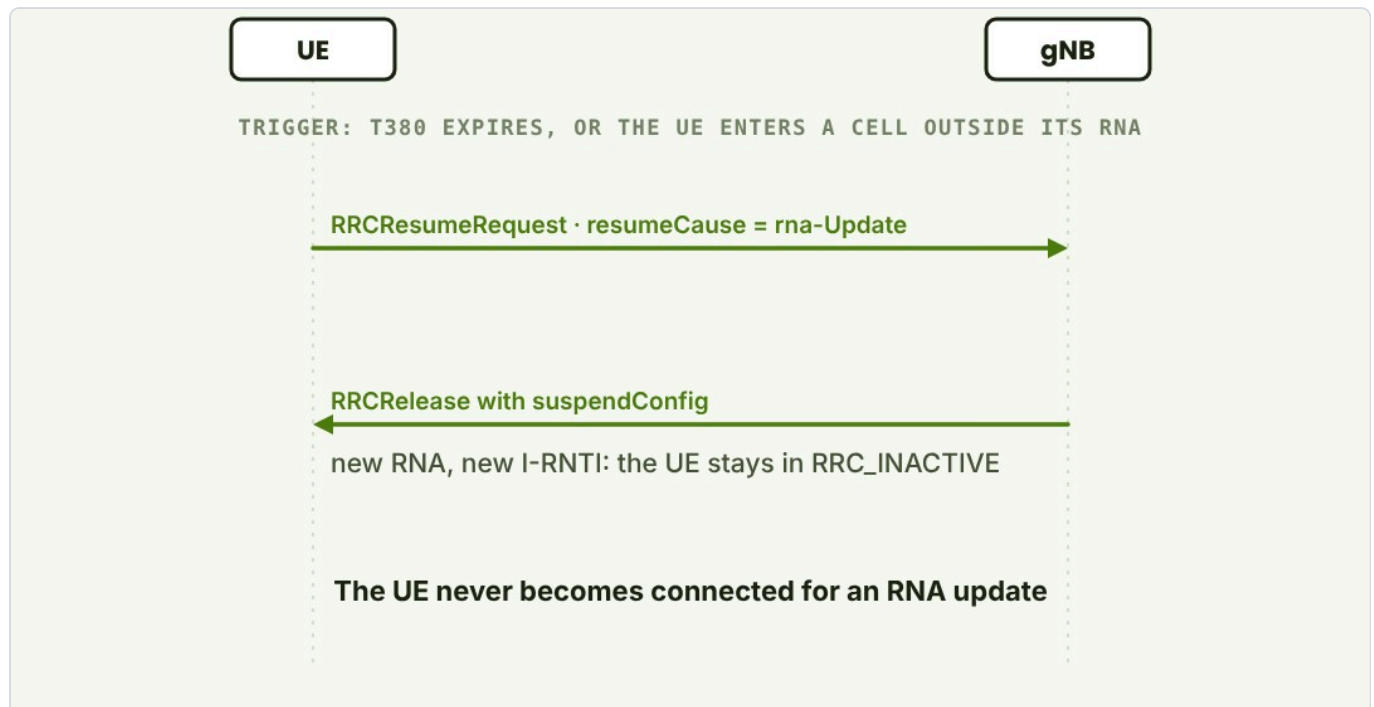
11.5 RAN paging



The core still thinks the UE is connected, so downlink data simply arrives at the last serving gNB. That gNB pages the UE in its own cells and asks its neighbours over Xn to page in theirs. The UE answers with a resume.

TS 38.300 §9.2.2.4.2 · TS 38.423

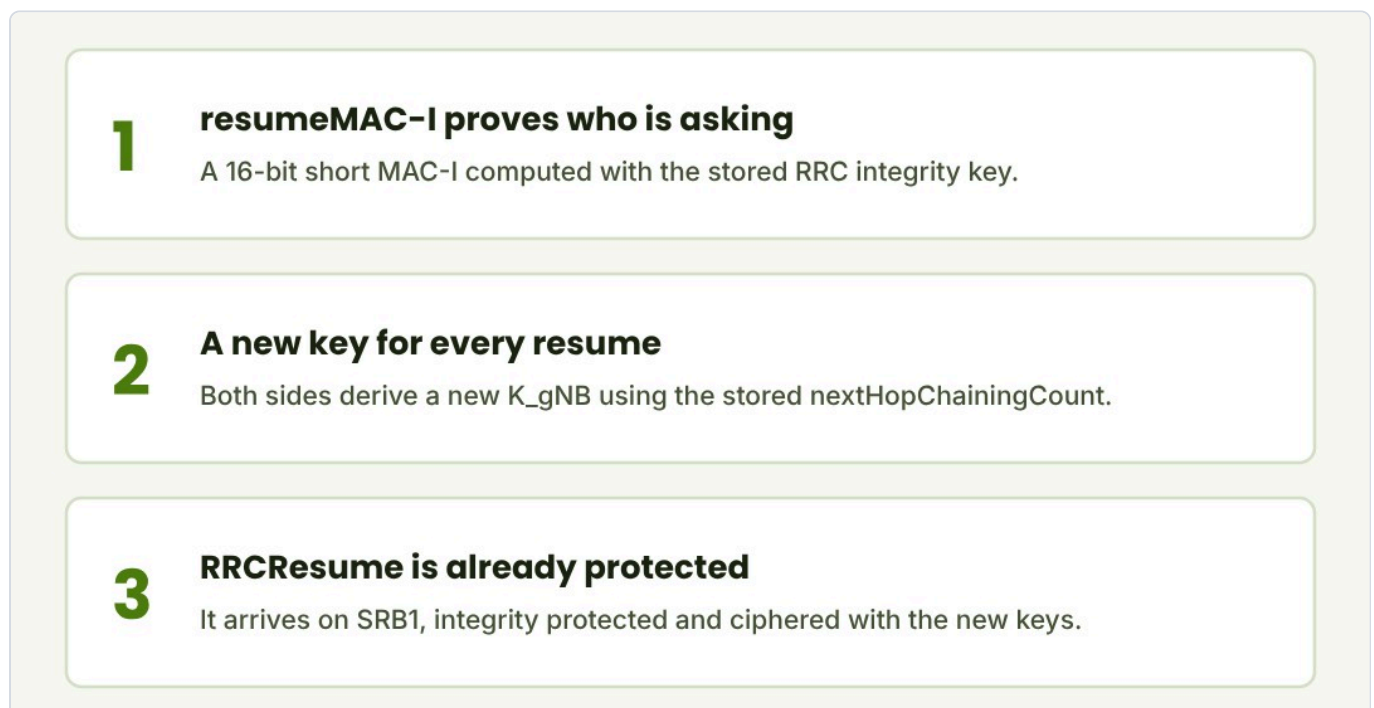
11.6 RNA update



The UE reports in when timer T380 expires or when it enters a cell outside its notification area. It sends a resume request with the cause rna-Update. The gNB can answer with a new suspend, so the UE stays inactive.

TS 38.300 §9.2.2.5 · TS 38.331 §5.3.13.8

11.7 Security at resume



The resume request goes out unprotected on SRB0, so it carries a short MAC-I to prove the UE is genuine. Both sides then derive a fresh key, and everything from RRCResume onwards is ciphered and integrity protected.

TS 33.501 §6.8.2 · TS 38.331 §5.3.13.3

11.8 Idle and inactive side by side

	RRC_IDLE	RRC_INACTIVE
UE identity	5G-S-TMSI	I-RNTI
Paging by	AMF, in the tracking areas	gNB, in the RNA
Area update	registration update (NAS)	RNA update (RRC)
Periodic timer	T3512	T380
Back to data	RRC setup + NAS service request	RRC resume
Core connection	released	kept: CM-CONNECTED

RRC_INACTIVE moves the standby job from the core to the RAN: the gNB keeps the context, the gNB pages, and the gNB is told when the UE leaves its area.

TS 38.300 §9.2.2 · TS 38.331 §4.2.1

5G Access Control

How a congested network stops access attempts before they are sent: cell barring and Unified Access Control.

12.1 Two mechanisms

Cell barring and reservation

MIB and SIB1

Decides whether the UE may camp on the cell at all.

Affects cell selection and reselection

Unified Access Control (UAC)

SIB1: uac-BarringInfo

Decides whether one access attempt may be sent.

Applies in RRC idle, inactive and connected

When a cell is overloaded, the cheapest request is the one that is never sent. 5G has two tools for that. Cell barring keeps UEs off a cell altogether. Unified Access Control filters single access attempts, by who is asking and what for.

TS 22.261 §6.22 · TS 38.304 §5.3 · TS 38.331 §5.3.14

12.2 Cell barring and reservation

MIB

cellBarred

barred or notBarred. Read before SIB1.

intraFreqReselection

if barred: may the UE pick another cell on this frequency?

SIB1

cellReservedForOperatorUse

only UEs with access identity 11 or 15 may camp

cellReservedForOtherUse

the UE treats the cell as barred

A barred cell is excluded for up to 300 seconds.

The same applies when the UE cannot read MIB or SIB1.

The MIB carries cellBarred, so the UE learns the status before it reads anything else. SIB1 can reserve the cell for operator use or for other use. A UE that finds a cell barred leaves it out of cell selection for up to 300 seconds.

TS 38.331 MIB, SIB1 · TS 38.304 §5.3.1

12.3 Who is asking, and what for

Access identity

who is asking

- 0 · no special configuration
- 1 · Multimedia Priority Service
- 2 · Mission Critical Service
- 11–15 · access classes 11 to 15

FROM THE USIM AND THE NETWORK

Access category

what the attempt is for

- 0 · answering paging
- 2 · emergency
- 3 · signalling
- 4 · voice call 7 · data

CHOSEN BY NAS FOR EACH ATTEMPT

Every access attempt has one category and one or more identities

UAC describes every access attempt with two things. The access identity says who the subscriber is, for example an ordinary user or a priority service user. The access category says what the attempt is for.

TS 24.501 §4.5.2 · TS 22.261 §6.22

12.4 Access categories

CAT	ACCESS ATTEMPT	RRC ESTABLISHMENT CAUSE
0	Mobile-terminated access (paging)	mt-Access
1	Delay-tolerant service	not applicable
2	Emergency	emergency
3	Mobile-originated signalling	mo-Signalling
4	MMTel voice	mo-VoiceCall
5	MMTel video	mo-VideoCall
6	SMS	mo-SMS
7	Mobile-originated data	mo-Data
32–63	Operator-defined	by operator rules

The standardised categories cover paging response, emergency, signalling, voice, video, SMS and data. Operators can define their own in the range 32 to 63. For most categories NAS also sets the matching RRC establishment cause.

TS 24.501 Table 4.5.2.2 · Table 4.5.6.1

12.5 What the cell broadcasts

SIB1 · uac-BarringInfo · ONE SET PER ACCESS CATEGORY

uac-BarringFactor

p00 to p95: the probability that an attempt is allowed

uac-BarringTime

4 to 512 s: base for the wait after a barred attempt

uac-BarringForAccessIdentity

7 bits: identities 1, 2 and 11 to 15. Bit 0 = not barred

A category with no barring set is not barred.

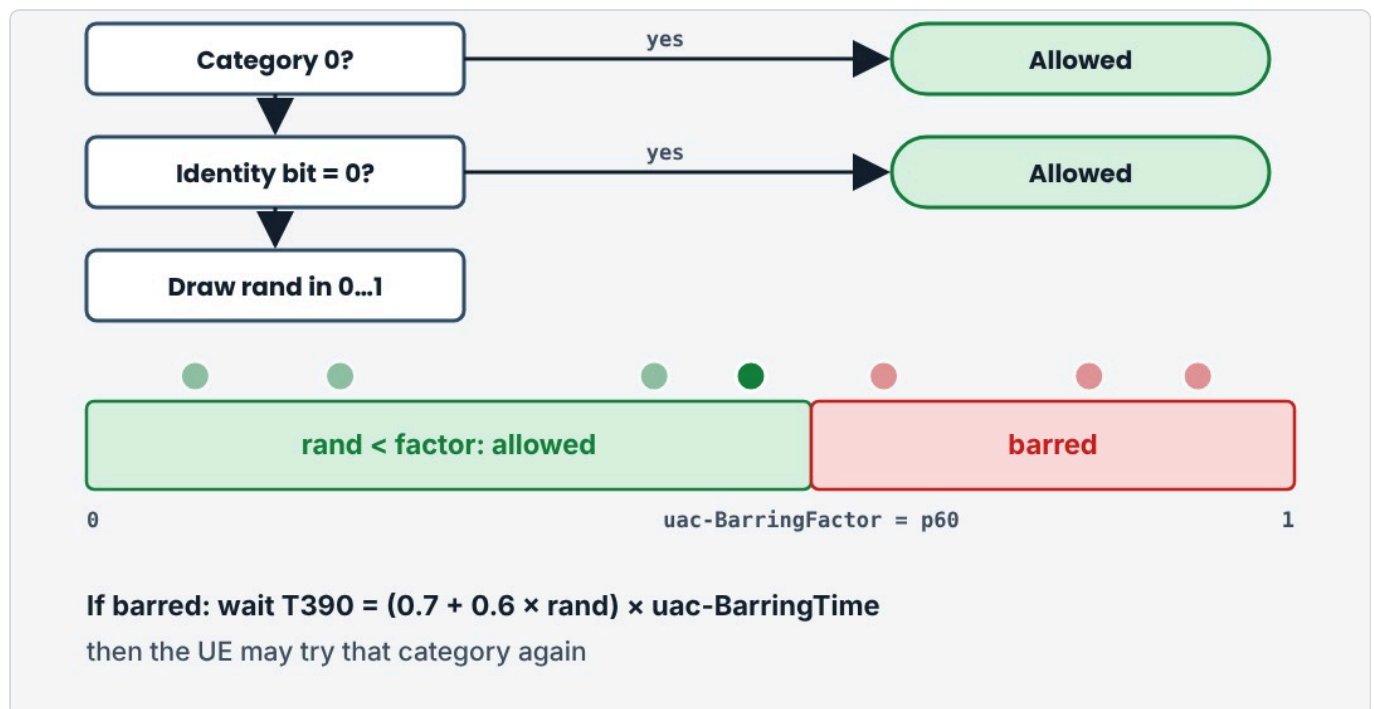
Barring can be set per PLMN, so operators sharing a cell can differ.

SIB1 can also carry uac-AccessCategory1-SelectionAssistanceInfo for delay-tolerant devices.

SIB1 lists the barring for each access category as a set of three values: a barring factor, a barring time, and seven bits that say which special access identities are barred.

TS 38.331 §6.3.2 · UAC-BarringInfoSetList

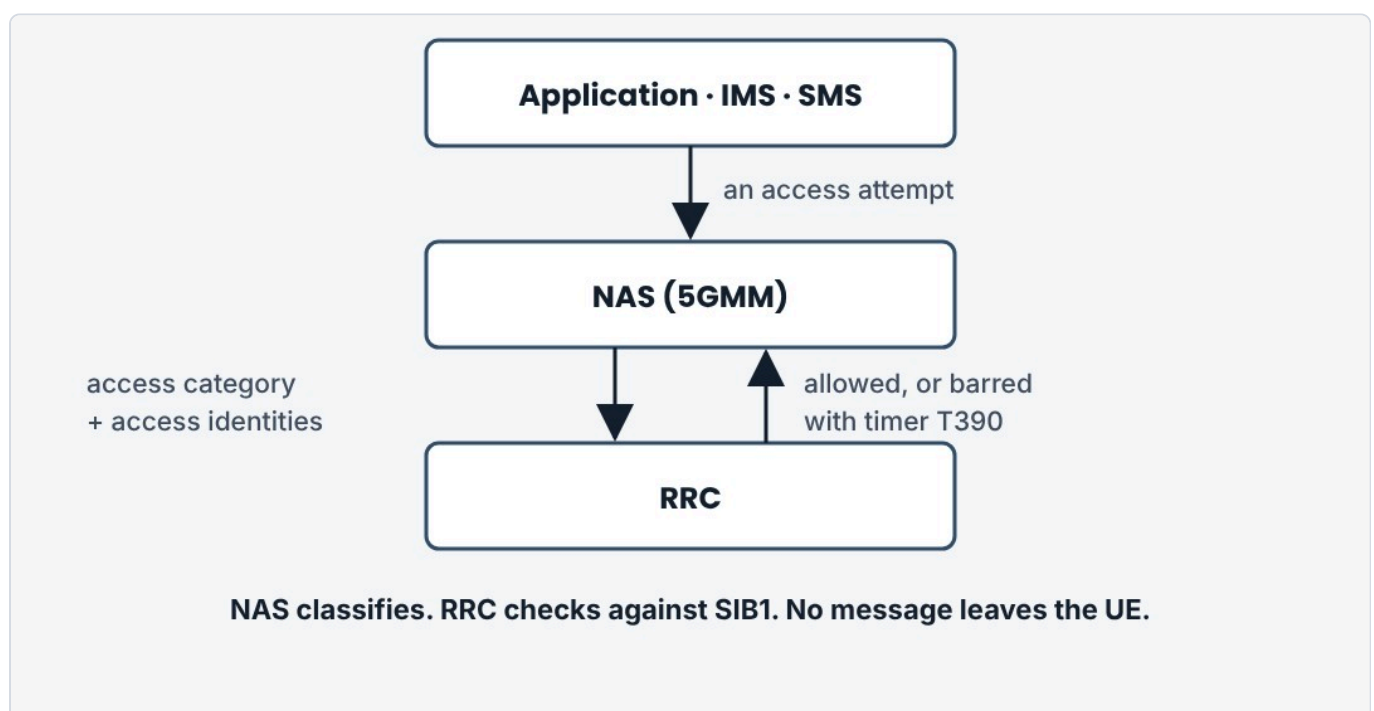
12.6 The barring check



Category 0, the response to paging, is never barred. A UE with a special identity whose bit is zero is allowed. Everyone else draws a random number. If it is below the barring factor the attempt goes ahead. If not, the UE waits for timer T390.

TS 38.331 §5.3.14.5

12.7 NAS classifies, RRC checks



The work is shared inside the UE. NAS maps the request to an access category and hands it to RRC with the access identities. RRC runs the check against the latest SIB1 and reports back. The network is not asked.

TS 24.501 §4.5 · TS 38.331 §5.3.14

12.8 Three things to remember

1

Cell barring comes first

A barred or reserved cell is not camped on, so no attempt starts there.

2

UAC = identity + category

NAS picks the category; the barring set for that category is in SIB1.

3

A random draw decides

Below the barring factor the attempt goes out. Otherwise T390 runs.

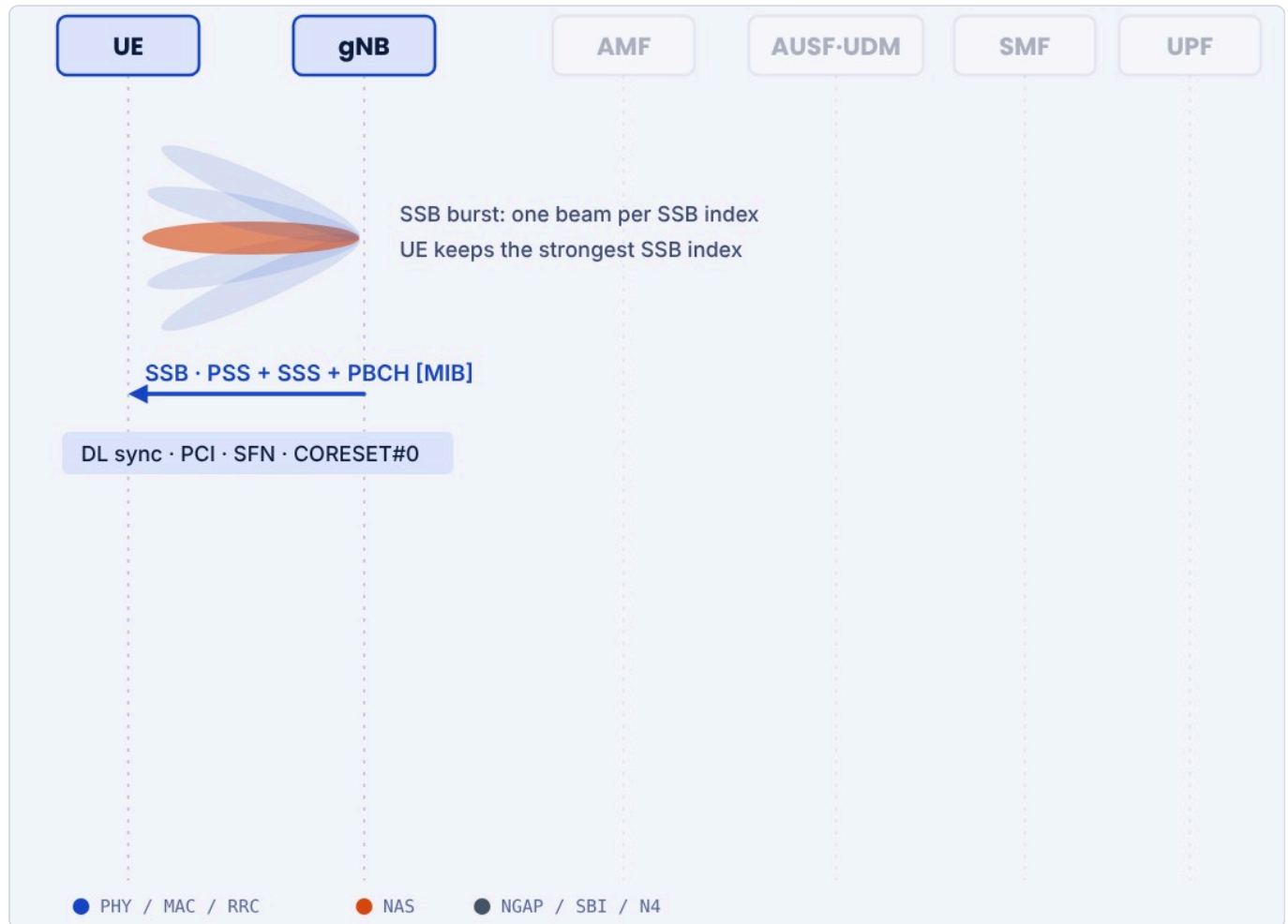
One framework replaces the separate barring schemes of LTE. The operator turns a few values in SIB1, and each UE decides for itself whether to send.

TS 22.261 · TS 24.501 §4.5 · TS 38.331 §5.3.14

5G SA Initial Registration

From the first SSB beam to the first IP packet: cell search, random access, registration and PDU session establishment, message by message.

13.1 Find the cell: SSB



The gNB sweeps SSB beams. PSS and SSS give timing and the physical cell ID. The MIB on PBCH gives the SFN and where to find SIB1. The UE keeps the strongest SSB index. That beam choice decides which PRACH occasion it will use.

TS 38.213 §4.1 · TS 38.331 MIB

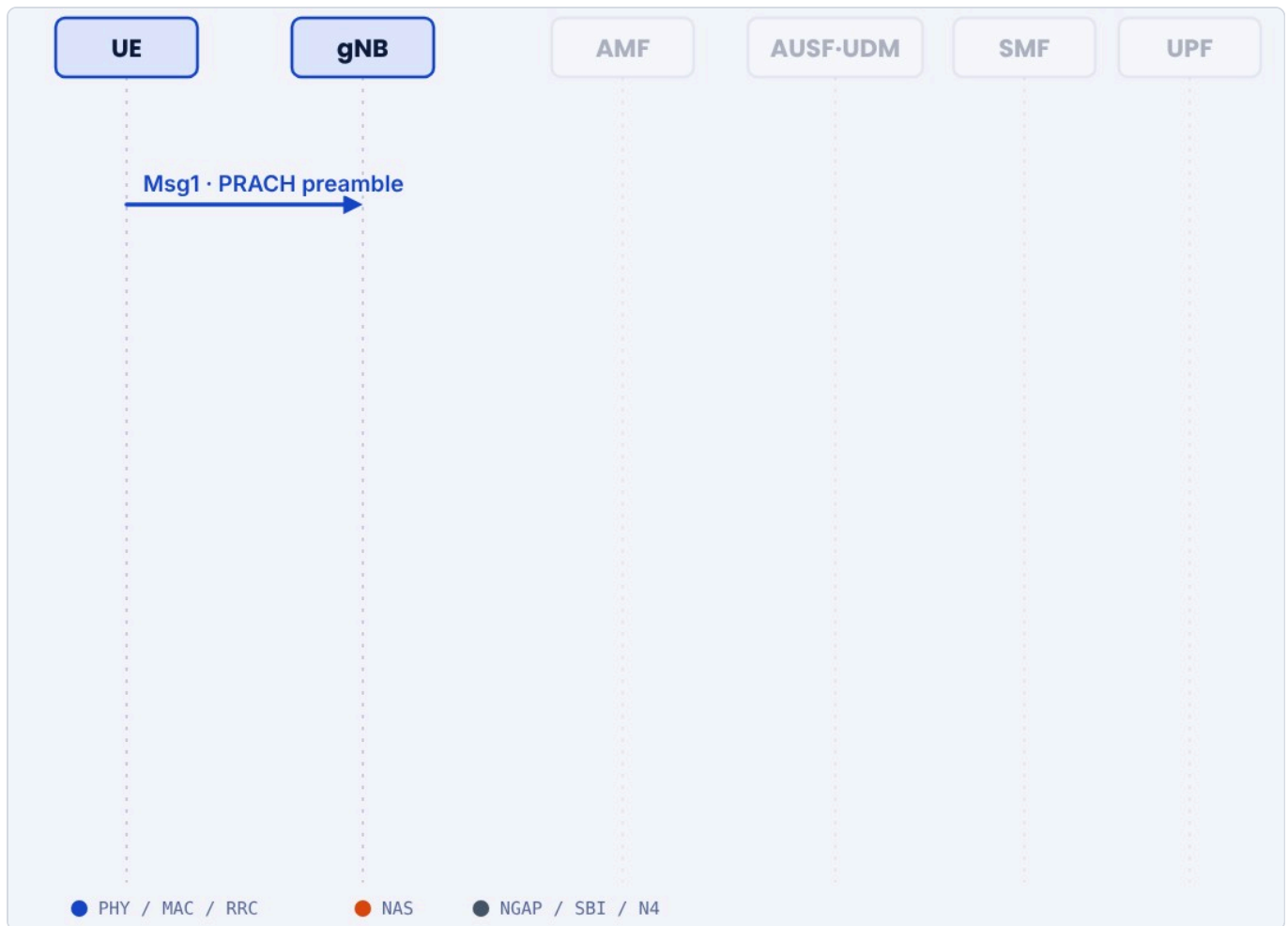
13.2 Read SIB1



DCI 1_0 scrambled with SI-RNTI in CORESET#0 schedules SIB1 on PDSCH. SIB1 carries the PLMN list, tracking area code, cell selection thresholds and the RACH configuration. The UE now camps on the cell.

TS 38.331 §5.2.2 · TS 38.304

13.3 Msg1: PRACH preamble



The UE sends a randomly chosen preamble in the PRACH occasion mapped to its SSB. It has no identity on the cell yet. The gNB measures the timing offset. The UE ramps its power on each retry.

TS 38.321 §5.1 · TS 38.213 §8.1

13.4 Msg2: Random Access Response



DCI 1_0 with RA-RNTI schedules the RAR. It gives the timing advance, an uplink grant for Msg3 and a Temporary C-RNTI. The UE computes the RA-RNTI itself from the PRACH occasion it used, and listens inside the RA response window.

TS 38.321 §5.1.4 · TS 38.213 §8.2

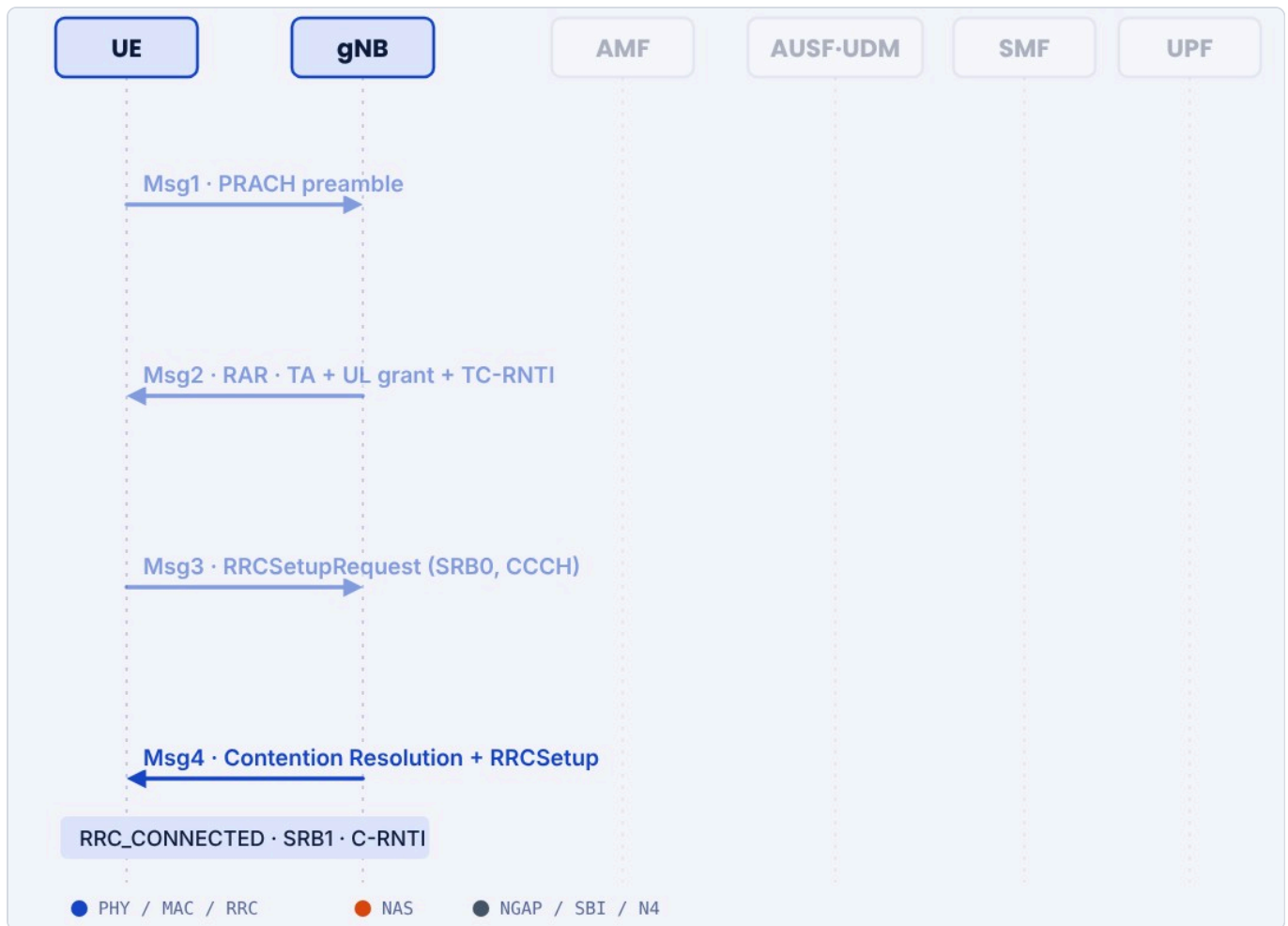
13.5 Msg3: RRCSetupRequest



On the granted PUSCH the UE sends ue-Identity and establishmentCause, here mo-Signalling. ue-Identity is ng-5G-S-TMSI-Part1 if the UE has one, otherwise a 39-bit random value.

TS 38.331 §5.3.3.3

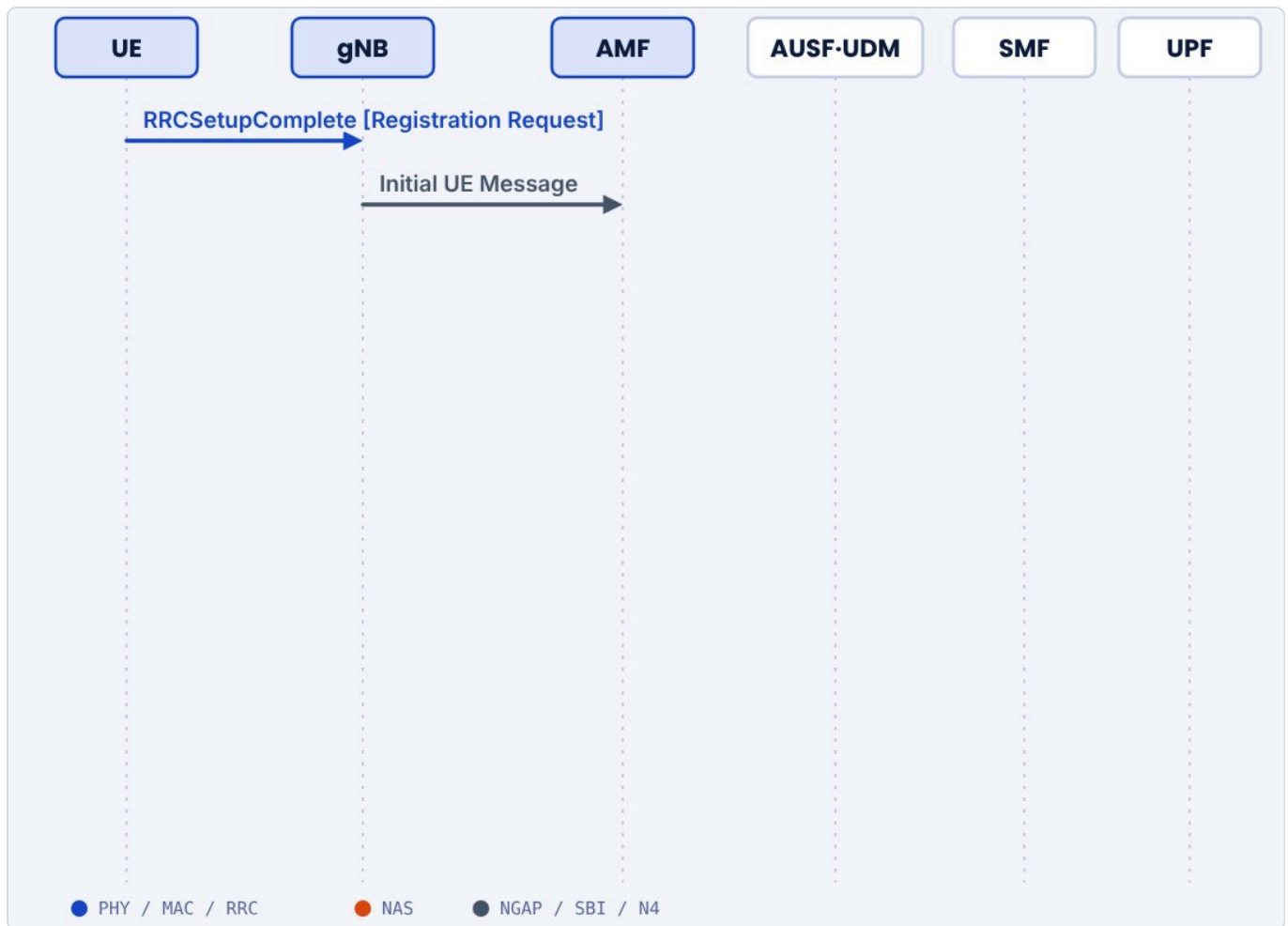
13.6 Msg4: RRCSetup



The Contention Resolution MAC CE echoes the Msg3 identity. The UE that sees its own identity has won. RRCSetup configures SRB1. The Temporary C-RNTI becomes the C-RNTI and the UE is RRC_CONNECTED.

TS 38.321 §5.1.5 · TS 38.331 §5.3.3.4

13.7 Registration Request reaches the AMF



RRCSetupComplete carries the NAS Registration Request: type initial, SUCI or 5G-GUTI, UE security capability, requested NSSAI. The gNB selects an AMF and forwards the NAS PDU in NGAP Initial UE Message. The UE starts T3510.

TS 24.501 §5.5.1.2.2 · TS 23.502 §4.2.2.2.2 · TS 38.413

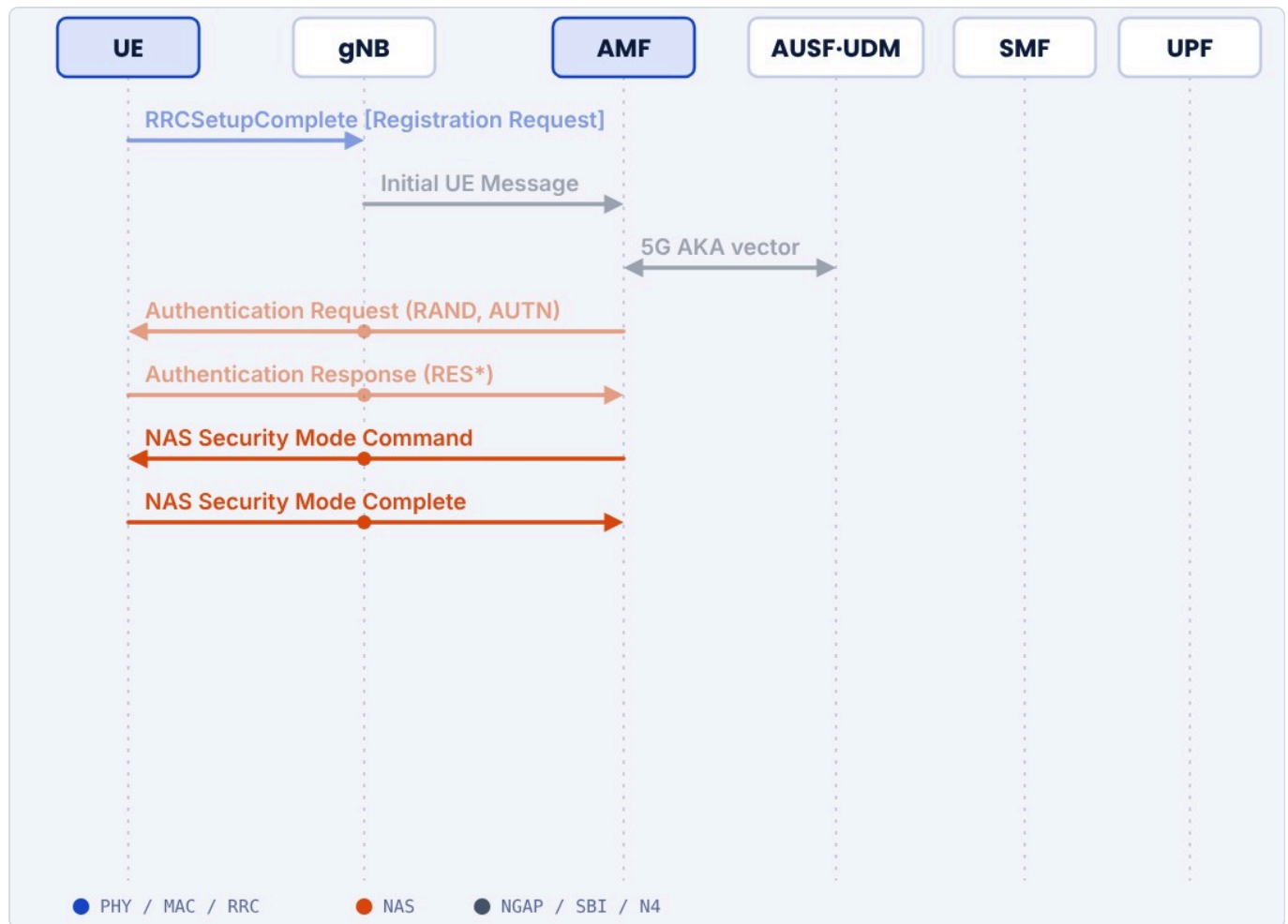
13.8 5G AKA authentication



UDM resolves the SUCI and creates the vector. The AMF sends RAND, AUTN, ngKSI and ABBA. The USIM checks AUTN and the UE returns RES*. The AMF checks HRES* against HXRES*, the AUSF checks RES* against XRES*. K_SEAF and K_AMF follow.

TS 33.501 §6.1.3.2

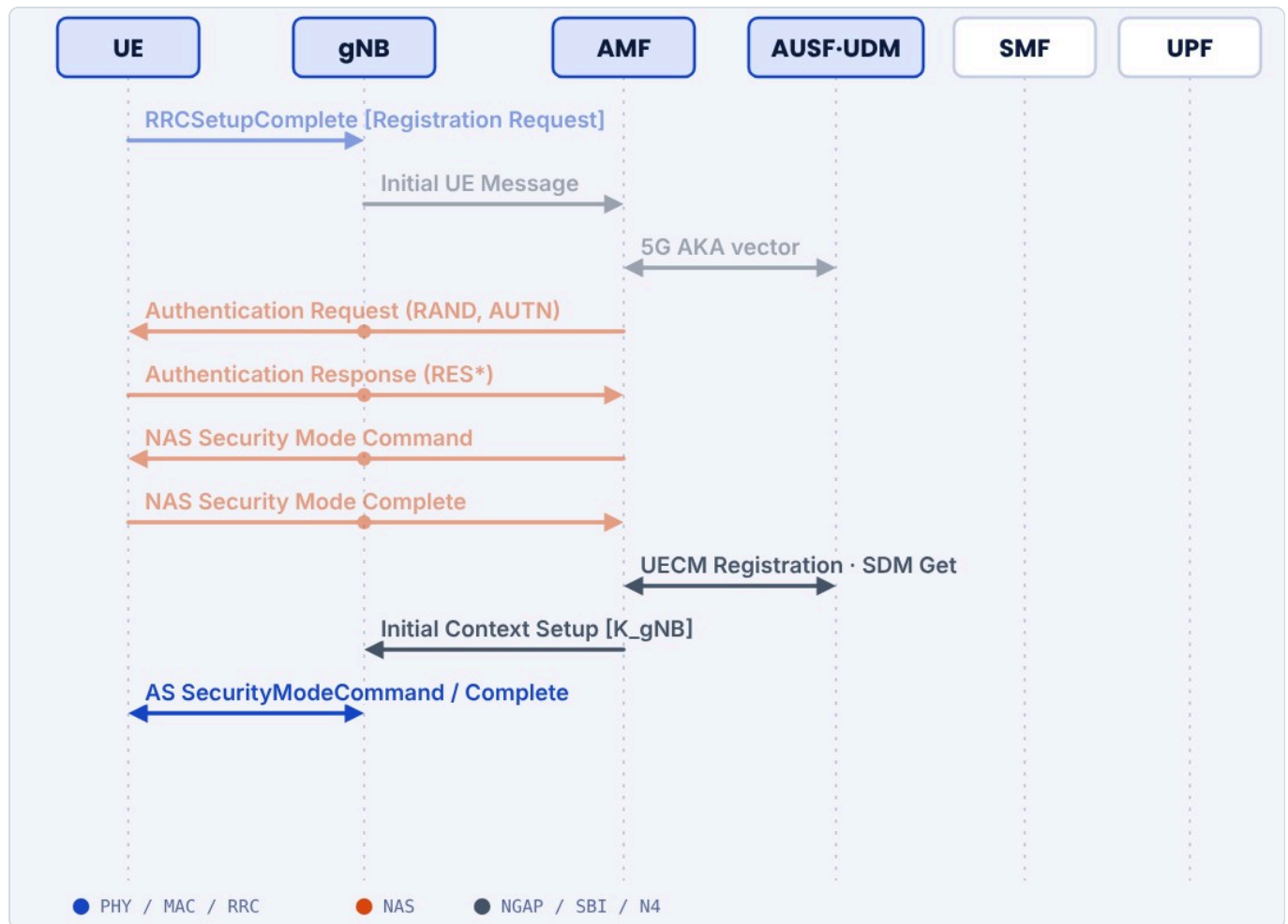
13.9 NAS security



The AMF selects the NAS ciphering and integrity algorithms and replays the UE security capabilities to stop bidding down. Security Mode Complete carries the full Registration Request in the NAS message container. From here NAS is ciphered and integrity protected.

TS 24.501 §5.4.2 · §5.5.1.2.2 · TS 33.501 §6.7.2

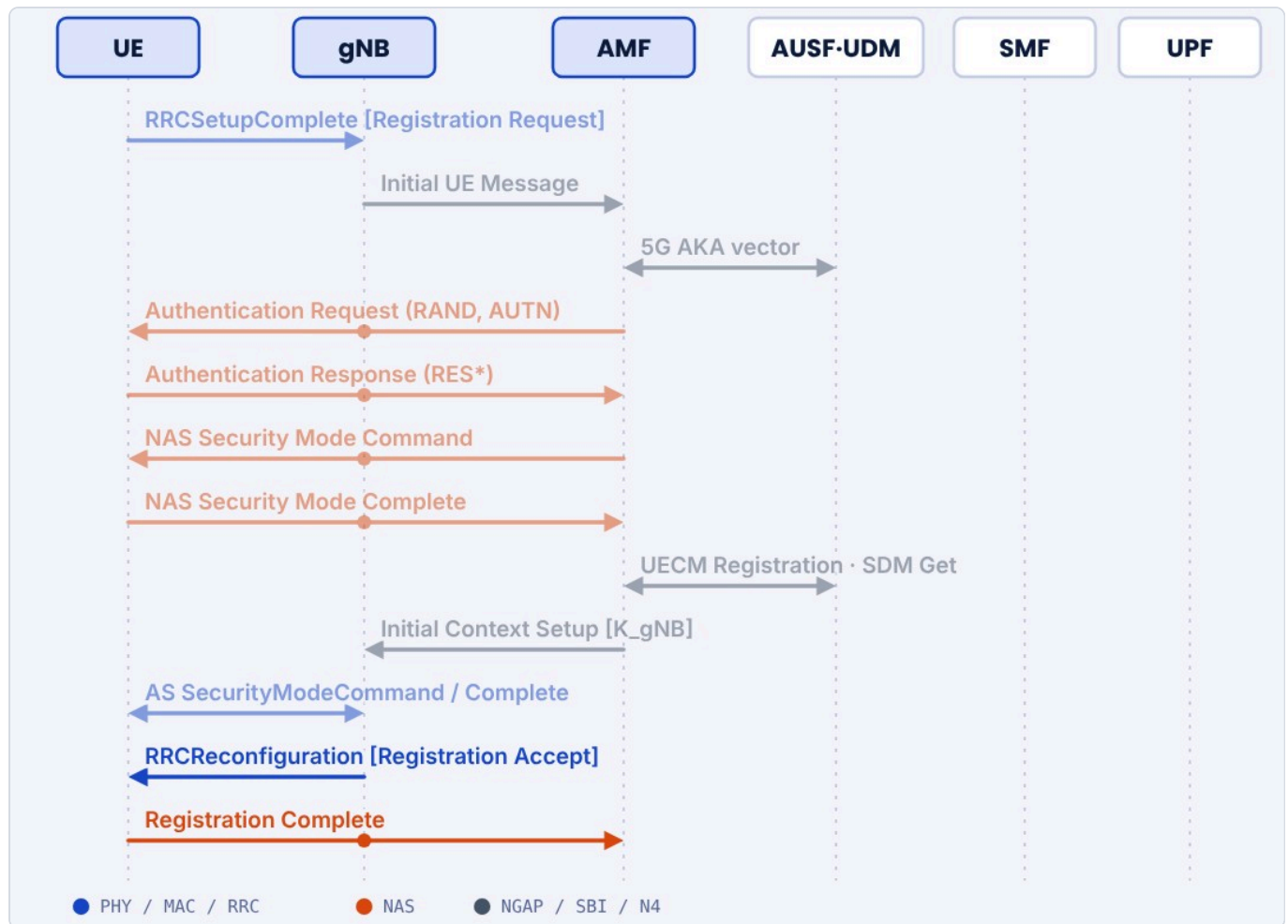
13.10 UE context and AS security



The AMF registers in UDM and fetches subscription data. Initial Context Setup gives the gNB K_gNB, the UE security capabilities and the allowed NSSAI. The gNB activates AS security with keys derived from K_gNB. It can also ask for the UE radio capabilities.

TS 23.502 §4.2.2.2.2 steps 14–21 · TS 38.413 · TS 38.331 §5.3.4

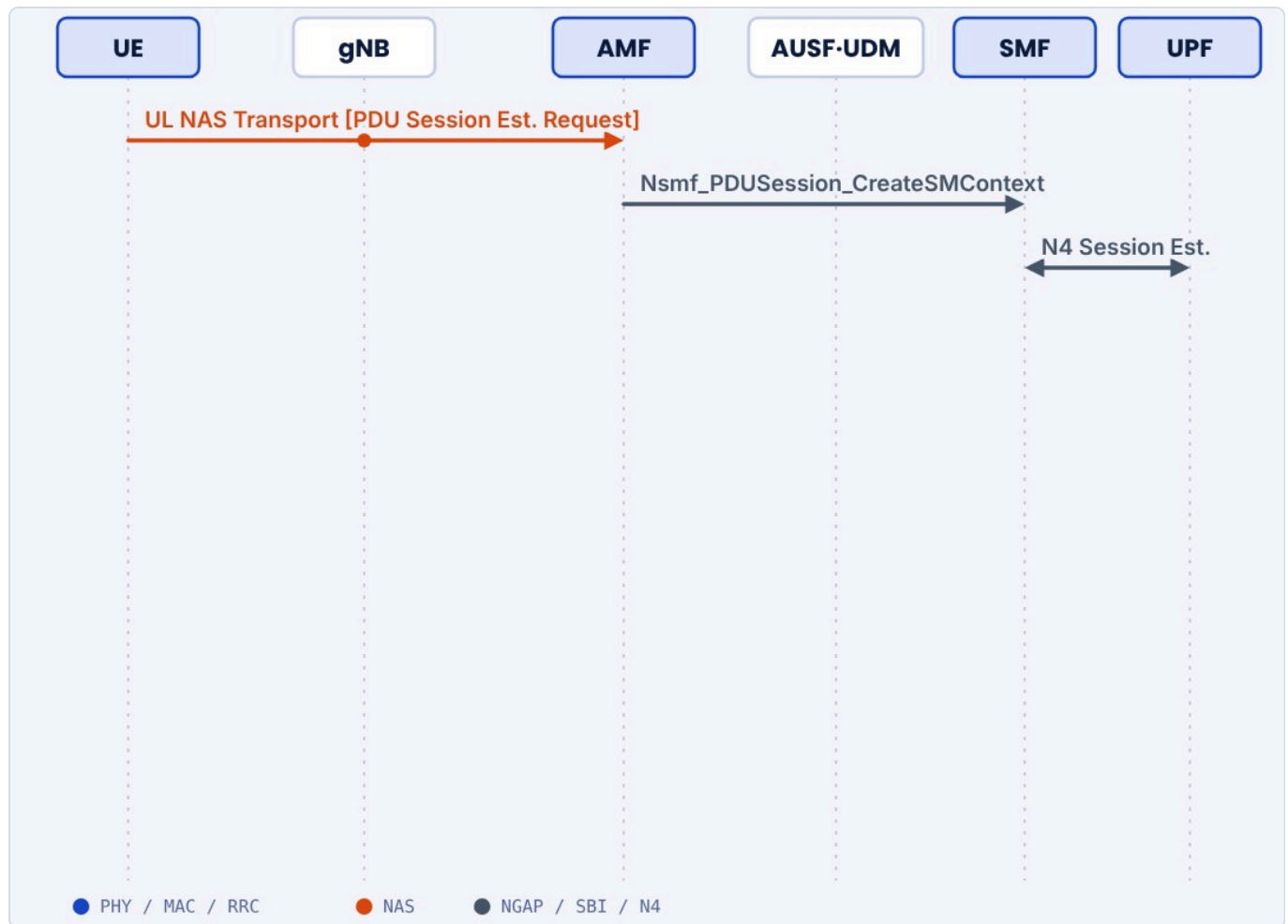
13.11 Registration Accept



Registration Accept assigns the 5G-GUTI, the TAI list, the allowed NSSAI and 5GS network feature support, such as IMS voice over PS. RRCReconfiguration sets up SRB2. The UE answers Registration Complete. It is registered, but still has no data path.

TS 24.501 §5.5.1.2.4 · TS 38.331 §5.3.5

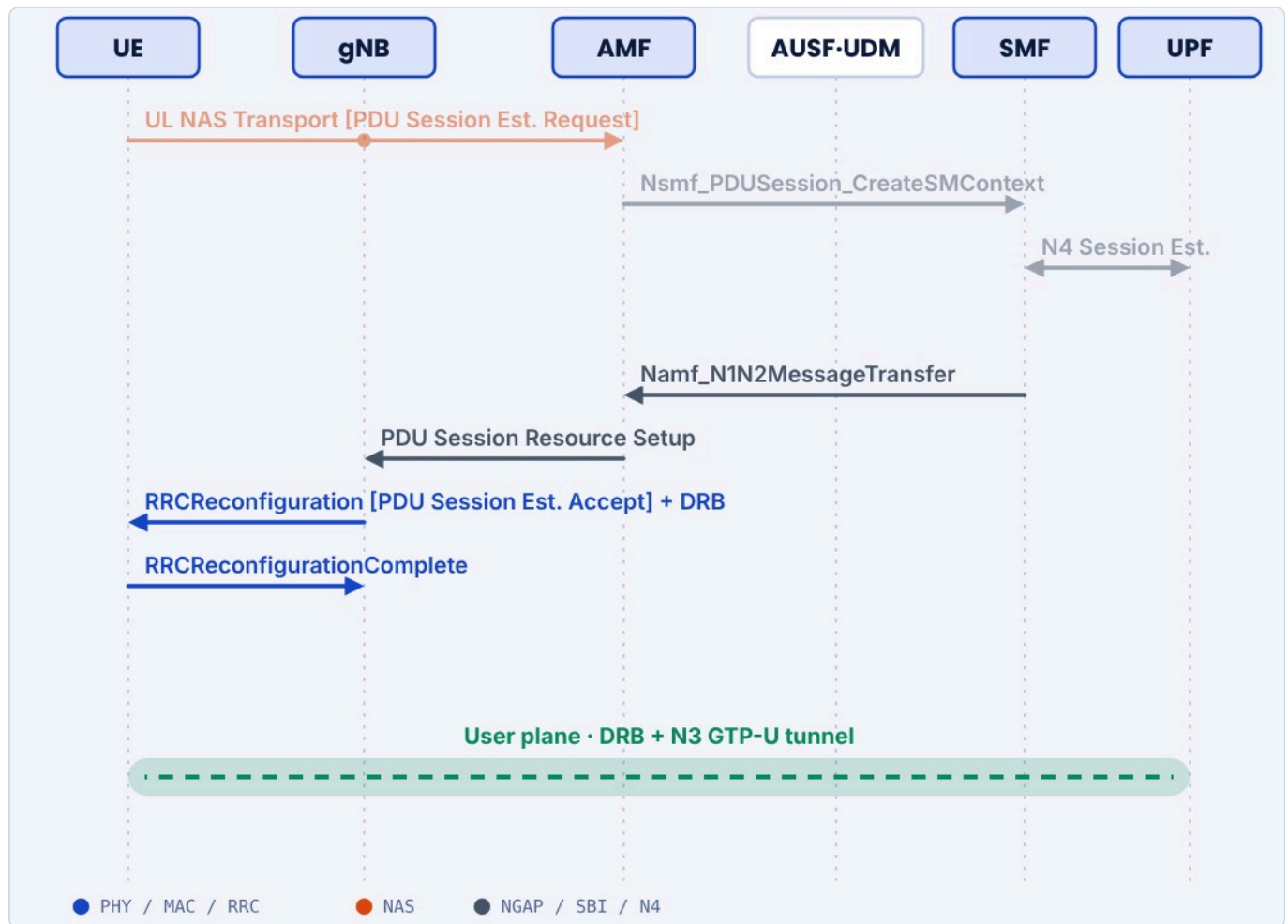
13.12 PDU Session Establishment Request



The UE asks for a PDU session: session ID, type, SSC mode and PCO, with DNN and S-NSSAI in the UL NAS Transport. The AMF selects an SMF. The SMF reads subscription data, gets policy from the PCF, selects a UPF, allocates the IP address and sets up N4.

TS 23.502 §4.3.2.2.1 steps 1–10 · TS 24.501 §6.4.1

13.13 Accept and user plane



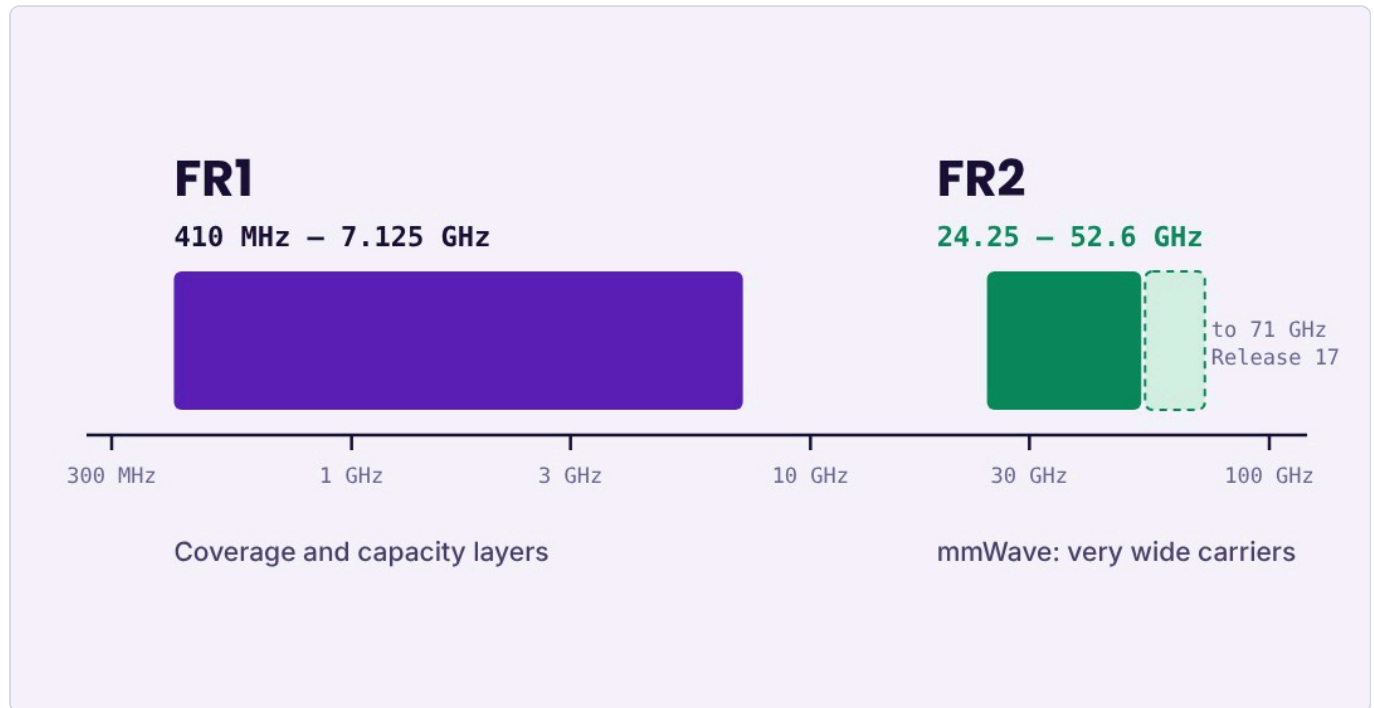
The SMF returns the Accept with the UE IP address, QoS rules and Session-AMBR, plus N2 information with the QoS flows and the UPF tunnel. The gNB sets up the DRB and returns its tunnel endpoint. The SMF updates the UPF over N4, and data can flow.

TS 23.502 §4.3.2.2.1 steps 11–17 · TS 38.413

5G NR Bands, NR-ARFCN and Bandwidth

Where a 5G carrier sits in the spectrum, how its channel number works, and how wide it can be.

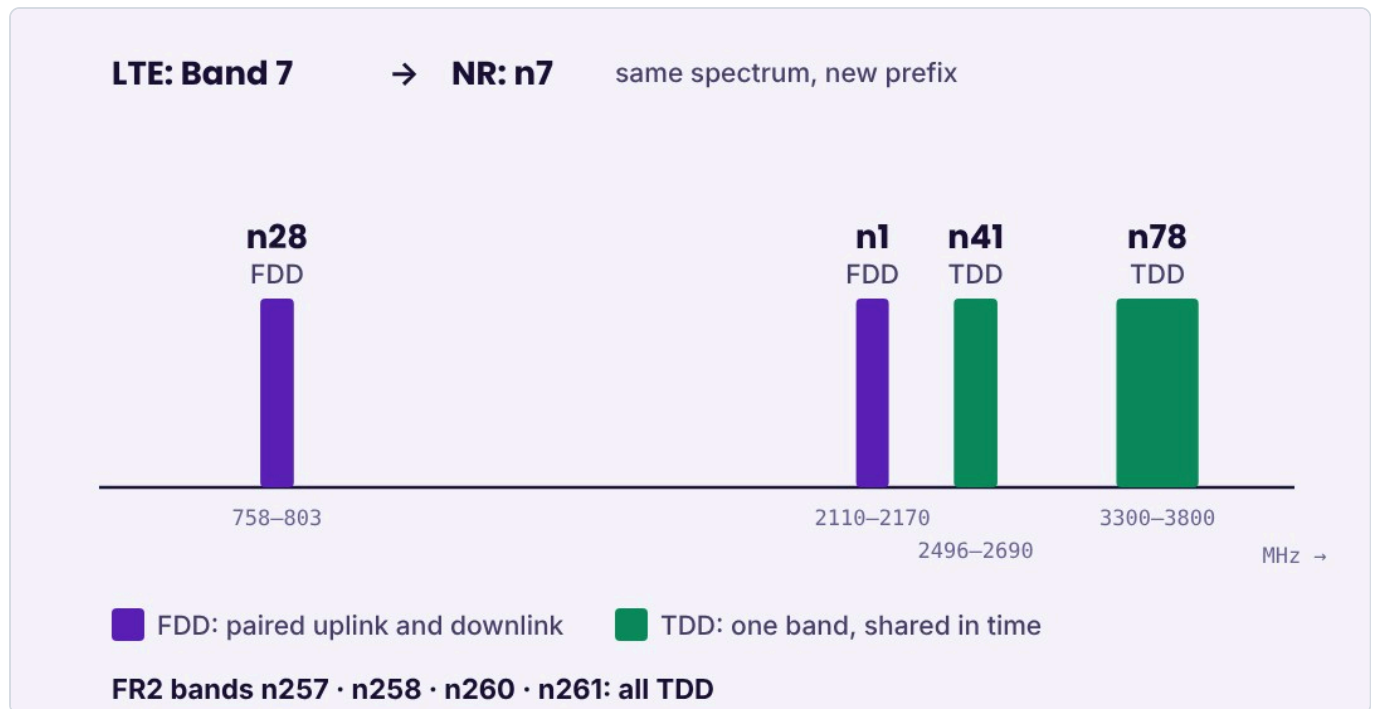
14.1 Two frequency ranges



NR works in two frequency ranges. FR1 runs from 410 MHz to 7.125 GHz. FR2, the millimetre wave range, runs from 24.25 to 52.6 GHz. Release 17 extends it to 71 GHz.

TS 38.104 §5.1

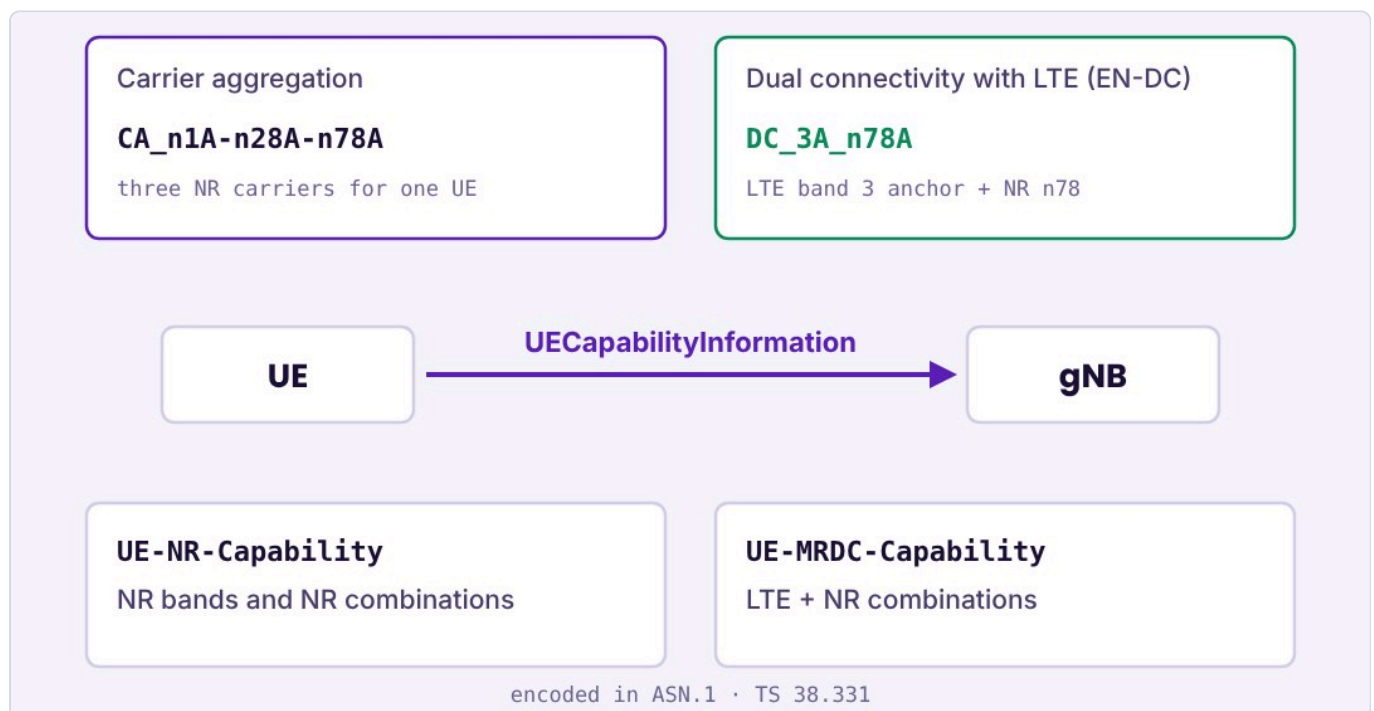
14.2 Operating bands



A network does not pick any frequency it likes. It uses operating bands, each with a number, the letter n in front, and a duplex mode. n28 and n1 are FDD. n41 and n78 are TDD. All FR2 bands are TDD.

TS 38.104 §5.2 · TS 38.101-1 Table 5.2-1

14.3 Band combinations



Bands can be combined, for carrier aggregation or for dual connectivity with LTE. Only combinations listed in the specification are allowed. The phone reports the ones it supports in its UE capability message, and the network configures only those.

TS 38.101-1 §5.5A · TS 38.101-3 · TS 38.331 §5.6.1

14.4 The channel number: NR-ARFCN

$$F = F_offs + \Delta F \times (N - N_offs)$$

Frequency range	Step ΔF	F_offs	N_offs	N range
0 – 3000 MHz	5 kHz	0 MHz	0	0 – 599999
3000 – 24250 MHz	15 kHz	3000 MHz	600000	600000 – 2016666
24250 – 100000 MHz	60 kHz	24250.08 MHz	2016667	2016667 – 3279165

The exact carrier position is given by a channel number, the NR-ARFCN. Frequency = offset + step $\times$ (channel number – its offset). The step is 5 kHz below 3 GHz, 15 kHz up to 24.25 GHz, and 60 kHz above that.

TS 38.104 §5.4.2.1

14.5 Try it on a real log

SCell 1 in the log

N = 154570

154570 $\times$ 5 kHz

= 772.85 MHz

band n28

PCell in the log

N = 641760

≥ 600000 , so the 15 kHz range

3000 MHz + (641760 – 600000) $\times$ 15 kHz

= 3626.40 MHz

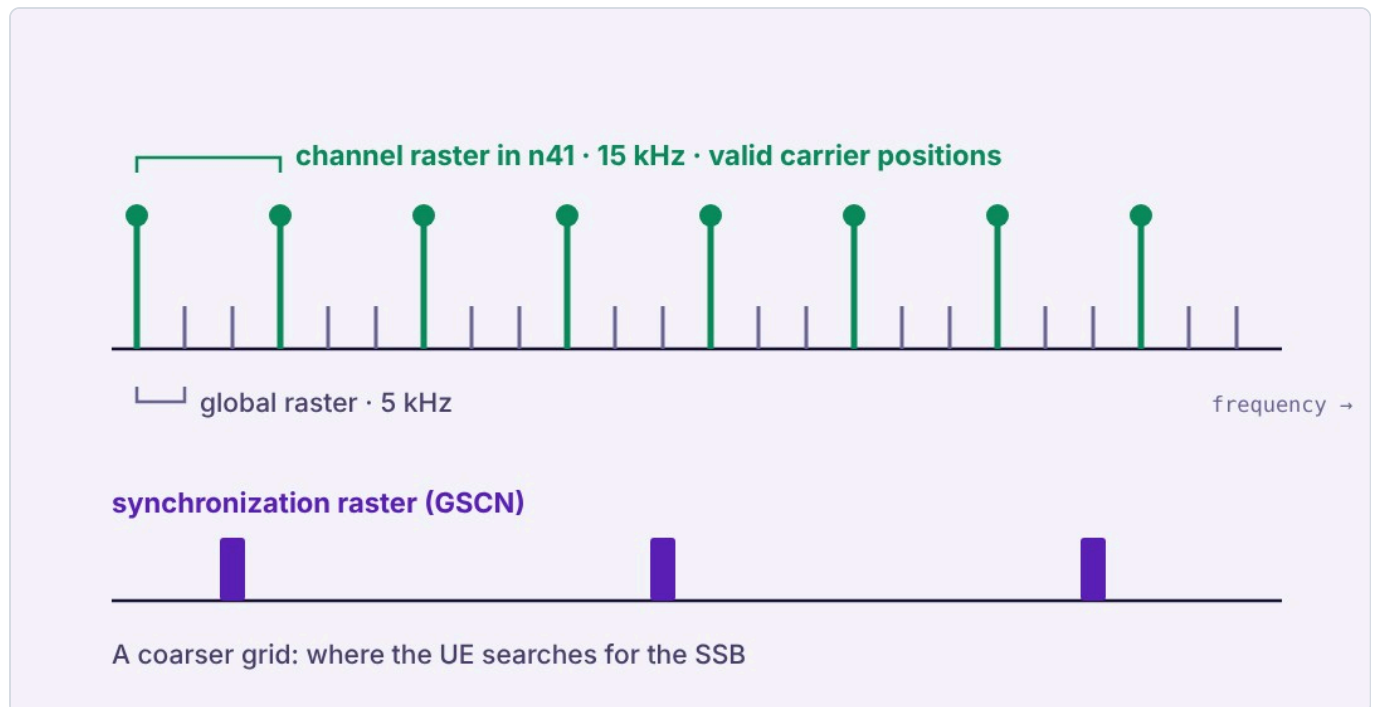
band n78

Also in the log: 431070 $\times$ 5 kHz = 2155.35 MHz, band n1

Channel 154570 $\times$ 5 kHz gives 772.85 MHz, in band n28. Channel 641760 is above 600000, so we use 15 kHz steps starting at 3 GHz. The result is 3626.4 MHz, in band n78.

NR-ARFCN values from a live-network log · TS 38.104 §5.4.2.1

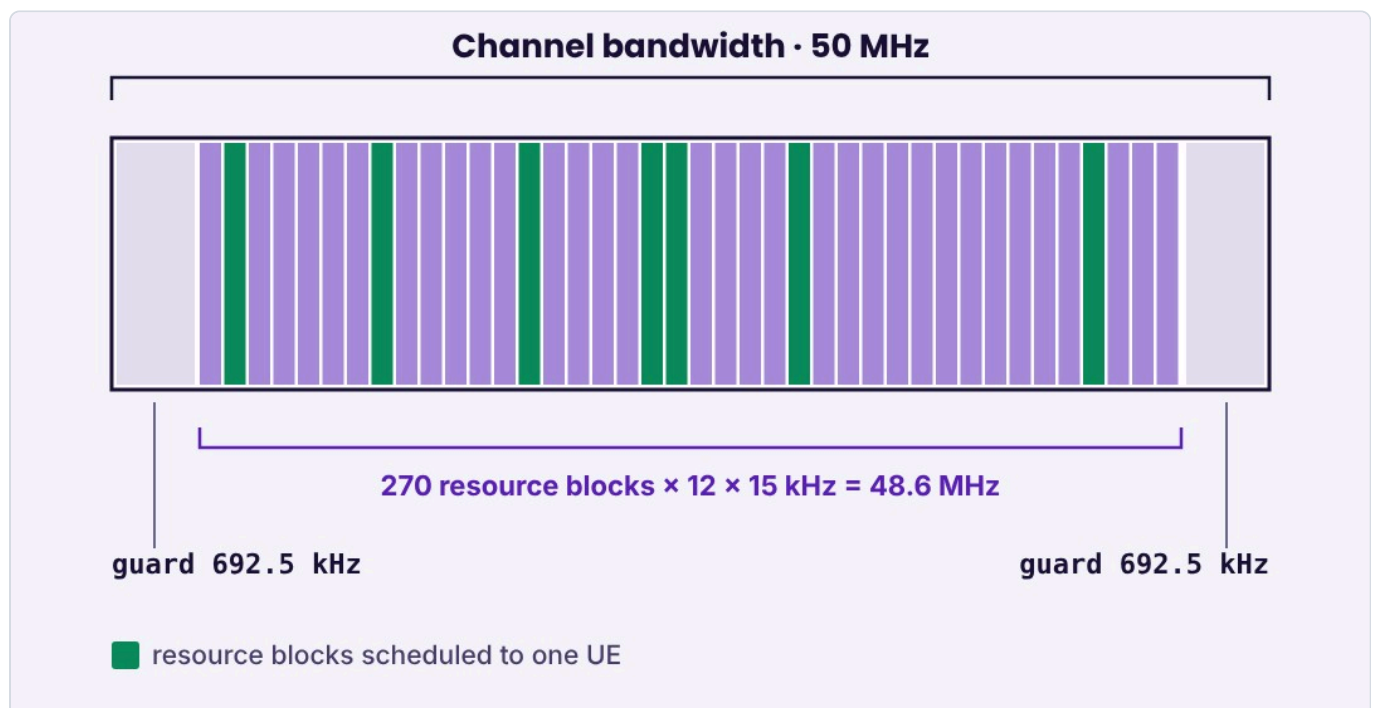
14.6 Channel raster



Not every point of the global grid can be used. Each band has its own channel raster. In n41 it is 15 kHz, so only every third point of the 5 kHz grid is a valid carrier position. A coarser synchronization raster tells the phone where to search for the SSB.

TS 38.104 §5.4.2.3 · §5.4.3

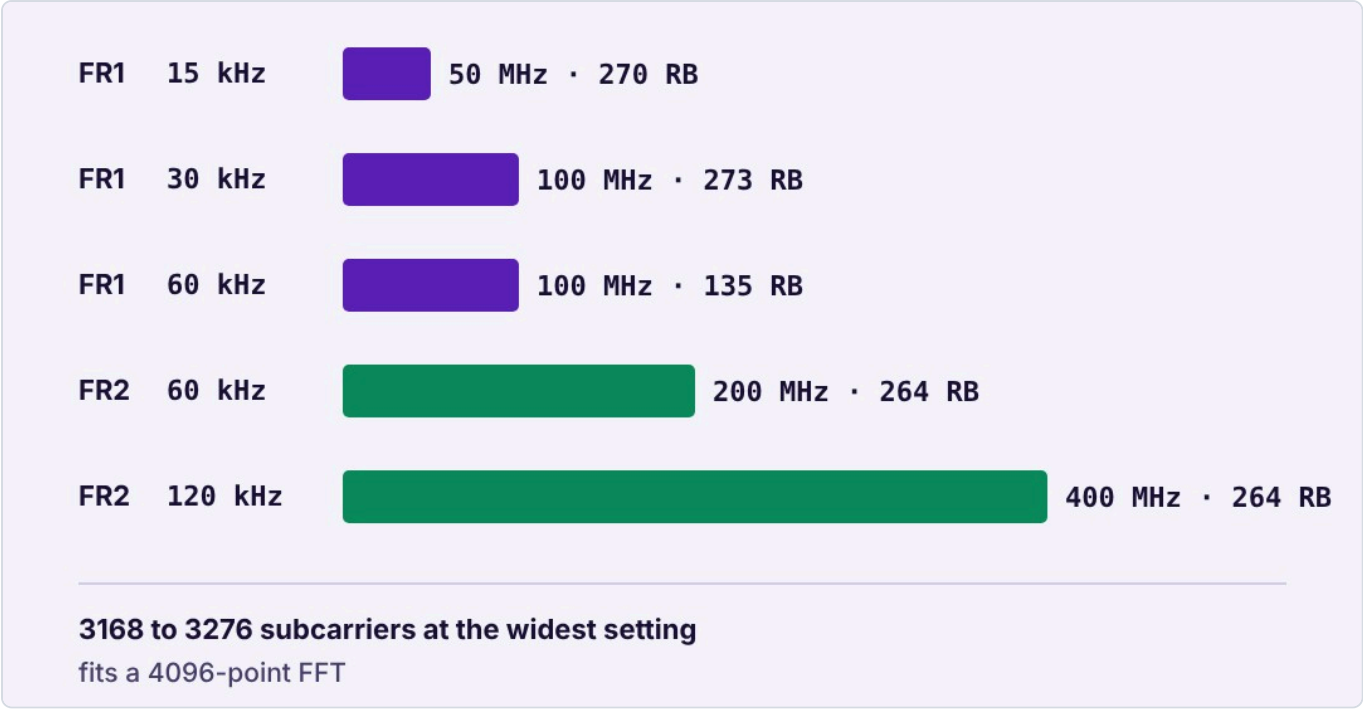
14.7 Channel bandwidth and guard bands



The channel bandwidth is not all usable. Resource blocks fill the middle and a guard band is left on each side. At 15 kHz spacing, a 50 MHz channel carries 270 resource blocks: 48.6 MHz, with about 700 kHz of guard on each side.

TS 38.101-1 §5.3.2 · §5.3.3

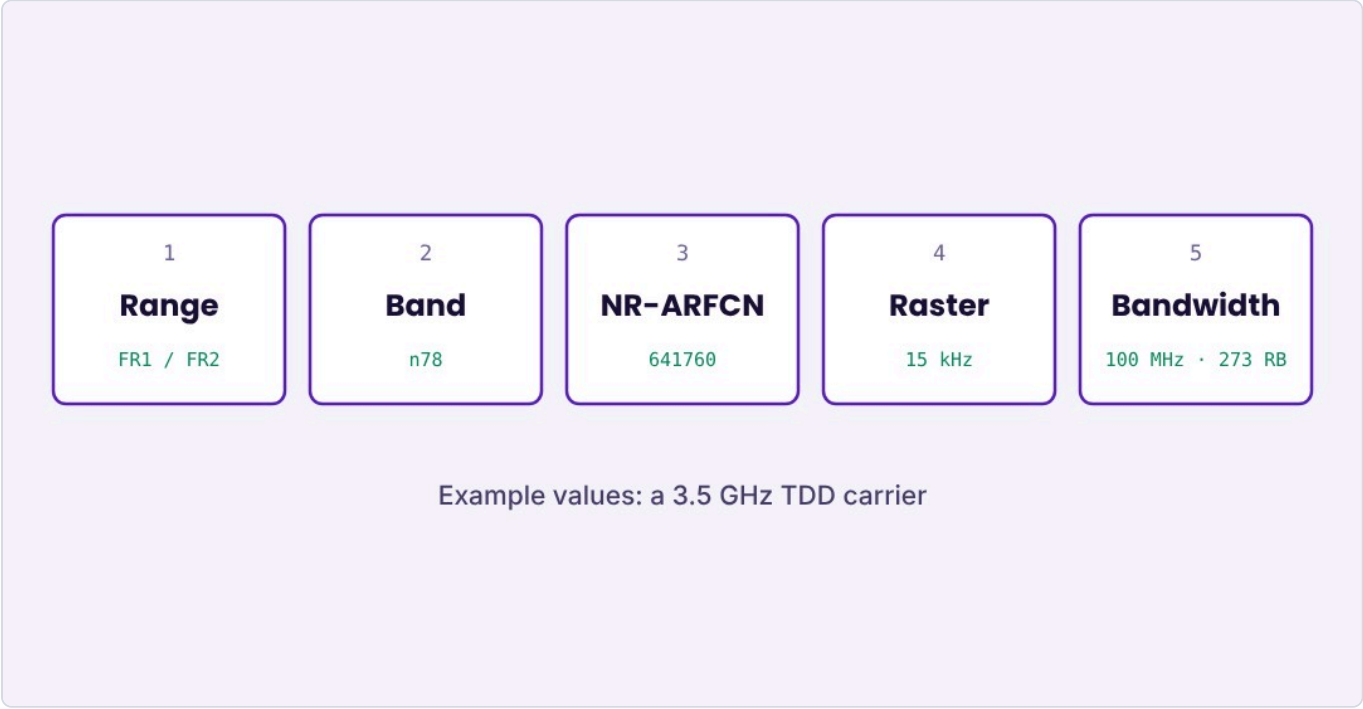
14.8 Maximum bandwidth per subcarrier spacing



The widest channel depends on the subcarrier spacing. In FR1, 15 kHz reaches 50 MHz and 30 kHz reaches 100 MHz with 273 resource blocks. In FR2, 120 kHz reaches 400 MHz. Each is about 3200 subcarriers, which fits a 4096-point FFT.

TS 38.101-1 Table 5.3.2-1 · TS 38.101-2 Table 5.3.2-1

14.9 Five ideas to read any log



To sum up: range, then band, then channel number, then raster, then bandwidth. With these five ideas you can read the frequency part of any 5G log.

TS 38.104 · TS 38.101-1 · TS 38.101-2